

(12) **United States Patent**
Thakkar et al.

(10) **Patent No.:** **US 12,403,312 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) CIRCUITRY FOR MEDICAL STIMULATION SYSTEMS	(56) References Cited
(71) Applicant: Lungpacer Medical Inc. , Vancouver (CA)	U.S. PATENT DOCUMENTS
(72) Inventors: Viral S. Thakkar , Burnaby (CA); Douglas G. Evans , Downingtown, PA (US)	1,002,254 A 9/1911 Forse 1,003,501 A 9/1911 Podlesak (Continued)
(73) Assignee: Lungpacer Medical Inc. , Vancouver (CA)	FOREIGN PATENT DOCUMENTS
(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.	CN 1652839 A 8/2005 CN 102143781 A 8/2011 (Continued)
(21) Appl. No.: 18/454,550	OTHER PUBLICATIONS
(22) Filed: Aug. 23, 2023	Antonica, A., et al., "Vagal Control of Lymphocyte Release from Rat Thymus," Journal of the Autonomic Nervous System, Elsevier, Aug. 1994, vol. 48(3), pp. 187-197. (Continued)
(65) Prior Publication Data US 2024/0157147 A1 May 16, 2024	Primary Examiner — Nicole F Johnson (74) <i>Attorney, Agent, or Firm</i> — Bookoff McAndrews, PLLC
Related U.S. Application Data	(57) ABSTRACT
(63) Continuation of application No. 16/898,157, filed on Jun. 10, 2020, now Pat. No. 11,771,900. (Continued)	A medical device that includes a flexible body having a proximal end and a distal end, and an electrode positioned on the body proximate the distal end. The electrode is configured to provide an electrical charge for stimulating tissue. The medical device includes an electrical connection positioned on the body proximate the proximal end. The electrical connection is configured to electrically couple the electrode to a power source. The medical device includes an electrical lead connecting the electrode to the electrical connection. The lead is on or in the flexible body. The body is configured to have a first configuration prior to being secured to an exterior of a tube and a second configuration having a shape that conforms to a profile of the tube with the electrode secured to an exterior of the tube.
(51) Int. Cl. <i>A61N 1/00</i> (2006.01) <i>A61N 1/04</i> (2006.01) (Continued)	20 Claims, 19 Drawing Sheets
(52) U.S. Cl. CPC <i>A61N 1/3611</i> (2013.01); <i>A61N 1/0452</i> (2013.01); <i>A61N 1/0456</i> (2013.01); (Continued)	
(58) Field of Classification Search CPC A61N 1/3611; A61N 1/0452 (Continued)	

Related U.S. Application Data					
(60)	Provisional application No. 62/860,452, filed on Jun. 12, 2019.		4,905,698	A	3/1990 Strohl, Jr. et al.
			4,911,174	A	3/1990 Pederson et al.
			4,934,049	A	6/1990 Kiekhäfer et al.
			4,944,088	A	7/1990 Doan et al.
(51)	Int. Cl. <i>A61N 1/05</i> (2006.01) <i>A61N 1/36</i> (2006.01)		4,951,682	A	8/1990 Petre
			4,957,110	A	9/1990 Vogel et al.
			4,989,617	A	2/1991 Memberg et al.
			5,005,587	A	4/1991 Scott
(52)	U.S. Cl. CPC <i>A61N 1/0476</i> (2013.01); <i>A61N 1/05</i> (2013.01); <i>A61N 1/3601</i> (2013.01)		5,010,895	A	4/1991 Maurer et al.
			5,029,585	A	7/1991 Lieber et al.
			5,036,848	A	8/1991 Hewson
			5,042,143	A	8/1991 Holleman et al.
(58)	Field of Classification Search USPC 607/42 See application file for complete search history.		5,056,519	A	10/1991 Vince
			5,115,818	A	5/1992 Holleman et al.
			5,146,918	A	9/1992 Kallok et al.
			5,170,802	A	12/1992 Mehra
(56)	References Cited U.S. PATENT DOCUMENTS		5,184,621	A	2/1993 Vogel et al.
			5,224,491	A	7/1993 Mehra
			5,243,995	A	9/1993 Maier
			5,265,604	A	11/1993 Vince
			5,267,569	A	12/1993 Lienhard
			5,314,463	A	5/1994 Camps et al.
			5,316,009	A	5/1994 Yamada
			5,324,322	A	6/1994 Grill, Jr. et al.
			5,330,522	A	7/1994 Kreyenhagen
			5,335,657	A	8/1994 Terry, Jr. et al.
			5,345,936	A	9/1994 Pomeranz et al.
			5,383,923	A	1/1995 Webster, Jr.
			5,397,338	A	3/1995 Grey et al.
			5,411,025	A	5/1995 Webster, Jr.
			5,417,208	A	5/1995 Winkler
			5,451,206	A	9/1995 Young
			5,456,254	A	10/1995 Pietroski et al.
			5,465,717	A	11/1995 Imran et al.
			5,476,498	A	12/1995 Ayers
			5,486,159	A	1/1996 Mahurkar
			5,507,725	A	4/1996 Savage et al.
			5,524,632	A	6/1996 Stein et al.
			5,527,358	A	6/1996 Mehmanesh et al.
			5,531,686	A	7/1996 Lundquist et al.
			5,549,655	A	8/1996 Erickson
			5,555,618	A	9/1996 Winkler
			5,567,724	A	10/1996 Kelleher et al.
			5,584,873	A	12/1996 Shoberg et al.
			5,604,231	A	2/1997 Smith et al.
			5,617,876	A	4/1997 Van Duyl
			5,647,870	A	7/1997 Kordis et al.
			5,665,103	A	9/1997 Lafontaine et al.
			5,678,535	A	10/1997 DiMarco
			5,683,370	A	11/1997 Luther et al.
			5,709,853	A	1/1998 Iino et al.
			5,716,392	A	2/1998 Bourgeois et al.
			5,733,255	A	3/1998 Dinh et al.
			5,755,765	A	5/1998 Hyde et al.
			5,776,111	A	7/1998 Tesio
			5,779,732	A	7/1998 Amundson
			5,782,828	A	7/1998 Chen et al.
			5,785,706	A	7/1998 Bednarek
			5,788,681	A	8/1998 Weaver et al.
			5,813,399	A	9/1998 Isaza et al.
			5,814,086	A	9/1998 Hirschberg et al.
			RE35,924	E	10/1998 Winkler
			5,824,027	A	10/1998 Hoffer et al.
			5,827,192	A	10/1998 Gopakumaran et al.
			5,916,163	A	6/1999 Panescu et al.
			5,944,022	A	8/1999 Nardella et al.
			5,954,761	A	9/1999 Machek et al.
			5,967,977	A	10/1999 Mullis et al.
			5,967,978	A	10/1999 Littmann et al.
			5,971,933	A	10/1999 Gopakumaran et al.
			5,983,126	A	11/1999 Wittkamp
			6,006,134	A	12/1999 Hill et al.
			6,024,702	A	2/2000 Iversen
			6,096,728	A	8/2000 Collins et al.
			6,120,476	A	9/2000 Fung et al.
			6,123,699	A	9/2000 Webster, Jr.
			6,126,649	A	10/2000 Vantassel et al.
			6,136,021	A	10/2000 Tockman et al.
			6,157,862	A	12/2000 Brownlee et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

6,161,029	A	12/2000	Spreigl et al.	7,225,019	B2	5/2007	Jahns et al.
6,165,133	A	12/2000	Rapoport et al.	7,229,429	B2	6/2007	Martin et al.
6,166,048	A	12/2000	Bencherif	7,231,260	B2	6/2007	Wallace et al.
6,171,277	B1	1/2001	Ponzi	7,235,070	B2	6/2007	Vanney
6,183,463	B1	2/2001	Webster, Jr.	7,269,459	B1	9/2007	Koh
6,198,970	B1	3/2001	Freed et al.	7,277,757	B2	10/2007	Casavant et al.
6,198,974	B1	3/2001	Webster, Jr.	7,283,875	B2	10/2007	Larsson et al.
6,201,994	B1	3/2001	Warman et al.	7,340,302	B1	3/2008	Falkenberg et al.
6,208,881	B1	3/2001	Champeau	7,363,085	B1	4/2008	Benser et al.
6,210,339	B1	4/2001	Kiepen et al.	7,363,086	B1	4/2008	Koh et al.
6,212,435	B1	4/2001	Lattner et al.	7,371,220	B1	5/2008	Koh et al.
6,216,045	B1	4/2001	Black et al.	7,416,552	B2	8/2008	Paul et al.
6,236,892	B1	5/2001	Feler	7,421,296	B1	9/2008	Benser et al.
6,240,320	B1	5/2001	Spehr et al.	7,454,244	B2	11/2008	Kassab et al.
6,249,708	B1	6/2001	Nelson et al.	7,519,425	B2	4/2009	Benser et al.
6,251,126	B1	6/2001	Ottenhoff et al.	7,519,426	B1	4/2009	Koh et al.
6,269,269	B1	7/2001	Ottenhoff et al.	7,522,953	B2	4/2009	Gharib et al.
6,292,695	B1	9/2001	Webster, Jr. et al.	7,553,305	B2	6/2009	Honebrink et al.
6,295,475	B1	9/2001	Morgan	7,555,349	B2	6/2009	Wessman et al.
6,322,559	B1	11/2001	Daulton et al.	7,569,029	B2	8/2009	Clark et al.
6,360,740	B1	3/2002	Ward et al.	7,591,265	B2	9/2009	Lee et al.
6,397,108	B1	5/2002	Camps et al.	7,593,760	B2	9/2009	Rodriguez et al.
6,400,976	B1	6/2002	Champeau	7,613,524	B2	11/2009	Jordan
6,415,183	B1	7/2002	Scheiner et al.	7,636,600	B1	12/2009	Koh
6,415,187	B1	7/2002	Kuzma et al.	7,670,284	B2	3/2010	Padget et al.
6,438,427	B1	8/2002	Rexhausen et al.	7,672,728	B2	3/2010	Libbus et al.
6,445,953	B1	9/2002	Bulkes et al.	7,672,729	B2	3/2010	Koh et al.
6,449,507	B1	9/2002	Hill et al.	7,676,275	B1	3/2010	Farazi et al.
6,463,327	B1	10/2002	Lurie et al.	7,676,910	B2	3/2010	Kiepen et al.
6,493,590	B1	12/2002	Wessman et al.	7,697,984	B2	4/2010	Hill et al.
6,501,991	B1	12/2002	Honeck et al.	7,747,323	B2	6/2010	Libbus et al.
6,508,802	B1	1/2003	Rosengart et al.	7,771,388	B2	8/2010	Olsen et al.
6,526,321	B1	2/2003	Spehr	7,783,362	B2	8/2010	Whitehurst et al.
6,569,114	B2	5/2003	Ponzi et al.	7,794,407	B2	9/2010	Rothenberg
6,584,362	B1	6/2003	Scheiner et al.	7,797,050	B2	9/2010	Libbus et al.
6,585,718	B2	7/2003	Hayzelden et al.	7,809,442	B2	10/2010	Bolea et al.
6,587,726	B2	7/2003	Lurie et al.	7,813,805	B1	10/2010	Farazi
6,602,242	B1	8/2003	Fung et al.	7,819,883	B2	10/2010	Westlund et al.
6,610,713	B2	8/2003	Tracey	7,840,270	B2	11/2010	Ignagni et al.
6,615,085	B1	9/2003	Boveja	7,853,302	B2	12/2010	Rodriguez et al.
6,630,611	B1	10/2003	Malowanec	7,869,865	B2	1/2011	Govari et al.
6,643,552	B2	11/2003	Edell et al.	7,891,085	B1	2/2011	Kuzma et al.
6,651,652	B1	11/2003	Waard	7,925,352	B2	4/2011	Stack et al.
6,682,526	B1	1/2004	Jones et al.	7,949,409	B2	5/2011	Bly et al.
6,702,780	B1	3/2004	Gilboa et al.	7,949,412	B1	5/2011	Harrison et al.
6,718,208	B2	4/2004	Hill et al.	7,962,214	B2	6/2011	Byerman et al.
6,721,603	B2	4/2004	Zabara et al.	7,962,215	B2	6/2011	Ignagni et al.
6,757,970	B1	7/2004	Kuzma et al.	7,970,475	B2	6/2011	Tehrani et al.
6,778,854	B2	8/2004	Puskas	7,972,323	B1	7/2011	Bencini et al.
6,779,257	B2	8/2004	Kiepen et al.	7,974,693	B2	7/2011	David et al.
6,844,713	B2	1/2005	Steber et al.	7,974,705	B2	7/2011	Zdeblick et al.
RE38,705	E	2/2005	Hill et al.	7,979,128	B2	7/2011	Tehrani et al.
6,881,211	B2	4/2005	Schweikert et al.	7,994,655	B2	8/2011	Bauer et al.
6,885,888	B2	4/2005	Rezai	8,000,765	B2	8/2011	Rodriguez et al.
6,907,285	B2	6/2005	Denker et al.	8,019,439	B2	9/2011	Kuzma et al.
6,934,583	B2	8/2005	Weinberg et al.	8,021,327	B2	9/2011	Selkee
6,981,314	B2	1/2006	Black et al.	8,036,750	B2	10/2011	Caparso et al.
6,999,820	B2	2/2006	Jordan	8,050,765	B2	11/2011	Lee et al.
7,018,374	B2	3/2006	Schon et al.	8,052,607	B2	11/2011	Byrd
7,047,627	B2	5/2006	Black et al.	8,104,470	B2	1/2012	Lee et al.
7,071,194	B2	7/2006	Teng	8,116,872	B2	2/2012	Tehrani et al.
7,072,720	B2	7/2006	Puskas	8,121,692	B2	2/2012	Haefner et al.
7,077,823	B2	7/2006	McDaniel	8,135,471	B2	3/2012	Zhang et al.
7,080,645	B2	7/2006	Genger et al.	8,140,164	B2	3/2012	Tehrani et al.
7,082,331	B1	7/2006	Park et al.	8,147,486	B2	4/2012	Honour et al.
7,130,700	B2	10/2006	Gardeski et al.	8,160,701	B2	4/2012	Zhao et al.
7,142,903	B2	11/2006	Rodriguez et al.	8,160,711	B2	4/2012	Tehrani et al.
7,149,585	B2	12/2006	Wessman et al.	8,195,297	B2	6/2012	Penner
7,155,278	B2	12/2006	King et al.	8,200,336	B2	6/2012	Tehrani et al.
7,167,751	B1	1/2007	Whitehurst et al.	8,206,343	B2	6/2012	Racz
7,168,429	B2	1/2007	Matthews et al.	8,224,456	B2	7/2012	Daglow et al.
7,184,829	B2	2/2007	Hill et al.	8,233,987	B2	7/2012	Gelfand et al.
7,206,636	B1	4/2007	Turcott	8,233,993	B2	7/2012	Jordan
7,212,867	B2	5/2007	Van Venrooij et al.	8,239,037	B2	8/2012	Glenn et al.
7,225,016	B1	5/2007	Koh	8,244,358	B2	8/2012	Tehrani et al.
				8,244,359	B2	8/2012	Gelfand et al.
				8,244,378	B2	8/2012	Bly et al.
				8,255,056	B2	8/2012	Tehrani
				8,256,419	B2	9/2012	Sinderby et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

8,265,736 B2	9/2012	Sathaye et al.	8,858,455 B2	10/2014	Rothenberg
8,265,759 B2	9/2012	Tehrani et al.	8,863,742 B2	10/2014	Blomquist et al.
8,275,440 B2	9/2012	Rodriguez et al.	8,886,277 B2	11/2014	Kim et al.
8,280,513 B2	10/2012	Tehrani et al.	8,897,879 B2	11/2014	Karamanoglu et al.
8,315,713 B2	11/2012	Burnes et al.	8,903,507 B2	12/2014	Desai et al.
8,321,808 B2	11/2012	Goetz et al.	8,903,509 B2	12/2014	Tockman et al.
8,335,567 B2	12/2012	Tehrani et al.	8,909,341 B2	12/2014	Gelfand et al.
8,340,783 B2	12/2012	Sommer et al.	8,914,113 B2	12/2014	Zhang et al.
8,348,941 B2	1/2013	Tehrani	8,918,169 B2	12/2014	Kassab et al.
8,369,954 B2	2/2013	Stack et al.	8,918,987 B2	12/2014	Kuzma et al.
8,374,704 B2	2/2013	Desai et al.	8,923,971 B2	12/2014	Haefner et al.
8,388,541 B2	3/2013	Messerly et al.	8,942,823 B2	1/2015	Desai et al.
8,388,546 B2	3/2013	Rothenberg	8,942,824 B2	1/2015	Yoo et al.
8,391,956 B2	3/2013	Zellers et al.	8,948,884 B2	2/2015	Ramachandran et al.
8,401,640 B2	3/2013	Zhao et al.	8,968,299 B2	3/2015	Kauphusman et al.
8,401,651 B2	3/2013	Caparso et al.	8,972,015 B2	3/2015	Stack et al.
8,406,883 B1	3/2013	Barker	8,983,602 B2	3/2015	Sathaye et al.
8,406,885 B2	3/2013	Ignagni et al.	9,008,775 B2	4/2015	Sathaye et al.
8,412,331 B2	4/2013	Tehrani et al.	9,026,231 B2	5/2015	Hoffer
8,412,350 B2	4/2013	Bly	9,037,264 B2	5/2015	Just et al.
8,428,711 B2	4/2013	Lin et al.	9,042,981 B2	5/2015	Yoo et al.
8,428,726 B2	4/2013	Ignagni et al.	9,072,864 B2	7/2015	Putz
8,428,730 B2	4/2013	Stack et al.	9,072,896 B2	7/2015	Dar et al.
8,433,412 B1	4/2013	Westlund et al.	9,072,899 B1	7/2015	Nickloes
8,442,638 B2	5/2013	Libbus et al.	9,108,058 B2	8/2015	Hoffer
8,457,764 B2	6/2013	Ramachandran et al.	9,108,059 B2	8/2015	Hoffer
8,467,876 B2	6/2013	Tehrani	9,125,578 B2	9/2015	Grunwald
8,473,068 B2	6/2013	Farazi	9,138,580 B2	9/2015	Ignagni et al.
8,478,412 B2	7/2013	Ignagni et al.	9,138,585 B2	9/2015	Saha et al.
8,478,413 B2	7/2013	Karamanoglu et al.	9,144,680 B2	9/2015	Kaula et al.
8,478,426 B2	7/2013	Barker	9,149,642 B2	10/2015	Mccabe et al.
8,483,834 B2	7/2013	Lee et al.	9,168,377 B2	10/2015	Hoffer
8,504,158 B2	8/2013	Karamanoglu et al.	9,174,046 B2	11/2015	Francois et al.
8,504,161 B1	8/2013	Kornet et al.	9,199,075 B1	12/2015	Westlund
8,509,901 B2	8/2013	Tehrani	9,205,258 B2	12/2015	Simon et al.
8,509,902 B2	8/2013	Cho et al.	9,216,291 B2	12/2015	Lee et al.
8,509,919 B2	8/2013	Yoo et al.	9,220,898 B2	12/2015	Hoffer
8,511,303 B2	8/2013	Djupesland	9,226,688 B2	1/2016	Jacobsen et al.
8,512,256 B2	8/2013	Rothenberg	9,226,689 B2	1/2016	Jacobsen et al.
8,522,779 B2	9/2013	Lee et al.	9,242,088 B2	1/2016	Thakkar et al.
8,527,036 B2	9/2013	Jalde et al.	9,259,573 B2	2/2016	Tehrani et al.
8,532,793 B2	9/2013	Morris et al.	9,295,846 B2	3/2016	Westlund et al.
8,554,323 B2	10/2013	Haefner et al.	9,314,618 B2	4/2016	Imran et al.
8,560,072 B2	10/2013	Caparso et al.	9,333,363 B2	5/2016	Hoffer et al.
8,560,086 B2	10/2013	Just et al.	9,345,422 B2	5/2016	Rothenberg
8,571,662 B2	10/2013	Hoffer	9,370,657 B2	6/2016	Tehrani et al.
8,571,685 B2	10/2013	Daglow et al.	9,398,931 B2	7/2016	Wittenberger et al.
8,615,297 B2	12/2013	Sathaye et al.	9,415,188 B2	8/2016	He et al.
8,617,228 B2	12/2013	Wittenberger et al.	9,427,566 B2	8/2016	Reed et al.
8,620,412 B2	12/2013	Griffiths et al.	9,427,588 B2	8/2016	Sathaye et al.
8,620,450 B2	12/2013	Tockman et al.	9,468,755 B2	10/2016	Westlund et al.
8,626,292 B2	1/2014	Mccabe et al.	9,474,894 B2	10/2016	Mercanzini et al.
8,630,707 B2	1/2014	Zhao et al.	9,485,873 B2	11/2016	Shah et al.
8,644,939 B2	2/2014	Wilson et al.	9,498,625 B2	11/2016	Bauer et al.
8,644,952 B2	2/2014	Desai et al.	9,498,631 B2	11/2016	Demmer et al.
8,646,172 B2	2/2014	Kuzma et al.	9,504,837 B2	11/2016	Demmer et al.
8,650,747 B2	2/2014	Kuzma et al.	9,526,449 B2	12/2016	Shaker
8,676,323 B2	3/2014	Ignagni et al.	9,532,724 B2	1/2017	Grunwald et al.
8,676,344 B2	3/2014	Desai et al.	9,533,160 B2	1/2017	Brooke et al.
8,694,123 B2	4/2014	Wahlstrand et al.	9,539,429 B2	1/2017	Brooke et al.
8,696,656 B2	4/2014	Abboud et al.	9,545,511 B2	1/2017	Thakkar et al.
8,706,223 B2	4/2014	Zhou et al.	9,561,369 B2	2/2017	Burnes et al.
8,706,235 B2	4/2014	Karamanoglu et al.	9,566,436 B2	2/2017	Hoffer et al.
8,706,236 B2	4/2014	Ignagni et al.	9,572,982 B2	2/2017	Burnes et al.
8,718,763 B2	5/2014	Zhou et al.	9,597,509 B2	3/2017	Hoffer et al.
8,725,259 B2	5/2014	Kornet et al.	9,615,759 B2	4/2017	Hurezan et al.
8,738,154 B2	5/2014	Zdeblick et al.	9,623,239 B2	4/2017	Francois et al.
8,755,889 B2	6/2014	Scheiner	9,623,252 B2	4/2017	Sathaye et al.
8,774,907 B2	7/2014	Rothenberg	9,662,494 B2	5/2017	Young et al.
8,781,578 B2	7/2014	Mccabe et al.	9,682,235 B1	6/2017	O'Mahony et al.
8,781,582 B2	7/2014	Ziegler et al.	9,694,185 B2	7/2017	Bauer
8,781,583 B2	7/2014	Cornelussen et al.	9,717,899 B2	8/2017	Kuzma et al.
8,801,693 B2	8/2014	He et al.	9,724,018 B2	8/2017	Cho et al.
8,805,511 B2	8/2014	Karamanoglu et al.	9,744,349 B2	8/2017	Westlund et al.
8,838,245 B2	9/2014	Lin et al.	9,744,351 B1	8/2017	Gelfand et al.
			9,776,005 B2	10/2017	Meyyappan et al.
			9,833,623 B2	12/2017	Gnanashanmugam et al.
			9,861,817 B2	1/2018	Cho et al.
			9,872,989 B2	1/2018	Jung et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

9,884,178	B2	2/2018	Bouton et al.	2005/0085868	A1	4/2005	Tehrani et al.
9,884,179	B2	2/2018	Bouton et al.	2005/0085869	A1	4/2005	Tehrani et al.
9,919,149	B2	3/2018	Imran et al.	2005/0090870	A1	4/2005	Hine et al.
9,931,504	B2	4/2018	Thakkar et al.	2005/0096710	A1	5/2005	Kieval
9,950,167	B2	4/2018	Hoffer et al.	2005/0109340	A1	5/2005	Tehrani
9,956,132	B2	5/2018	Francois et al.	2005/0113710	A1	5/2005	Stahmann et al.
9,956,396	B2	5/2018	Young et al.	2005/0115561	A1	6/2005	Stahmann et al.
9,968,785	B2	5/2018	Hoffer et al.	2005/0131485	A1	6/2005	Knudson et al.
9,968,786	B2	5/2018	Bauer et al.	2005/0138791	A1	6/2005	Black et al.
9,987,488	B1	6/2018	Gelfand et al.	2005/0138792	A1	6/2005	Black et al.
9,999,768	B2	6/2018	Gelfand et al.	2005/0143787	A1	6/2005	Boveja et al.
10,022,546	B2	7/2018	Hoffer	2005/0159799	A1	7/2005	Daglow et al.
10,035,017	B2	7/2018	Thakkar et al.	2005/0165457	A1	7/2005	Benser et al.
10,039,920	B1 *	8/2018	Thakkar	2005/0182454	A1	8/2005	Gharib et al.
10,195,429	B1	2/2019	Thakkar et al.	2005/0187584	A1	8/2005	Denker et al.
10,293,164	B2	5/2019	Nash et al.	2005/0192655	A1	9/2005	Black et al.
10,300,270	B2	5/2019	Gelfand et al.	2005/0240240	A1	10/2005	Park et al.
10,315,035	B2	6/2019	Bauer	2005/0251238	A1	11/2005	Wallace et al.
10,335,592	B2	7/2019	Bauer et al.	2005/0251239	A1	11/2005	Wallace et al.
10,369,361	B2	8/2019	Bauer et al.	2005/0288728	A1	12/2005	Libbus et al.
10,391,314	B2	8/2019	Hoffer et al.	2005/0288729	A1	12/2005	Libbus et al.
10,406,366	B2	9/2019	Westlund et al.	2005/0288730	A1	12/2005	Deem et al.
10,406,367	B2	9/2019	Meyyappan et al.	2006/0024222	A1	2/2006	Dachs et al.
10,413,203	B2	9/2019	Saha et al.	2006/0030894	A1	2/2006	Tehrani
10,448,995	B2	10/2019	Olson	2006/0035849	A1	2/2006	Spiegelman et al.
10,493,271	B2	12/2019	Bauer et al.	2006/0058852	A1	3/2006	Koh et al.
10,512,772	B2	12/2019	Hoffer et al.	2006/0074449	A1	4/2006	Denker et al.
10,966,738	B2	4/2021	Humayun	2006/0079936	A1	4/2006	Boveja et al.
11,045,614	B2	6/2021	Andreas et al.	2006/0095106	A1	5/2006	Maxfield et al.
11,122,972	B2	9/2021	Cohen et al.	2006/0122661	A1	6/2006	Mandell
11,154,229	B2	10/2021	Das et al.	2006/0122662	A1	6/2006	Tehrani et al.
11,154,547	B2	10/2021	Bright et al.	2006/0130833	A1	6/2006	Younes
11,161,306	B2	11/2021	Kidmose et al.	2006/0142815	A1	6/2006	Tehrani et al.
11,172,859	B2	11/2021	Connor	2006/0149334	A1	7/2006	Tehrani et al.
11,179,541	B2	11/2021	Metzger	2006/0155222	A1	7/2006	Sherman et al.
2001/0052345	A1	12/2001	Niazi	2006/0167523	A1	7/2006	Tehrani et al.
2002/0026228	A1	2/2002	Schauerte	2006/0188325	A1	8/2006	Dolan
2002/0056454	A1	5/2002	Samzelius	2006/0195159	A1	8/2006	Bradley et al.
2002/0065544	A1	5/2002	Smits et al.	2006/0217791	A1	9/2006	Spinka et al.
2002/0087156	A1	7/2002	Maguire et al.	2006/0224209	A1	10/2006	Meyer
2002/0091419	A1	7/2002	Firlik et al.	2006/0224222	A1	10/2006	Bradley et al.
2002/0128546	A1	9/2002	Silver	2006/0229677	A1	10/2006	Moffitt et al.
2002/0188325	A1	12/2002	Hill et al.	2006/0229687	A1	10/2006	Goetz et al.
2002/0188332	A1	12/2002	Lurie et al.	2006/0247729	A1	11/2006	Tehrani et al.
2003/0074039	A1	4/2003	Puskas	2006/0253161	A1	11/2006	Libbus et al.
2003/0078623	A1	4/2003	Weinberg et al.	2006/0253182	A1	11/2006	King
2003/0125786	A1	7/2003	Gliner et al.	2006/0258667	A1	11/2006	Teng
2003/0195571	A1	10/2003	Burnes et al.	2006/0259107	A1	11/2006	Caparso et al.
2004/0003813	A1	1/2004	Banner et al.	2006/0276848	A1	12/2006	Min et al.
2004/0010303	A1	1/2004	Bolea et al.	2006/0282131	A1	12/2006	Caparso et al.
2004/0030362	A1	2/2004	Hill et al.	2006/0287679	A1	12/2006	Stone
2004/0044377	A1	3/2004	Larsson et al.	2007/0005053	A1	1/2007	Dando
2004/0064069	A1	4/2004	Reynolds et al.	2007/0021795	A1	1/2007	Tehrani
2004/0077936	A1	4/2004	Larsson	2007/0027448	A1	2/2007	Paul et al.
2004/0088015	A1	5/2004	Casavant et al.	2007/0043400	A1	2/2007	Donders et al.
2004/0111139	A1	6/2004	McCreery	2007/0087314	A1	4/2007	Gomo
2004/0172075	A1	9/2004	Shafer et al.	2007/0093875	A1	4/2007	Chavan et al.
2004/0186543	A1	9/2004	King et al.	2007/0106339	A1	5/2007	Errico et al.
2004/0193231	A1	9/2004	David et al.	2007/0106357	A1	5/2007	Denker et al.
2004/0210261	A1	10/2004	King et al.	2007/0112402	A1	5/2007	Grill et al.
2004/0211417	A1	10/2004	Lurie	2007/0112403	A1	5/2007	Moffitt et al.
2004/0254612	A1	12/2004	Ezra et al.	2007/0118183	A1	5/2007	Gelfand et al.
2005/0004565	A1	1/2005	Vanney	2007/0150006	A1	6/2007	Libbus et al.
2005/0013879	A1	1/2005	Lin et al.	2007/0150026	A1	6/2007	Bourget et al.
2005/0021102	A1	1/2005	Ignagni et al.	2007/0156199	A1	7/2007	Koh et al.
2005/0027338	A1	2/2005	Hill	2007/0168007	A1	7/2007	Kuzma et al.
2005/0033136	A1	2/2005	Govari et al.	2007/0173900	A1	7/2007	Siegel et al.
2005/0033137	A1	2/2005	Oral et al.	2007/0191908	A1	8/2007	Jacob et al.
2005/0043765	A1	2/2005	Williams et al.	2007/0196780	A1	8/2007	Ware et al.
2005/0065567	A1	3/2005	Lee et al.	2007/0203549	A1	8/2007	Demarais et al.
2005/0070981	A1	3/2005	Verma	2007/0208388	A1	9/2007	Jahns et al.
2005/0075578	A1	4/2005	Gharib et al.	2007/0221224	A1	9/2007	Pittman et al.
2005/0085865	A1	4/2005	Tehrani	2007/0240718	A1	10/2007	Daly
2005/0085866	A1	4/2005	Tehrani	2007/0250056	A1	10/2007	Vanney
2005/0085867	A1	4/2005	Tehrani et al.	2007/0250162	A1	10/2007	Royalty
				2007/0255379	A1	11/2007	Williams et al.
				2007/0265611	A1	11/2007	Ignagni et al.
				2007/0277832	A1	12/2007	Doshi et al.
				2007/0288076	A1	12/2007	Bulkes et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2008/0039916 A1	2/2008	Colliou et al.	2012/0035684 A1	2/2012	Thompson et al.
2008/0065002 A1	3/2008	Lobl et al.	2012/0035691 A1	2/2012	Tockman et al.
2008/0121231 A1	5/2008	Sinderby et al.	2012/0053654 A1	3/2012	Tehrani et al.
2008/0125828 A1	5/2008	Ignagni et al.	2012/0078320 A1	3/2012	Schotzko et al.
2008/0161878 A1	7/2008	Tehrani et al.	2012/0130217 A1	5/2012	Kauphusman et al.
2008/0167695 A1	7/2008	Tehrani et al.	2012/0158091 A1	6/2012	Tehrani et al.
2008/0177190 A1	7/2008	Libbus et al.	2012/0209284 A1	8/2012	Westlund et al.
2008/0177347 A1	7/2008	Tehrani et al.	2012/0215278 A1	8/2012	Penner
2008/0183186 A1	7/2008	Bly et al.	2012/0323293 A1	12/2012	Tehrani et al.
2008/0183187 A1	7/2008	Bly	2013/0018247 A1	1/2013	Glenn et al.
2008/0183239 A1	7/2008	Tehrani et al.	2013/0018427 A1	1/2013	Pham et al.
2008/0183240 A1	7/2008	Tehrani et al.	2013/0023972 A1	1/2013	Kuzma et al.
2008/0183253 A1	7/2008	Bly	2013/0030385 A1	1/2013	Schultz et al.
2008/0183254 A1	7/2008	Bly et al.	2013/0030496 A1	1/2013	Karamanoglu et al.
2008/0183255 A1	7/2008	Bly et al.	2013/0030497 A1	1/2013	Karamanoglu et al.
2008/0183259 A1	7/2008	Bly et al.	2013/0030498 A1	1/2013	Karamanoglu et al.
2008/0183264 A1	7/2008	Bly et al.	2013/0060245 A1	3/2013	Grunewald et al.
2008/0183265 A1	7/2008	Bly et al.	2013/0066392 A1	3/2013	Bruce et al.
2008/0188903 A1	8/2008	Tehrani et al.	2013/0085493 A1	4/2013	Bloom et al.
2008/0215106 A1	9/2008	Lee et al.	2013/0085550 A1	4/2013	Polefko et al.
2008/0243007 A1	10/2008	Liao et al.	2013/0116743 A1	5/2013	Karamanoglu et al.
2008/0269834 A1	10/2008	Byerman et al.	2013/0123891 A1	5/2013	Swanson
2008/0288010 A1	11/2008	Tehrani et al.	2013/0131743 A1	5/2013	Yamasaki et al.
2008/0288015 A1	11/2008	Tehrani et al.	2013/0150908 A1	6/2013	Lindenthaler
2008/0312712 A1	12/2008	Penner	2013/0158514 A1	6/2013	Elia et al.
2008/0312725 A1	12/2008	Penner	2013/0158625 A1	6/2013	Gelfand et al.
2009/0024047 A1	1/2009	Shipley et al.	2013/0165989 A1	6/2013	Gelfand et al.
2009/0024176 A1	1/2009	Yun et al.	2013/0167372 A1	7/2013	Black et al.
2009/0036947 A1	2/2009	Westlund et al.	2013/0197601 A1	8/2013	Tehrani et al.
2009/0118785 A1	5/2009	Ignagni et al.	2013/0237906 A1	9/2013	Park et al.
2009/0248122 A1	10/2009	Pianca	2013/0268018 A1	10/2013	Brooke et al.
2009/0275956 A1	11/2009	Burnes et al.	2013/0276787 A1	10/2013	Martin et al.
2009/0275996 A1	11/2009	Burnes et al.	2013/0282090 A1	10/2013	Decré et al.
2009/0275997 A1	11/2009	Faltys et al.	2013/0289686 A1	10/2013	Masson et al.
2009/0276022 A1	11/2009	Burnes et al.	2013/0296964 A1	11/2013	Tehrani
2009/0318993 A1	12/2009	Eidenschink et al.	2013/0296973 A1	11/2013	Tehrani et al.
2010/0022950 A1	1/2010	Anderson et al.	2013/0296977 A1	11/2013	Chiu et al.
2010/0034977 A1	2/2010	Kim et al.	2013/0317587 A1	11/2013	Barker
2010/0036451 A1	2/2010	Hoffer	2013/0333696 A1	12/2013	Lee et al.
2010/0042193 A1	2/2010	Slavin	2014/0012182 A1	1/2014	Shantha
2010/0057046 A1	3/2010	Stevens et al.	2014/0067032 A1	3/2014	Morris et al.
2010/0077606 A1	4/2010	Black et al.	2014/0088580 A1	3/2014	Wittenberger et al.
2010/0094376 A1	4/2010	Penner	2014/0114371 A1	4/2014	Westlund et al.
2010/0105977 A1	4/2010	De Taboada et al.	2014/0121716 A1	5/2014	Casavant et al.
2010/0114227 A1	5/2010	Cholette	2014/0128953 A1	5/2014	Zhao et al.
2010/0114254 A1	5/2010	Kornet	2014/0148780 A1	5/2014	Putz
2010/0114261 A1	5/2010	Errico et al.	2014/0276789 A1	9/2014	Dandler et al.
2010/0135553 A1	6/2010	Joglekar	2014/0288384 A1	9/2014	Mulrooney
2010/0198296 A1	8/2010	Ignagni et al.	2014/0288551 A1	9/2014	Rupinder et al.
2010/0204766 A1	8/2010	Zdeblick et al.	2014/0296738 A1	10/2014	Johnson et al.
2010/0249865 A1	9/2010	Zhang et al.	2014/0316486 A1	10/2014	Zhou et al.
2010/0268311 A1	10/2010	Cardinal et al.	2014/0316496 A1	10/2014	Masson et al.
2010/0269337 A1	10/2010	Dye et al.	2014/0324115 A1	10/2014	Ziegler et al.
2010/0312302 A1	12/2010	Zealear	2014/0378803 A1	12/2014	Geistert et al.
2010/0319691 A1	12/2010	Lurie et al.	2015/0018839 A1	1/2015	Morris et al.
2011/0034977 A1	2/2011	Janik et al.	2015/0034081 A1	2/2015	Tehrani et al.
2011/0060380 A1	3/2011	Gelfand et al.	2015/0045810 A1	2/2015	Hoffer et al.
2011/0060381 A1	3/2011	Ignagni et al.	2015/0045848 A1	2/2015	Cho et al.
2011/0077726 A1	3/2011	Westlund et al.	2015/0073232 A1	3/2015	Ahmad et al.
2011/0087301 A1	4/2011	Li et al.	2015/0119950 A1	4/2015	Demmer et al.
2011/0093032 A1	4/2011	Boggs, II et al.	2015/0126839 A1	5/2015	Li et al.
2011/0118815 A1	5/2011	Kuzma et al.	2015/0142082 A1	5/2015	Simon et al.
2011/0152967 A1	6/2011	Bruce et al.	2015/0142083 A1	5/2015	Yomtov
2011/0190845 A1	8/2011	Weisfeldt et al.	2015/0148877 A1	5/2015	Thakkar et al.
2011/0224750 A1	9/2011	Scheiner	2015/0165207 A1	6/2015	Karamanoglu
2011/0230932 A1	9/2011	Tehrani et al.	2015/0196354 A1	7/2015	Haverkost et al.
2011/0230935 A1	9/2011	Zdeblick	2015/0196356 A1	7/2015	Kauphusman et al.
2011/0230945 A1	9/2011	Ohtaka et al.	2015/0202448 A1	7/2015	Hoffer et al.
2011/0270358 A1	11/2011	Davis et al.	2015/0231348 A1	8/2015	Lee et al.
2011/0288609 A1	11/2011	Tehrani et al.	2015/0250982 A1	9/2015	Osyka et al.
2011/0301471 A1	12/2011	Kwok et al.	2015/0265833 A1	9/2015	Meyyappan et al.
2011/0313330 A1	12/2011	Loushin et al.	2015/0283340 A1	10/2015	Zhang et al.
2012/0004715 A1	1/2012	Ramachandran et al.	2015/0290476 A1	10/2015	Krocak et al.
2012/0029444 A1	2/2012	Anderson et al.	2015/0359487 A1	12/2015	Coulombe
2012/0029509 A1	2/2012	Smith	2015/0374252 A1	12/2015	De La Rama et al.
			2015/0374991 A1	12/2015	Morris et al.
			2016/0001072 A1	1/2016	Gelfand et al.
			2016/0059017 A1	3/2016	Moffitt et al.
			2016/0101280 A1	4/2016	Thakkar et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2016/0114165	A1	4/2016	Levine et al.	
2016/0129244	A1	5/2016	Westlund	
2016/0144078	A1	5/2016	Young et al.	
2016/0144166	A1 *	5/2016	Decré	A61B 5/24 607/116
2016/0166831	A9	6/2016	Cho et al.	
2016/0193460	A1	7/2016	Xu et al.	
2016/0228696	A1	8/2016	Imran et al.	
2016/0235324	A1	8/2016	Merishin et al.	
2016/0239627	A1	8/2016	Cerny et al.	
2016/0256692	A1	9/2016	Baru	
2016/0287877	A1	10/2016	Jung	
2016/0310730	A1	10/2016	Martins et al.	
2016/0331326	A1	11/2016	Xiang et al.	
2016/0354607	A1	12/2016	Masson et al.	
2016/0367815	A1	12/2016	Hoffer	
2017/0007825	A1	1/2017	Thakkar et al.	
2017/0013713	A1	1/2017	Shah et al.	
2017/0021163	A1	1/2017	Westlund et al.	
2017/0021166	A1	1/2017	Bauer et al.	
2017/0021175	A1	1/2017	Yu et al.	
2017/0028191	A1	2/2017	Mercanzini et al.	
2017/0036017	A1	2/2017	Tehrani et al.	
2017/0050033	A1	2/2017	Wechter	
2017/0080216	A1	3/2017	Pham	
2017/0143973	A1	5/2017	Tehrani	
2017/0143975	A1	5/2017	Hoffer et al.	
2017/0196503	A1	7/2017	Narayan et al.	
2017/0224993	A1	8/2017	Sathaye et al.	
2017/0232250	A1	8/2017	Kim et al.	
2017/0252558	A1	9/2017	O'Mahony et al.	
2017/0291023	A1	10/2017	Kuzma et al.	
2017/0296812	A1	10/2017	O'Mahony et al.	
2017/0312006	A1	11/2017	McFarlin et al.	
2017/0312507	A1	11/2017	Bauer et al.	
2017/0312508	A1	11/2017	Bauer et al.	
2017/0312509	A1	11/2017	Bauer et al.	
2017/0326354	A1	11/2017	Westlund et al.	
2017/0326359	A1	11/2017	Gelfand et al.	
2017/0347921	A1	12/2017	Haber et al.	
2018/0001086	A1	1/2018	Bartholomew et al.	
2018/0008821	A1	1/2018	Gonzalez et al.	
2018/0110562	A1	4/2018	Govari et al.	
2018/0117265	A1	5/2018	Armes	
2018/0117334	A1	5/2018	Jung	
2018/0169406	A1	6/2018	Shah et al.	
2018/0256440	A1	9/2018	Francois et al.	
2018/0280691	A1	10/2018	Ackermann et al.	
2018/0280692	A1	10/2018	Gelfand et al.	
2018/0326209	A1	11/2018	Gelfand et al.	
2018/0339156	A1 *	11/2018	Nash	A61N 1/36185
2019/0247656	A1	8/2019	Bauer	
2019/0255322	A1	8/2019	Bauer et al.	
2019/0294245	A1	9/2019	Ramer	
2019/0351229	A1	11/2019	Westlund et al.	

FOREIGN PATENT DOCUMENTS

CN	102762250	A	10/2012
EP	0993840	A1	4/2000
EP	1304135	A2	4/2003
EP	0605796	B1	8/2003
EP	0783346	B2	1/2007
EP	1833558	A1	9/2007
EP	1833558	B1	10/2011
EP	2489395	A1	8/2012
FR	2801509	A1	6/2001
JP	H08510677	A	11/1996
JP	2003503119	A	1/2003
JP	2010516353	A	5/2010
JP	2011200571	A	10/2011
JP	2012000195	A	1/2012
WO	WO-9407564	A2	4/1994
WO	WO-9508357	A1	3/1995

WO	WO-9732545	A1	9/1997
WO	WO-9964105	A1	12/1999
WO	WO-9965561	A1	12/1999
WO	WO-0100273	A1	1/2001
WO	WO-02058785	A1	8/2002
WO	WO-03005887	A2	1/2003
WO	WO-03094855	A1	11/2003
WO	WO-2005018524	A2	3/2005
WO	WO 2005053788	A1	6/2005
WO	WO-2006063339	A2	6/2006
WO	WO 2006066280	A1	6/2006
WO	WO-2006110338	A1	10/2006
WO	WO-2006115877	A1	11/2006
WO	WO-2007053508	A1	5/2007
WO	WO 2008027297	A2	3/2008
WO	WO-2008092246	A1	8/2008
WO	WO-2008094344	A1	8/2008
WO	WO-2009006337	A1	1/2009
WO	WO-2009134459	A2	11/2009
WO	WO-2010029842	A1	3/2010
WO	WO-2010148412	A1	12/2010
WO	WO-2011094631	A1	8/2011
WO	WO-2011158410	A1	12/2011
WO	WO-2012106533	A2	8/2012
WO	WO 2013086461	A1	6/2013
WO	WO-2013131187	A1	9/2013
WO	WO-2013188965	A1	12/2013
WO	WO-2014008171	A1	1/2014
WO	WO-2015075548	A1	5/2015
WO	WO-2015109401	A1	7/2015
WO	WO 2016025912	A1	2/2016
WO	WO-2019006239	A1	1/2019
WO	WO-2019154834	A1	8/2019
WO	WO-2019154837	A1	8/2019
WO	WO-2019154839	A1	8/2019

OTHER PUBLICATIONS

Ataya A., et al., "Temporary Transvenous Diaphragmatic Neurostimulation in Prolonged Mechanically Ventilated Patients: A Feasibility Trial (RESCUE 1)," Critical Care Explorations, 2020, vol. 2, pp. 1-7.

Ayas N.T., et al., "Prevention of Human Diaphragm Atrophy with Short periods of Electrical Stimulation," American Journal of Respiratory and Critical Care Medicine, Jun. 1999, vol. 159(6), pp. 2018-2020.

Bassi T., et al., "Increased Hippocampal Apoptotic Index After 50 Hours of Gold-Standard Mechanical Ventilation in a Pre-Clinical Pig Model," American Journal of Respiratory and Critical Care Medicine, 2020, vol. 201, pp. A5961-A5961.

Bassi T.G., et al., "Reply to Salimi et al.: Stimulating Neural Pathways to Reduce Mechanical Ventilation-associated Neurocognitive Dysfunction," American Journal of Respiratory and Critical Care Medicine, Mar. 2022, vol. 205(5), pp. 590-592.

Bassi T.G., et al., "Systematic review of cognitive impairment and brain insult after mechanical ventilation," Critical Care, 2021, vol. 25, Issue 99.

Bassi T.G., et al., "Transvenous Diaphragm Neurostimulation Mitigates Ventilation-associated Brain Injury," American Journal of Respiratory Critical Care Medicine, Dec. 2021, vol. 204(12), pp. 1391-1402.

Borovikova, et al., "Role of the Vagus Nerve in the Anti-Inflammatory Effects of CNI-1493," Proceedings of the Annual Meeting of Professional Research Scientists: Experimental Biology 2000, Abstract 97.9, Apr. 15-18, 2000.

Borovikova L.V., et al., "Role of Vagus Nerve Signaling in CNI-1493-Mediated Suppression of Acute Inflammation," Autonomic Neuroscience: Basic and Clinical, vol. 85 (1-3), Dec. 20, 2000, pp. 141-147.

Borovikova L.V., et al., "Vagus Nerve Stimulation Attenuates the Systemic Inflammatory Response to Endotoxin," Nature, Macmillan Magazines Ltd, vol. 405, May 25, 2000, pp. 458-462.

Castle M., et al., "Autonomic Brainstem Nuclei are Linked to the Hippocampus," Neuroscience, 2005, vol. 134(2), pp. 657-669.

(56)

References Cited**OTHER PUBLICATIONS**

- Chen, T., et al., "Toll-like receptor 4 knockout ameliorates neuroinflammation due to lung-brain interaction in mechanically ventilated mice," *Brain, Behavior, and Immunity*, 2016, vol. 56, pp. 42-55.
- Daggeti, W.M. et al., "Intracaval Electrophrenic Stimulation. I. Experimental Application during Barbiturate Intoxication Hemorrhage and Gang," *Journal of Thoracic and Cardiovascular Surgery*, 1966, vol. 51 (5), pp. 676-884.
- Daggeti, W.M. et al., "Intracaval electrophrenic stimulation. II. Studies on Pulmonary Mechanics Surface Tension Urine Flow and Bilateral Ph," *Journal of Thoracic and Cardiovascular Surgery*, 1970, vol. 60(1), pp. 98-107.
- De Gregorio, M.A. et al., "The Gunther Tulip Retrievable Filter: Prolonged Temporary Filtration by Repositioning within the Inferior Vena Cava," *Journal of Vascular and Interventional Radiology*, 2003, vol. 14, pp. 1259-1265.
- Deng Y-J et al., "The Effect of Positive Pressure Ventilation Combined with Diaphragm Pacing on Respiratory Mechanics in Patients with Respiratory Failure; Respiratory Mechanics," *Chinese critical care medicine*, Apr. 2011, vol. 23(4), pp. 213-215.
- Elkins M., et al., "Inspiratory Muscle Training Facilitates Weaning From Mechanical Ventilation Among Patients in the Intensive Care Unit: a Systematic Review," *Journal of Physiotherapy*, 2015, vol. 61, pp. 1-10.
- Escher D.J., et al., "Clinical Control of Respiration by Transvenous Phrenic Pacing," *Transactions—American Society for Artificial Internal Organs*, Apr. 1968, vol. 14 (1), pp. 192-197.
- Fleshner M., et al., "Thermogenic and Corticosterone Responses to Intravenous Cytokines (IL-1 β and TNF- α) are Attenuated by Subdiaphragmatic Vagotomy," *Journal of Neuroimmunology*, Jun. 1998, vol. 86, pp. 134-141.
- Fong T.G., et al., "Cerebral Perfusion Changes in Older Delirious Patients Using 99mTc HMPAO SPECT," *Journals of Gerontology—Series A: Biological Sciences and Medical Sciences*, 2006, vol. 61(12), pp. 1294-1299.
- Frisch S., "A Feasibility Study of a Novel Minimally Invasive Approach for Diaphragm Pacing," *Master of Science Thesis*, Simon Fraser University, 2009, p. 148.
- Furman, S., "Transvenous Stimulation of the Phrenic Nerves," *Journal of Thoracic and Cardiovascular Surgery*, 1971, vol. 62 (5), pp. 743-751.
- Gaykema R.P.A. et al., "Subdiaphragmatic Vagotomy Suppresses Endotoxin-Induced Activation of Hypothalamic Corticotropin-Releasing Hormone Neurons and ACTH Secretion," *Endocrinology*, The Endocrine Society, 1995, vol. 136 (10), pp. 4717-4720.
- Ghazvineh S., et al., "Rhythmic Air-puff into Nasal Cavity Modulates Activity Across Multiple Brain Areas: A Non-invasive Brain Stimulation Method to Reduce Ventilator-induced Memory Impairment," *Respiratory Physiology and Neurobiology*, 2021;287. doi:10.1016/j.resp.2021.103627.
- Giordano G., et al., "Neuroinflammation, Neuronal Damage or Cognitive Impairment Associated with Mechanical Ventilation: A Systematic Review of Evidence from Animal Studies," *Journal of Critical Care*. 2021;62:246-255. doi:10.1016/j.jcrc.2020.12.017.
- Gupta A.K., "Respiration Rate Measurement Based on Impedance Pneumography," *Data Acquisition Products*, Texas Instruments, Application Report, SBAA181, Feb. 2011, 11 pages.
- Guslandi M., "Nicotine Treatment for Ulcerative Colitis," *The British Journal of Clinical Pharmacology*, Blackwell Science Ltd, 1999, vol. 48, pp. 481-484.
- Hauglund N.L., et al., "Cleaning the Sleeping Brain—the Potential Restorative Function of the Glymphatic System," *Current Opinion in Physiology*, 2020, vol. 15, pp. 1-6.
- Hillman D.R., et al., "The effect of diaphragm contraction on upper airway collapsibility," *Journal of Applied Physiology*, 2013, vol. 115, pp. 337-345.
- Henkel C.K., "Chapter 21—The Auditory System," *Fundamental Neuroscience for Basic and Clinical Applications (Fifth Edition)*, 2018, pp. 306-319.
- Hoffer J.A. et al., "Diaphragm Pacing with Endovascular Electrodes", IFESS 2010—International Functional Electrical Stimulation Society, 15th Anniversary Conference, Vienna, Austria, Sep. 2010.
- Huffman W.J., "Modulation of Neuroinflammation and Memory Dysfunction Using Percutaneous Vagus Nerve Stimulation in Mice," *Brain Stimulation*, 2019, vol. 12 (1), pp. 19-29.
- Ikeda K., et al., "The Respiratory Control Mechanisms in the Brainstem and Spinal Cord: Integrative Views of the Neuroanatomy and neurophysiology," *Journal of Physiological Sciences*, Jan. 2017; vol. 67(1), pp. 45-62.
- International Search Report for Application No. PCT/IB2021/050214, mailed on Apr. 4, 2021, 3 pages.
- Ishii K., et al., "Effects of Bilateral Transvenous Diaphragm Pacing on Hemodynamic Function in Patients after Cardiac Operations. Experimental and Clinical Study," *The Journal of Thoracic and Cardiovascular Surgery*, Jul. 1990, vol. 100 (1), pp. 108-114.
- Johnson J.O., "13—Autonomic Nervous System: Physiology," *Pharmacology and Physiology for Anesthesia (Second Edition)*, 2019, pp. 270-281.
- Karalis N. and Sirota A., "Breathing Coordinates Limbic Network Dynamics Underlying Memory Consolidation," *bioRxiv*, Published online Jan. 1, 2018, 29 pages.
- Kawashima K., et al., "Extraneuronal Cholinergic System in Lymphocytes," *Pharmacology & Therapeutics*, Elsevier, 2000, vol. 86, pp. 29-48.
- Khan B.A., et al., "Biomarkers for Delirium—A review," *Journal of the American Geriatrics Society*, Nov. 2011, vol. 59(Suppl. 2), pp. S256-S261.
- Kosel M., "Chronic Vagus Nerve Stimulation for Treatment-resistant Depression Increases Regional Cerebral Blood Flow in the Dorsolateral Prefrontal Cortex," *Psychiatry Research—Neuroimaging*, 2011, vol. 191(3):153-159.
- Lee, C., et al., "The Association of Perioperative Serum Lactate Levels with Postoperative Delirium in Elderly Trauma Patients," *BioMed Research International*, 2019, vol. 2019, Article 3963780.
- Lee F.H.K., et al., "Dysfunctional Cerebral Autoregulation is Associated with Delirium in Critically Ill Adults," *Journal of Cerebral Blood Flow and Metabolism*, 2019, vol. 39(12), pp. 2512-2520.
- Levine S., et al., "Rapid disuse atrophy of diaphragm fibers in mechanically ventilated humans," *New England Journal of Medicine*, 2008, vol. 358, pp. 1327-1335.
- Lungpacer: Therapy, News. <<http://lungpacer.com>>. Accessed Dec. 27, 2016.
- Madretsma, G.S., et al., "Nicotine Inhibits the In-vitro Production of Interleukin 2 and Tumour Necrosis Factor- α by Human Mononuclear Cells," *Immunopharmacology*, Elsevier, Oct. 1996, vol. 35 (1), pp. 47-51.
- Mankowski R.T., et al., "Intraoperative Hemidiaphragm Electrical Stimulation Reduces Oxidative Stress and Upregulates Autophagy in Surgery Patients Undergoing Mechanical Ventilation: Exploratory Study," *Journal of Translational Medicine*, 2016, vol. 14, pp. 1-7.
- Marcy, T.W. et al., "Diaphragm Pacing for Ventilatory Insufficiency," *Journal of Intensive Care Medicine*, 1987, vol. 2 (6), pp. 345-353.
- Martin A.D., et al., "Inspiratory Muscle Strength Training Improves Weaning Outcome in Failure to Wean Patients: a Randomized Trial," *Critical Care*, 2011, vol. 15, pp. 1-12.
- Martinez-Canabal A., et al., "Hippocampal Neurogenesis Regulates Recovery of Defensive Responses by Recruiting Threat- and Extinction-signalling Brain Networks," *Scientific Reports*, 2019, vol. 9(1).
- McCorry, L.K., et al., "Physiology of the Autonomic Nervous System," *American Journal of Pharmaceutical Education*, 2007, vol. 71(4), Article 78.
- Meyyappan R., "Diaphragm Pacing during Controlled Mechanical Ventilation: Pre-Clinical Observations Reveal A Substantial Improvement In Respiratory Mechanics", 17th Biennial Canadian Biomechanics Society Meeting, Burnaby, BC, Jun. 6-9, 2012.
- Nabutovsky, Y., et al., "Lead Design and Initial Applications of a New Lead for Long-Term Endovascular Vagal Stimulation," *PACE*, Blackwell Publishing, Inc, Jan. 2007, vol. 30(1), pp. S215-S218.

(56)

References Cited**OTHER PUBLICATIONS**

- Nobis W.P., et al., "Amygdala Stimulation-induced Apnea is Attention and Nasal-breathing Dependent," *Annals of Neurology*, Mar. 2018, vol. 83 (3), pp. 460-471.
- Nobis L., et al., "Hippocampal volume across age: Nomograms derived from over 19,700 people in UK Biobank," *NeuroImage: Clinical*, 2019, vol. 23, Article 10194.
- Onders R., "A Diaphragm Pacing as a Short-Term Assist to Positive Pressure Mechanical Ventilation in Critical Care Patients," *Chest*, Oct. 24, 2007, vol. 132(4), pp. 5715-5728.
- Onders R., "Diaphragm Pacing for Acute Respiratory Failure," *Difficult Decisions in Thoracic Surgery*, Chapter 37, Springer-Verlag, 2011, M.K. Ferguson (ed.), pp. 329-335.
- Onders R., et al., "Diaphragm Pacing with Natural Orifice Transluminal Endoscopic Surgery: Potential for Difficult-To-Wean Intensive Care Unit Patients," *Surgical Endoscopy*, 2007, vol. 21, pp. 475-479.
- Pavlovic D., et al., "Diaphragm Pacing During Prolonged Mechanical Ventilation of the Lungs could Prevent from Respiratory Muscle Fatigue," *Medical Hypotheses*, 2003, vol. 60 (3), pp. 398-403.
- Peters, A.J., et al., Ketamine alters hippocampal cell proliferation and improves learning in mice after traumatic brain injury, *Anesthesiology*, 2018, vol. 129(2), pp. 278-295.
- Peters Van Ton, A.M., et al., "Systemic Inflammation and Cerebral Dysfunction," *Annual Update in Intensive Care and Emergency Medicine*, 2018, vol. 37, pp. 487-501.
- Planas R.F., et al., "Diaphragmatic Pressures: Transvenous vs. Direct Phrenic Nerve Stimulation," *Journal of Applied Physiology*, vol. 59(1), 1985, pp. 269-273.
- Plog B.A., et al., "Biomarkers of Traumatic Injury are Transported from Brain to Blood via the Glymphatic System," *Journal of Neuroscience*, 2015, vol. 35(2), pp. 518-526.
- Quilez M., et al., "Injurious Mechanical Ventilation Affects Neuronal Activation in Ventilated Rats," *Critical Care*, 2011, vol. 15(3), p. R124.
- Reynolds, S.C., et al., "Mitigation of Ventilator-induced Diaphragm Atrophy by Transvenous Phrenic Nerve Stimulation," *American Journal of Respiratory and Critical Care Medicine*, 2017, vol. 195, pp. 339-348.
- Romanovsky, A.A., et al., "The Vagus Nerve in the Thermoregulatory Response to Systemic Inflammation," *American Journal of Physiology*, 1997, vol. 273 (1 Pt 2), pp. R407-R413.
- Sachs M.C., et al., "Performance of Maximum Inspiratory Pressure Tests and Maximum Inspiratory Pressure Reference Equations for 4 Race/Ethnic Groups," *Respiratory Care*, Oct. 2009, vol. 54 (10), pp. 1321-1328.
- Sahu S. and Swain A., "Optic Nerve Sheath Diameter: A Novel Way to Monitor the Brain," *Journal of Neuroanaesthesiology and Critical Care*, Feb. 2017, vol. 4(4), pp. S13-S18.
- Salimi, M., et al., "Stimulating neural pathways to reduce mechanical ventilation-associated neuro-cognitive dysfunction," *American Journal of Respiratory and Critical Care Medicine*, Letter to the Editor, Oct. 25, 2021.
- Salmela L., et al., "Verification of the Position of a Central Venous Catheter by Intra-Atrial ECG. When does this method fail?," *Acta Anaesthesiol Scand*, 1993, vol. 37 (1), pp. 26-28.
- Sandborn W.J., "Transdermal Nicotine for Mildly to Moderately Active Ulcerative Colitis," *Annals of Internal Medicine*, vol. 126 (5), Mar. 1, 1997, pp. 364-371.
- Sandoval R., "A Catch/Ike Property-Based Stimulation Protocol for Diaphragm Pacing, Master of Science Coursework project", Simon Fraser University, Mar. 2013.
- Sarnoff, S.J. et al., "Electrophrenic Respiration," *Science*, 1948, vol. 108, p. 482.
- Sato E., et al., "Acetylcholine Stimulates Alveolar Macrophages to Release Inflammatory Cell Chemotactic Activity," *American Journal of Physiology*, vol. 274 (Lung Cellular and Molecular Physiology 18), 1998, pp. L970-L979.
- Sato, K.Z., et al., "Diversity of mRNA Expression for Muscarinic Acetylcholine Receptor Subtypes and Neuronal Nicotinic Acetylcholine Receptor Subunits in Human Mononuclear Leukocytes and Leukemic Cell Lines," *Neuroscience Letters*, vol. 266 (1), 1999, pp. 17-20.
- Schauerte P., et al., "Transvenous Parasympathetic Nerve Stimulation in the Inferior Vena Cava and Atrioventricular Conduction," *Journal of Cardiovascular Electrophysiology*, vol. 11 (1), Jan. 2000, pp. 64-69.
- Schauerte P.N., et al., "Transvenous Parasympathetic Cardiac Nerve Stimulation: An Approach for Stable Sinus Rate Control," *Journal of Cardiovascular Electrophysiology*, vol. 10 (11), Nov. 1999, pp. 1517-1524.
- Scheinman R.I., et al., "Role of Transcriptional Activation of IkB α in Mediation of Immunosuppression by Glucocorticoids," *Science*, vol. 270, Oct. 13, 1995, pp. 283-286.
- Sher, M.E., et al., "The Influence of Cigarette Smoking on Cytokine Levels in Patients with Inflammatory Bowel Disease," *Inflammatory Bowel Diseases*, vol. 5 (2), May 1999, pp. 73-78.
- Slutsky A.S., "History of Mechanical Ventilation. From Vesalius to Ventilator-induced Lung Injury," *American Journal of Respiratory and Critical Care Medicine*, May 2015, vol. 191(10), pp. 1106-1115.
- Steinlein, O., "New Functions for Nicotinic Acetylcholine Receptors?," *Behavioural Brain Research*, vol. 95, 1998, pp. 31-35.
- Sternberg E.M., (Series Editor) "Neural-Immune Interactions in Health and Disease," *The Journal of Clinical Investigation*, vol. 100 (11), Dec. 1997, pp. 2641-2647.
- Sykes, A.P., et al., "An Investigation into the Effect and Mechanisms of Action of Nicotine in Inflammatory Bowel Disease," *Inflammation Research*, vol. 49, 2000, pp. 311-319.
- Tang, H., et al., "Smad3 initiates oxidative stress and proteolysis that underlies diaphragm dysfunction during mechanical ventilation," *Nature: Scientific Reports*, vol. 7, issue 14530, 2017.
- Toyabe S., et al., "Identification of Nicotinic Acetylcholine Receptors on Lymphocytes in the Periphery as well as Thymus in Mice," *Immunology*, 1997, vol. 92, pp. 201-205.
- Van Dijk A.P.M., et al., "Transdermal Nicotine Inhibits Interleukin 2 Synthesis by Mononuclear Cells Derived from Healthy Volunteers," *European Journal of Clinical Investigation*, vol. 28, 1998, pp. 664-671.
- Vitacca M., et al., "Acute Effects of Deep Diaphragmatic Breathing in COPD Patients with Chronic Respiratory Insufficiency," *European Respiratory Journal*, 1998, vol. 11(2), pp. 408-415.
- Wanner, A. et al., "Transvenous Phrenic Nerve Stimulation in Anesthetized Dogs," *Journal of Applied Physiology*, 1973, vol. 34 (4), pp. 489-494.
- Watkins L.R., et al., "Blockade of Interleukin-1 Induced Hyperthermia by Subdiaphragmatic Vagotomy: Evidence for Vagal Mediation of Immune-Brain Communication," *Neuroscience Letters*, 1995, vol. 183, pp. 27-31.
- Watkins L.R., et al., "Implications of Immune-to-Brain Communication for Sickness and Pain," *PNAS (Proceedings of the National Academy of Sciences of the USA)*, vol. 96 (14), Jul. 6, 1999, pp. 7710-7713.
- Whaley K., et al., "C2 Synthesis by Human Monocytes is Modulated by a Nicotinic Cholinergic Receptor," *Nature*, vol. 293, Oct. 15, 1981, pp. 580-582 (and reference page).
- Yokota H., et al., "Regional Cerebral Blood Flow in Delirium Patients," *Psychiatry and Clinical Neurosciences*, Jun. 2003, vol. 57(3), pp. 337-339.
- Extended European Search Report for Application No. 22186152.9, mailed on Nov. 7, 2022, 8 pages.
- Extended European Search Report for Application No. 20806537, mailed on Dec. 15, 2022, 7 pages.
- International Search Report in International Application No. PCT/US2020/037017, mailed Sep. 24, 2020 (3 pages).

* cited by examiner

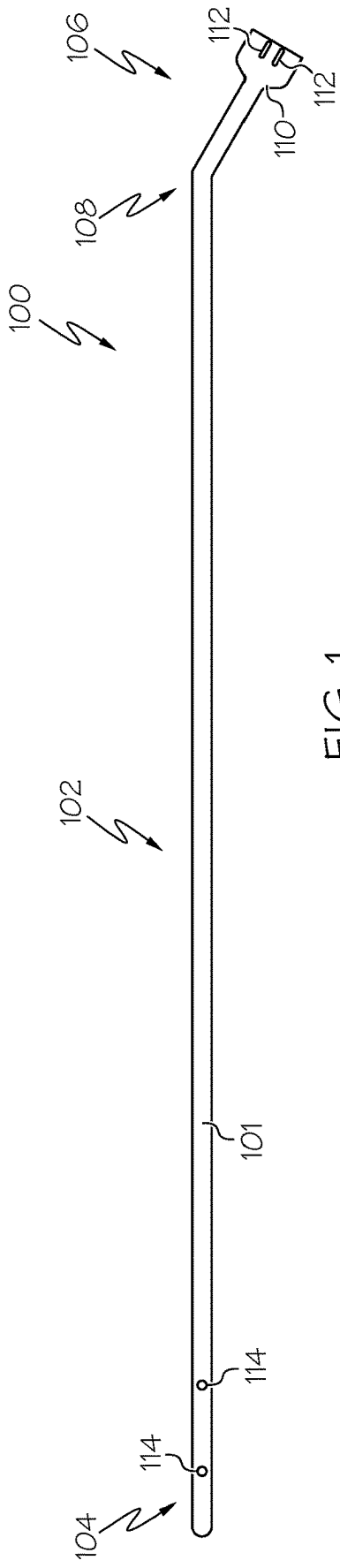


FIG. 1

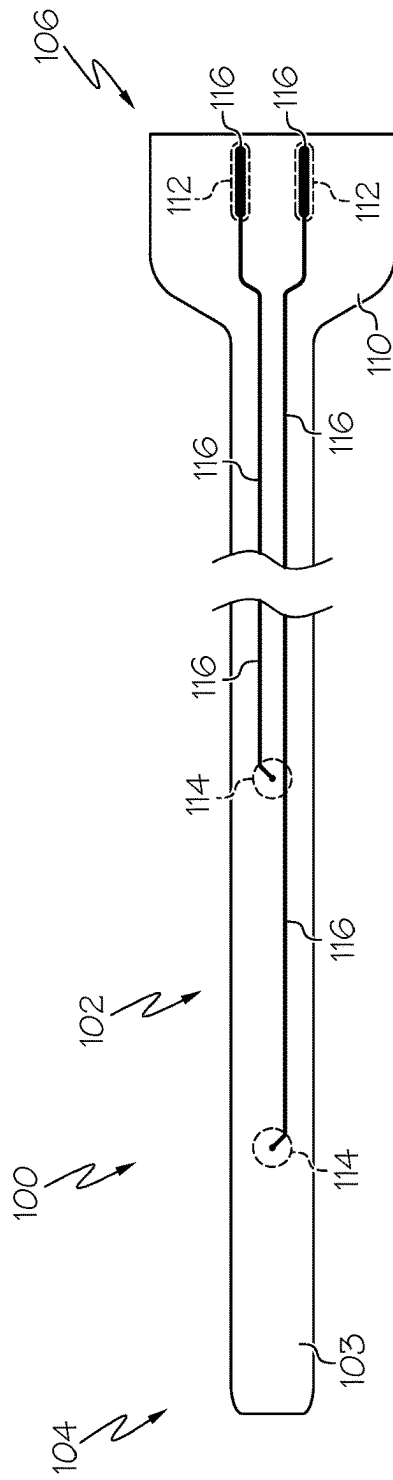


FIG. 2

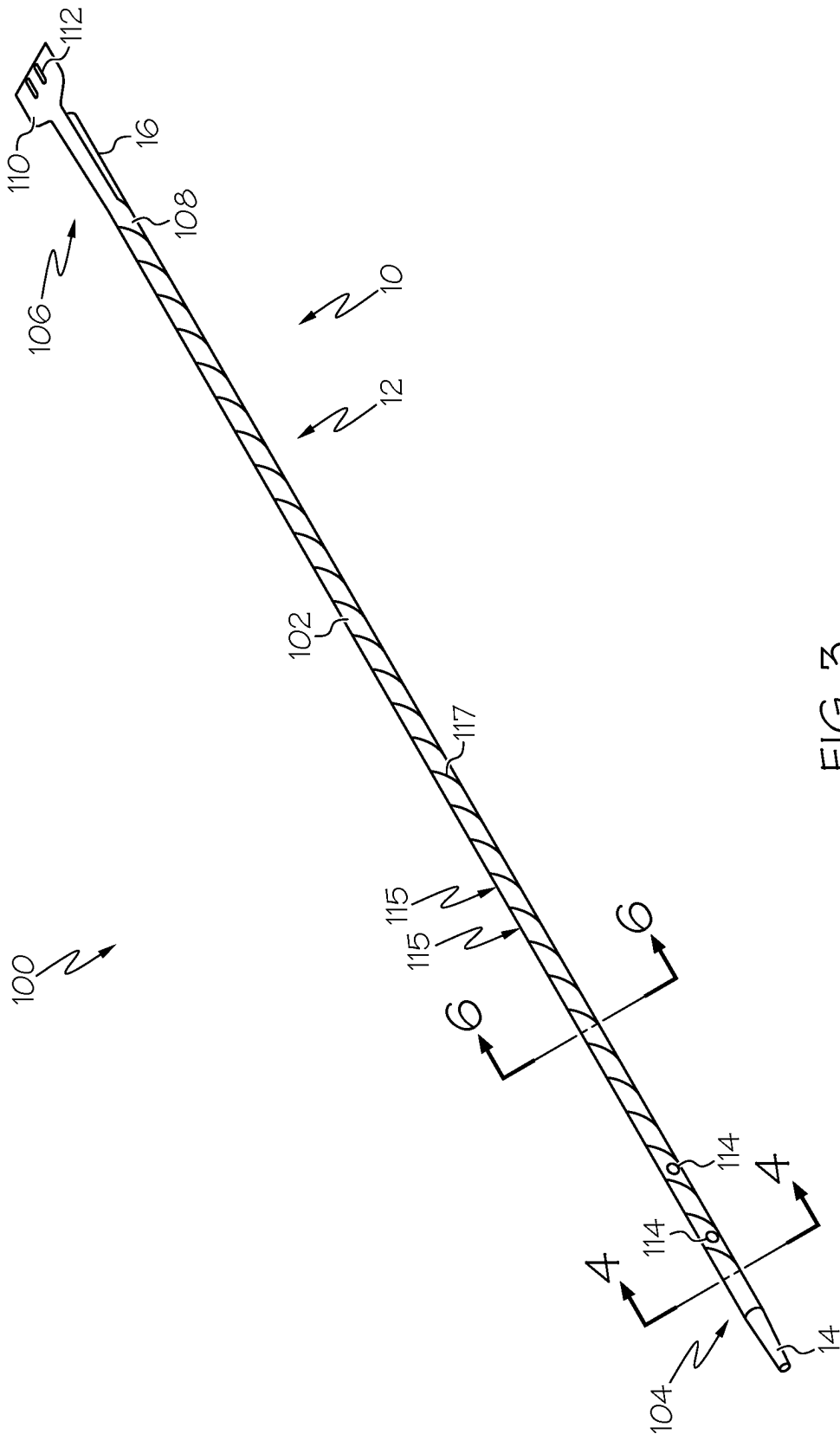
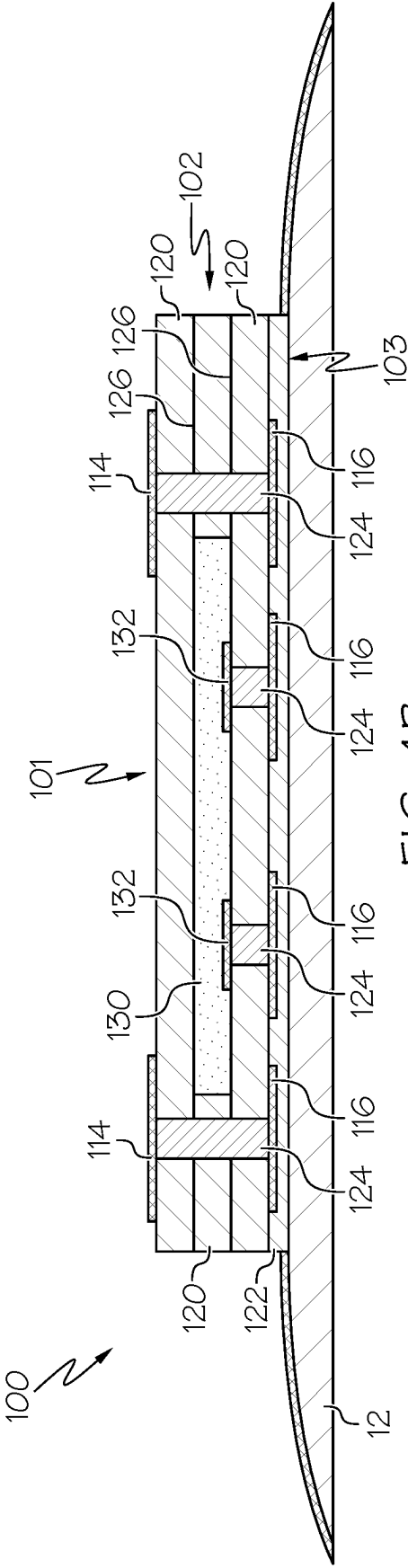
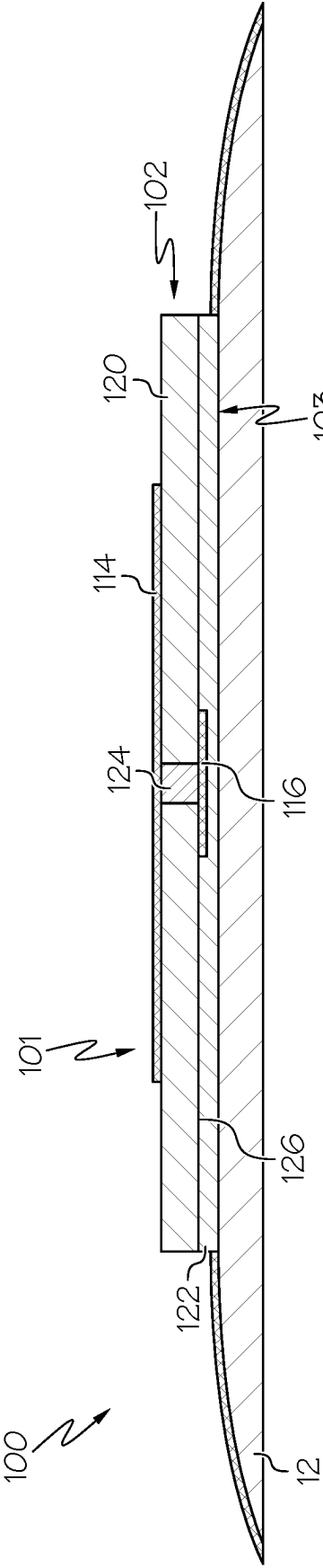
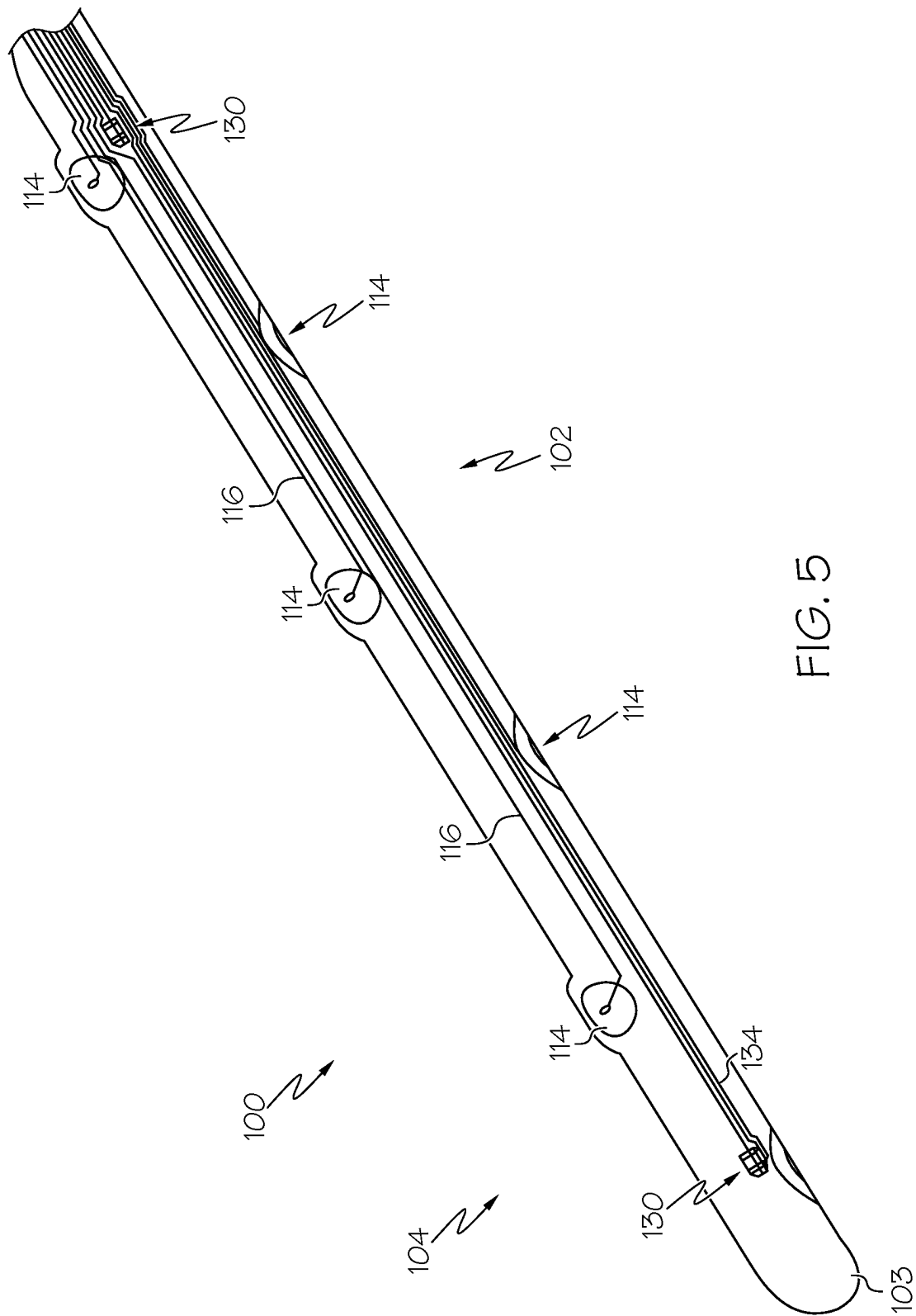


FIG. 3





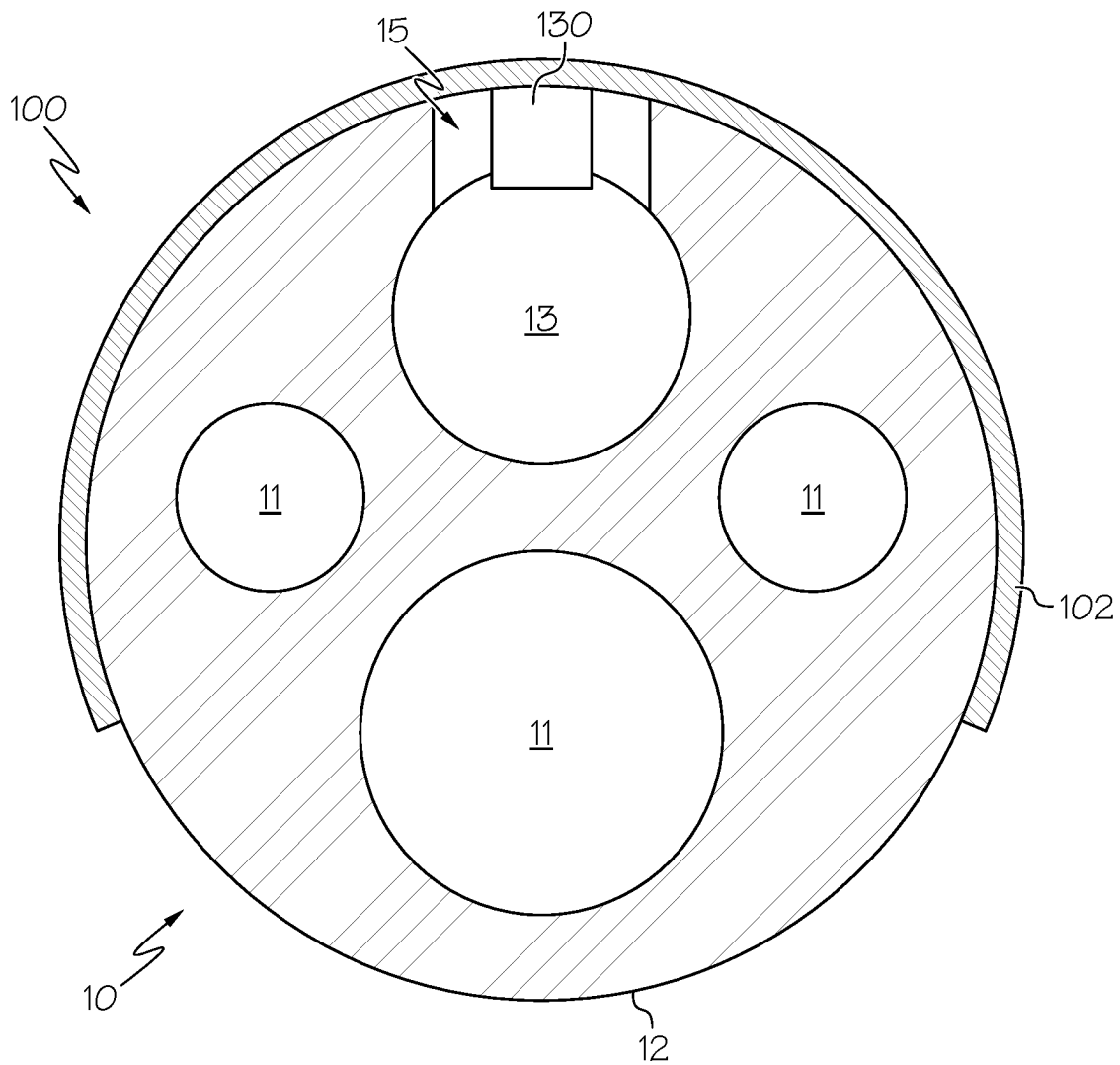


FIG. 6

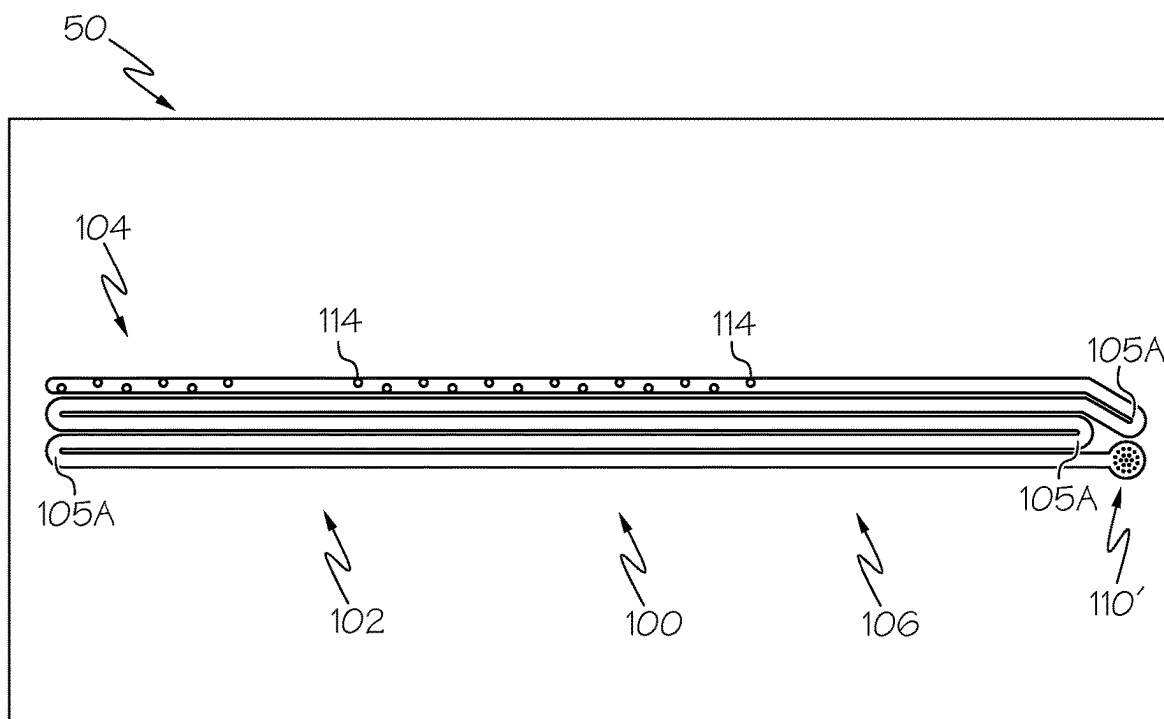


FIG. 7A

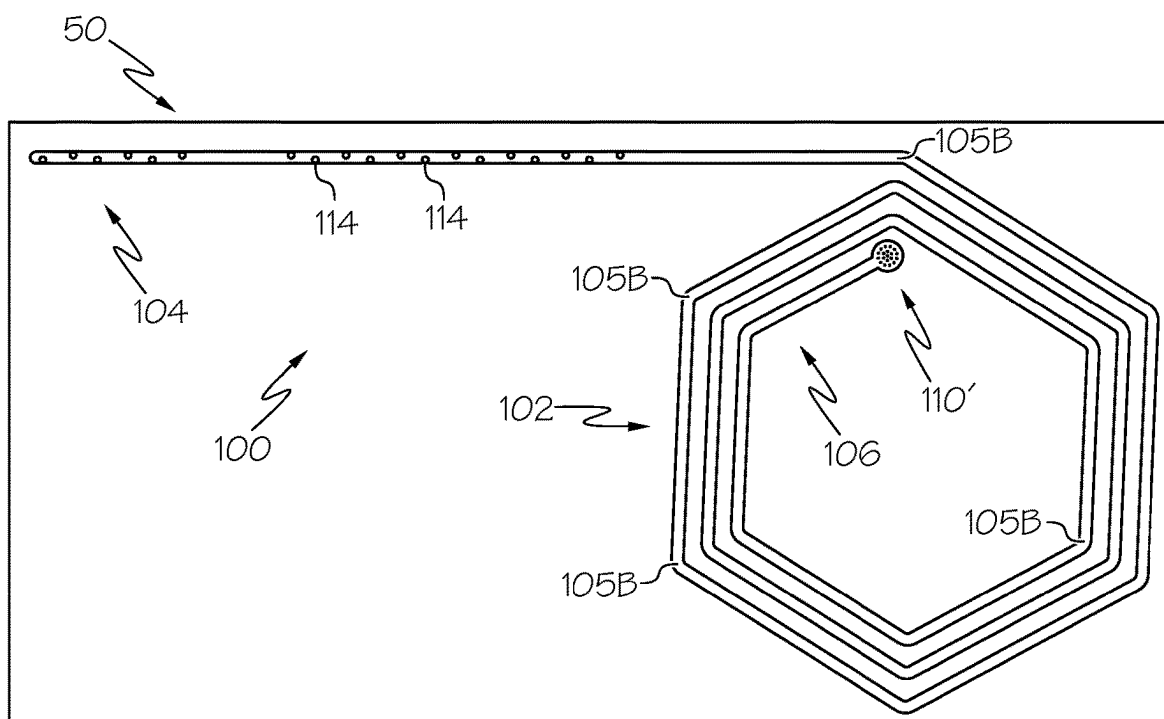
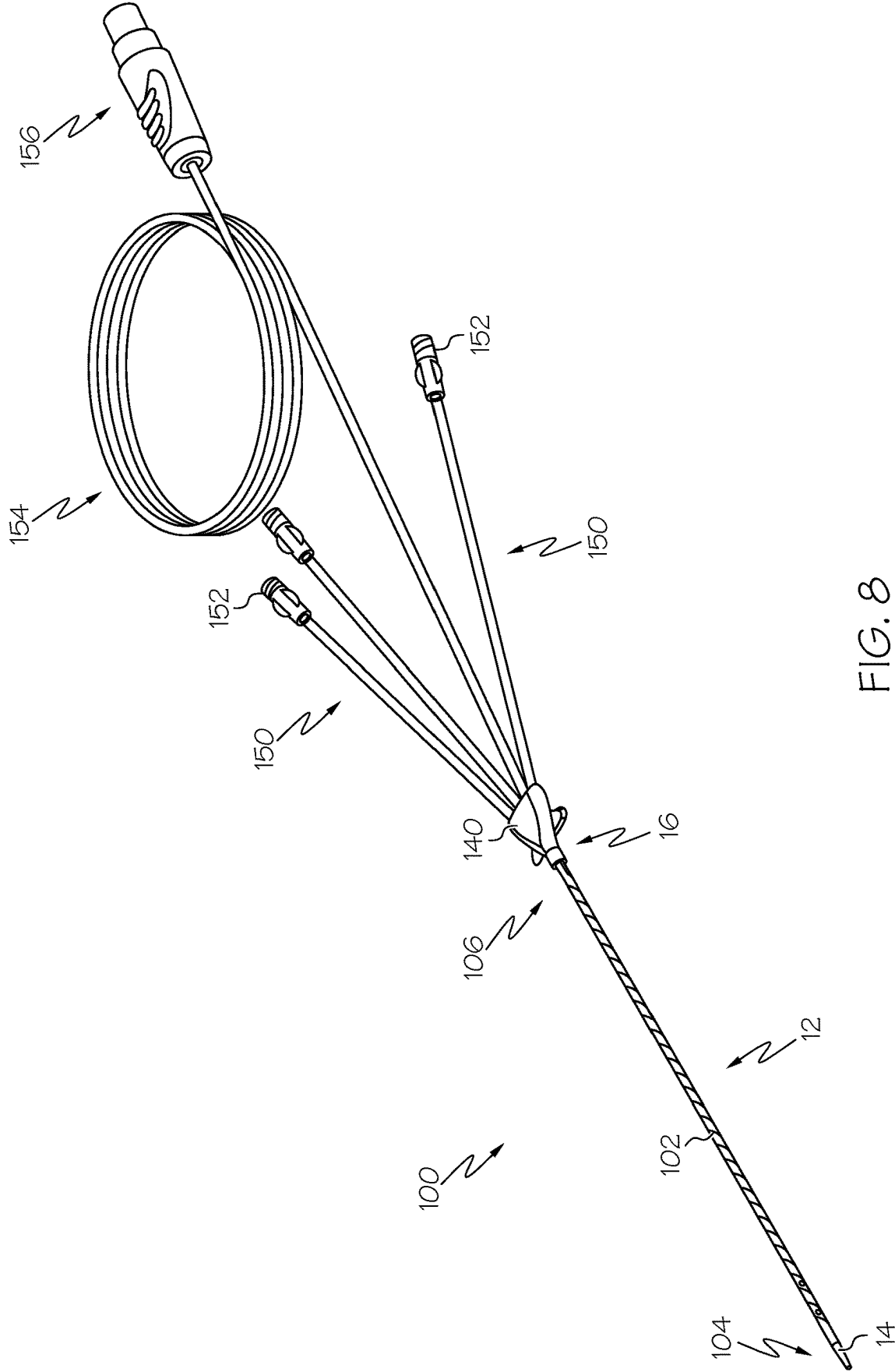


FIG. 7B



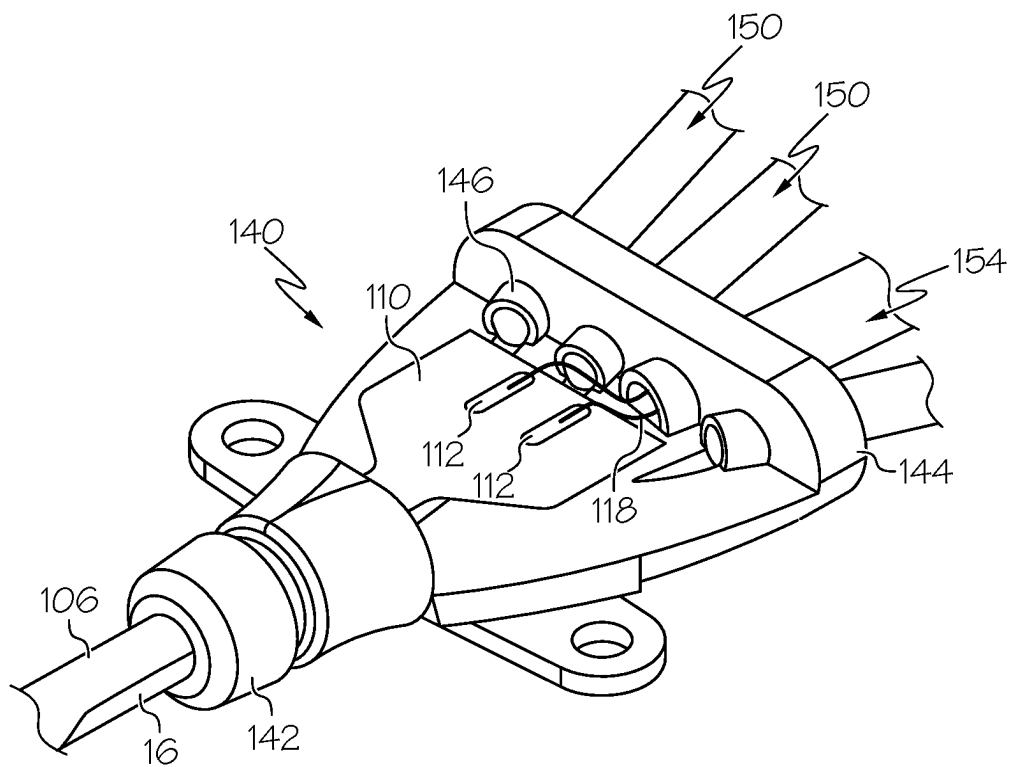


FIG. 9

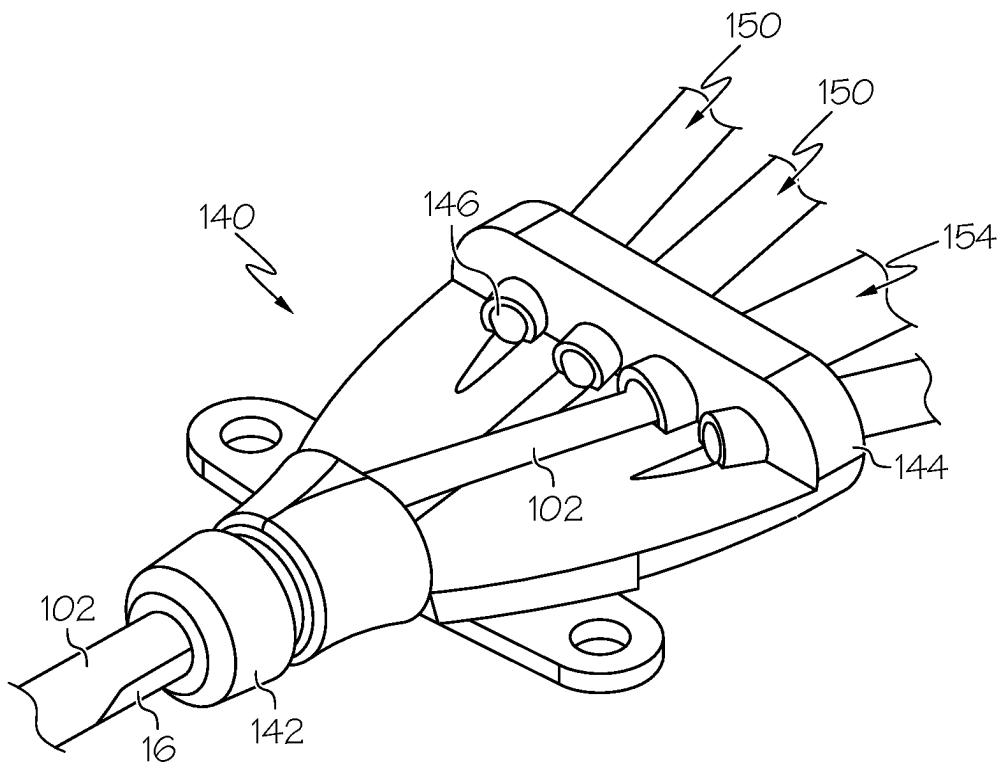


FIG. 10

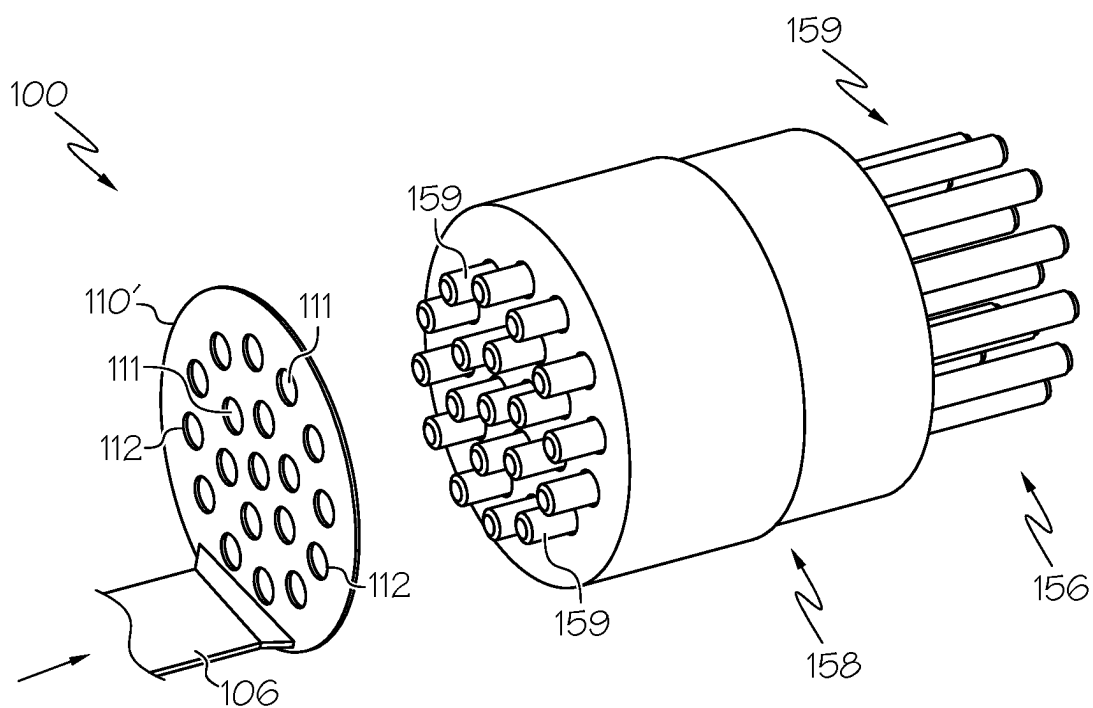


FIG. 11

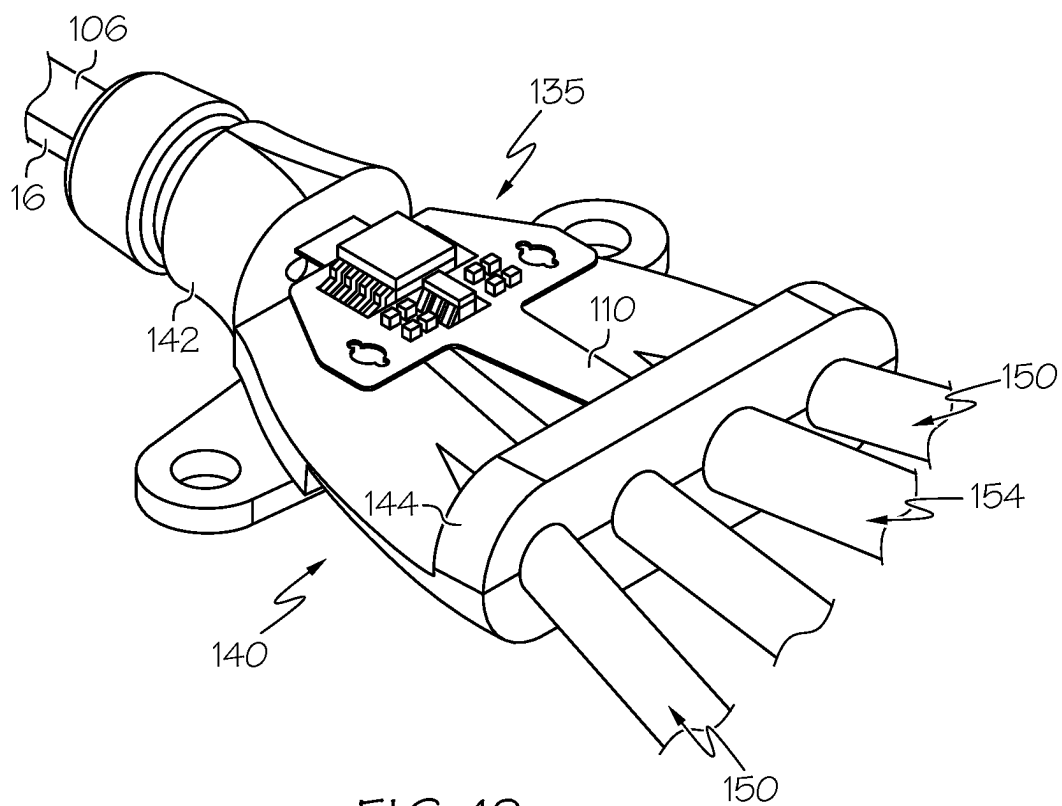
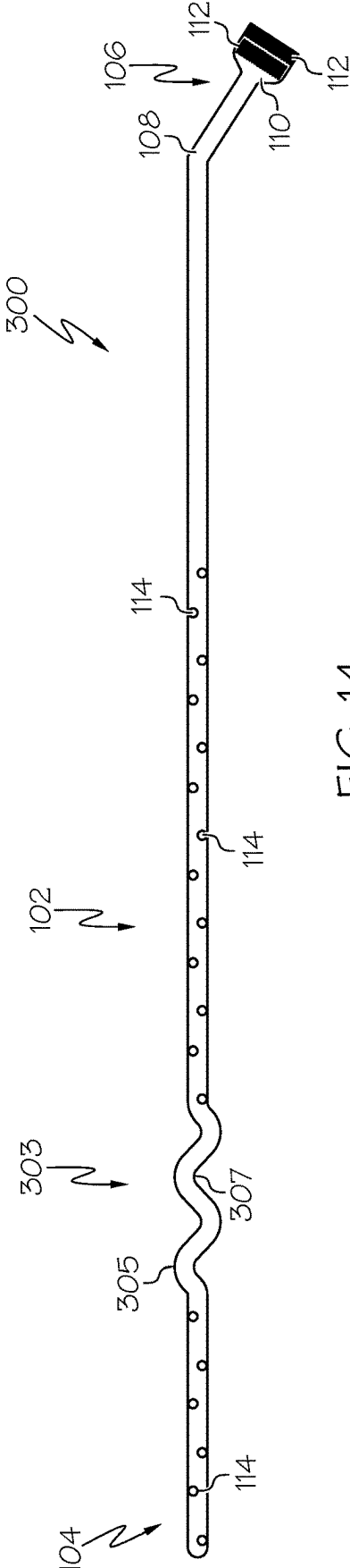
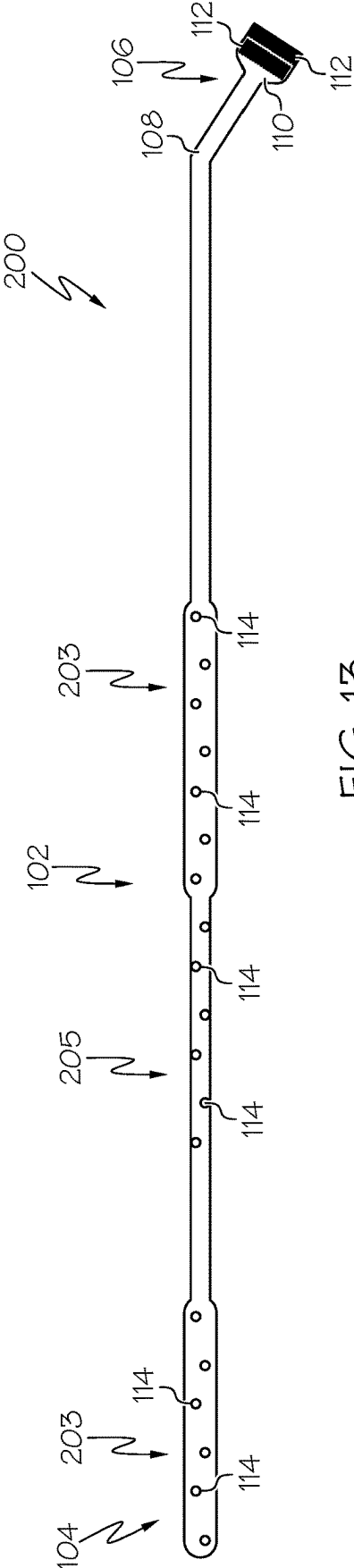
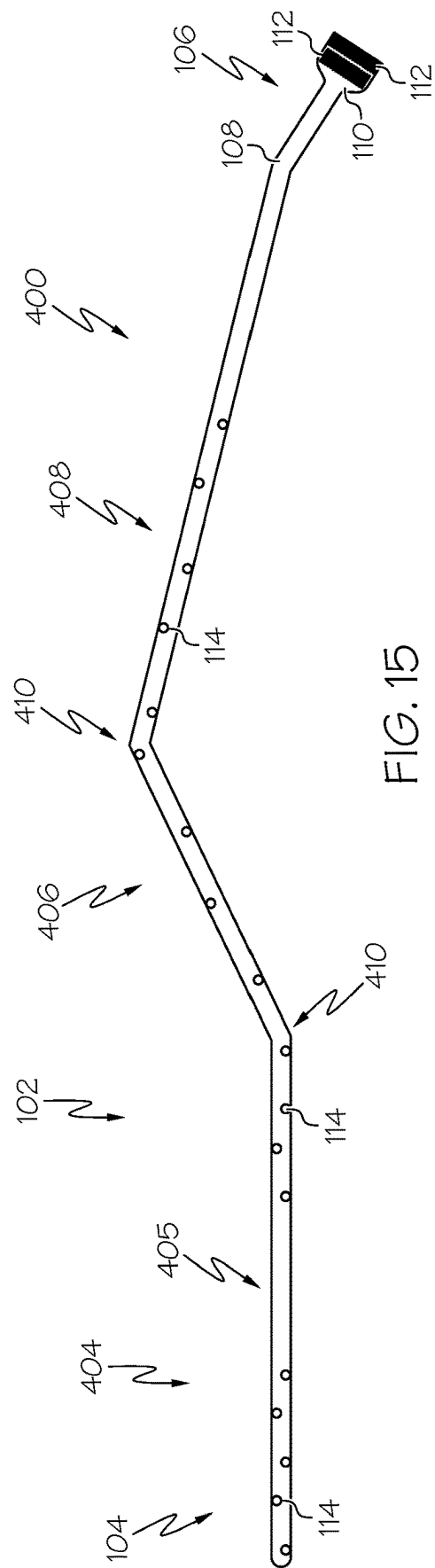


FIG. 12





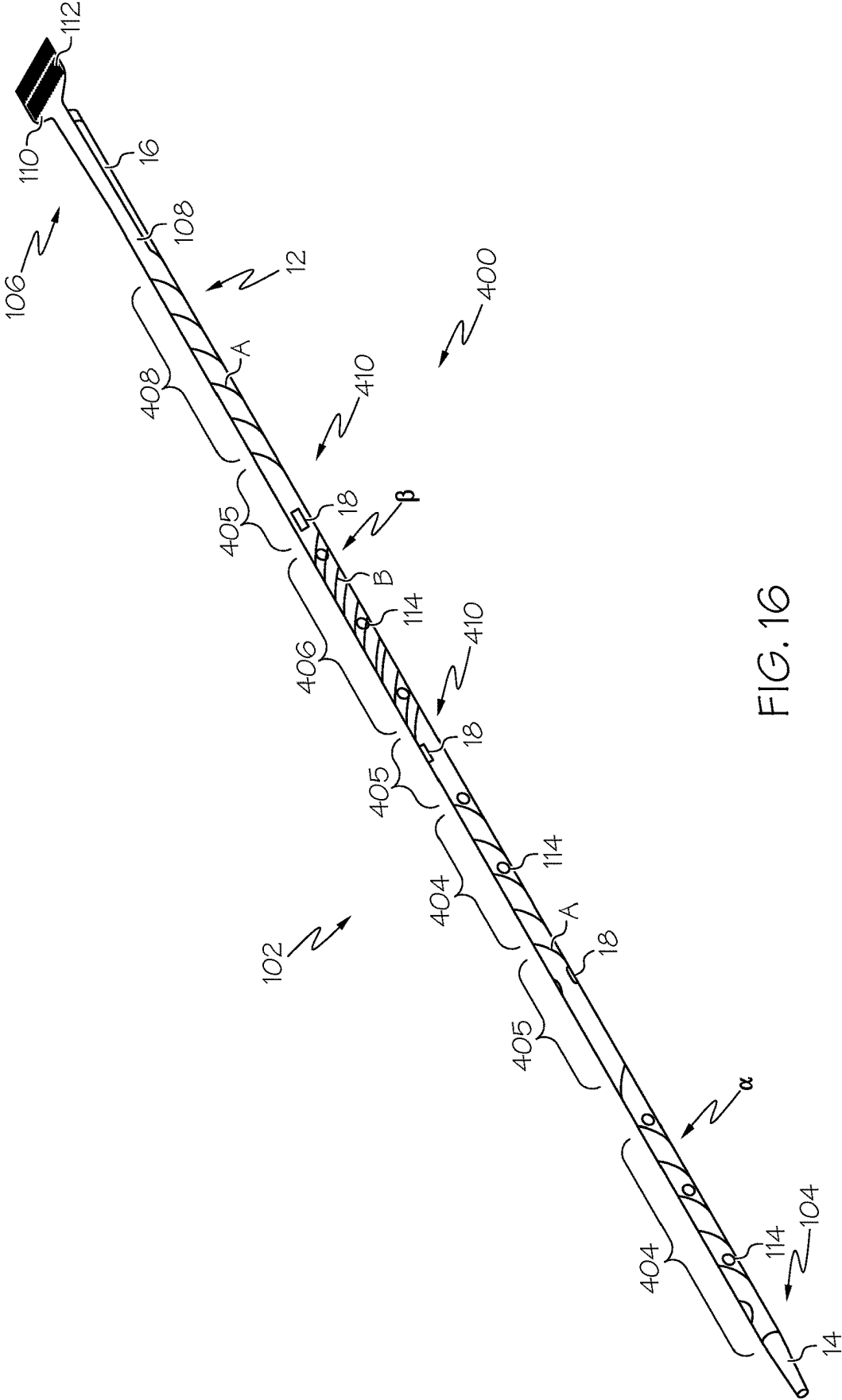
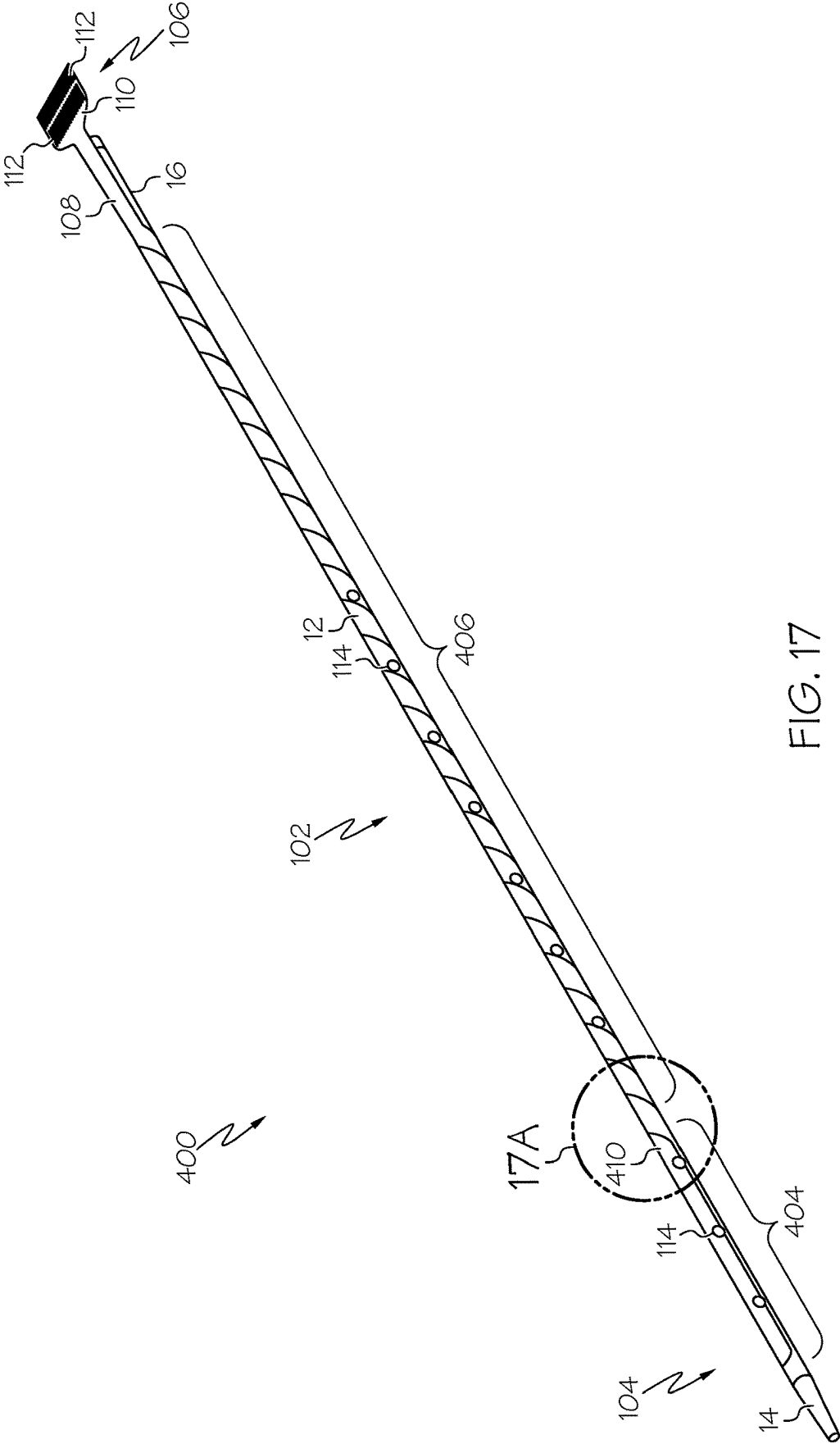


FIG. 16



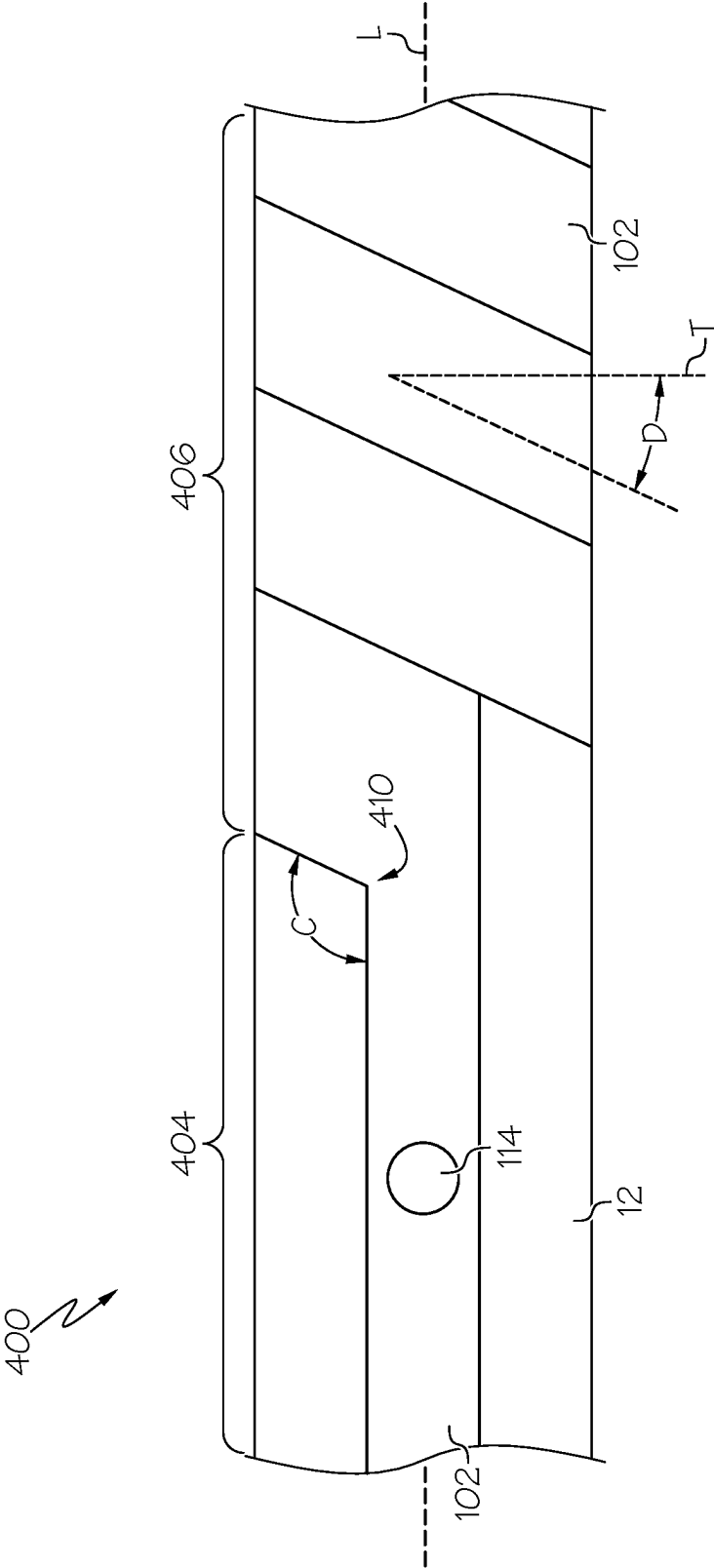
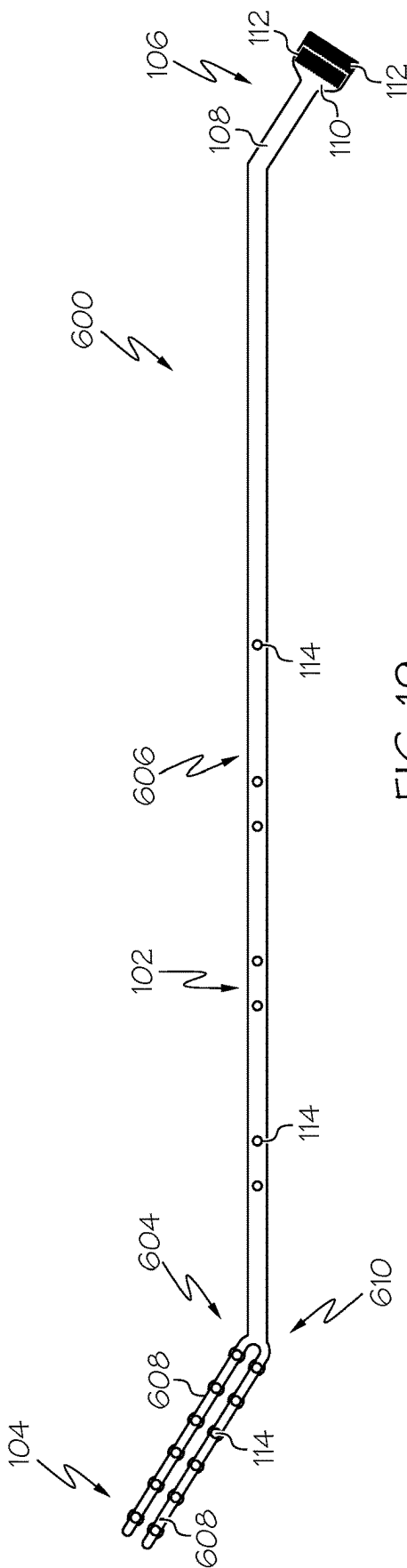
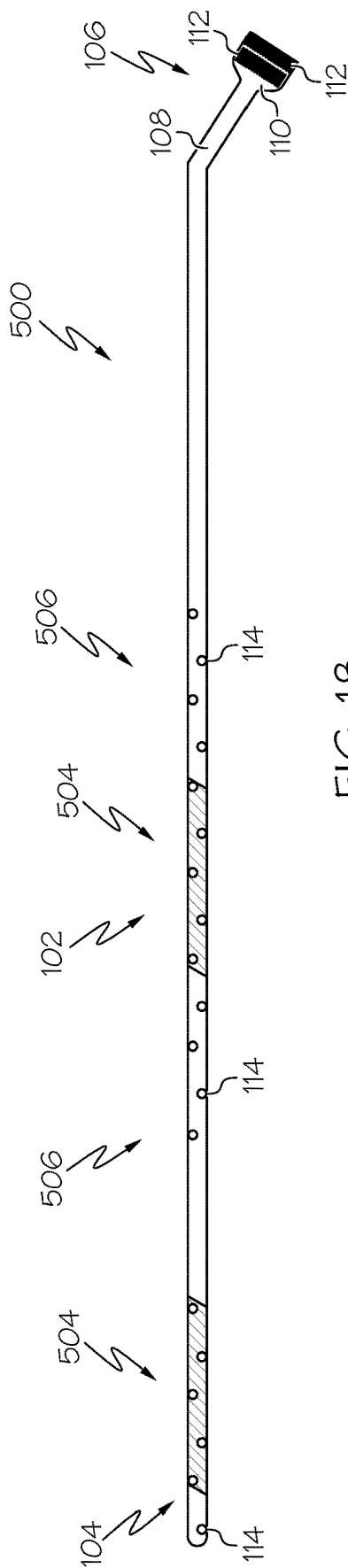
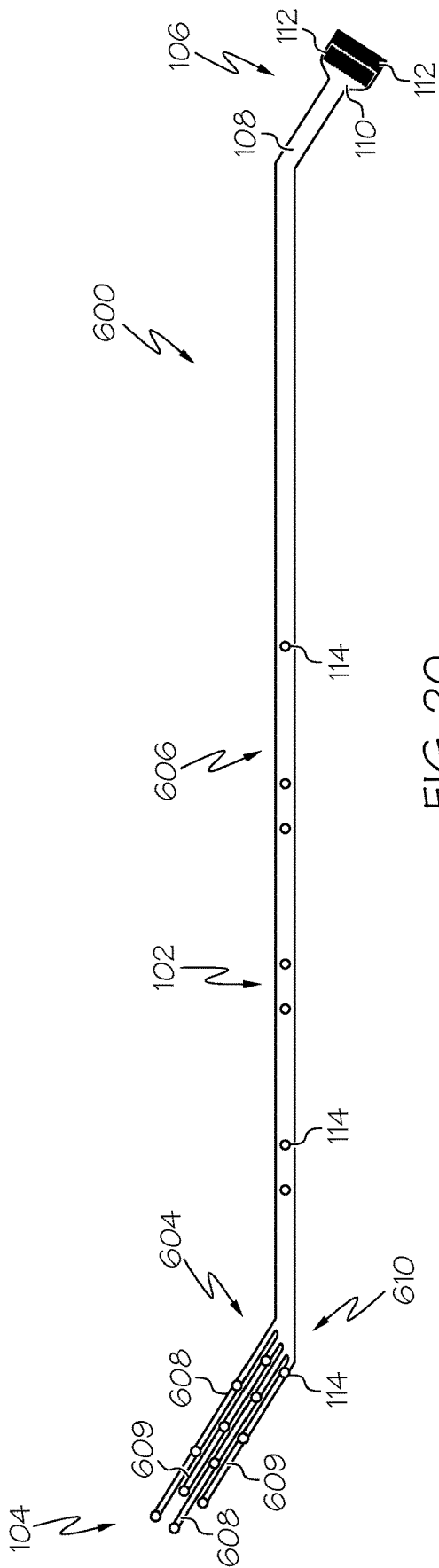


FIG. 17A





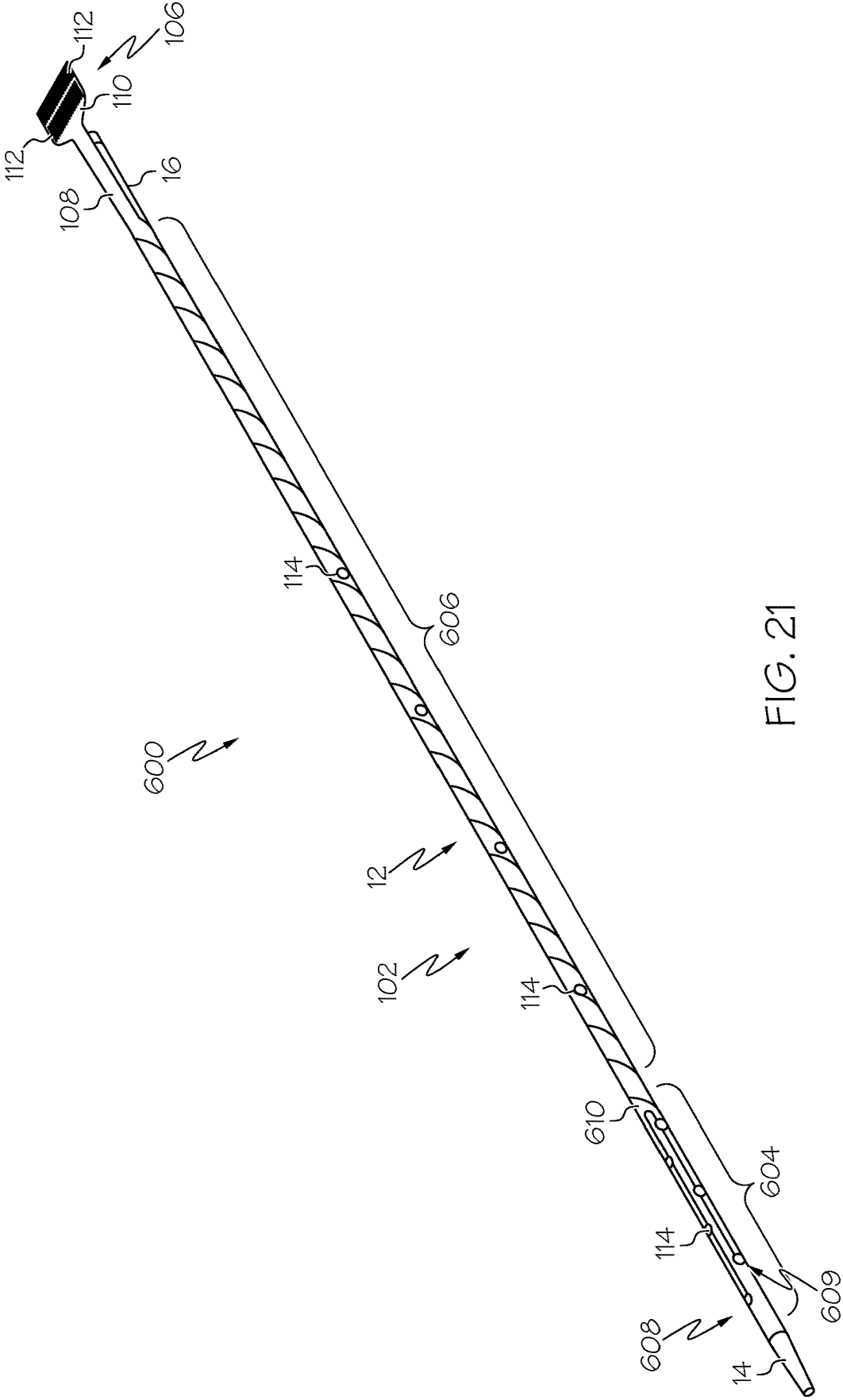
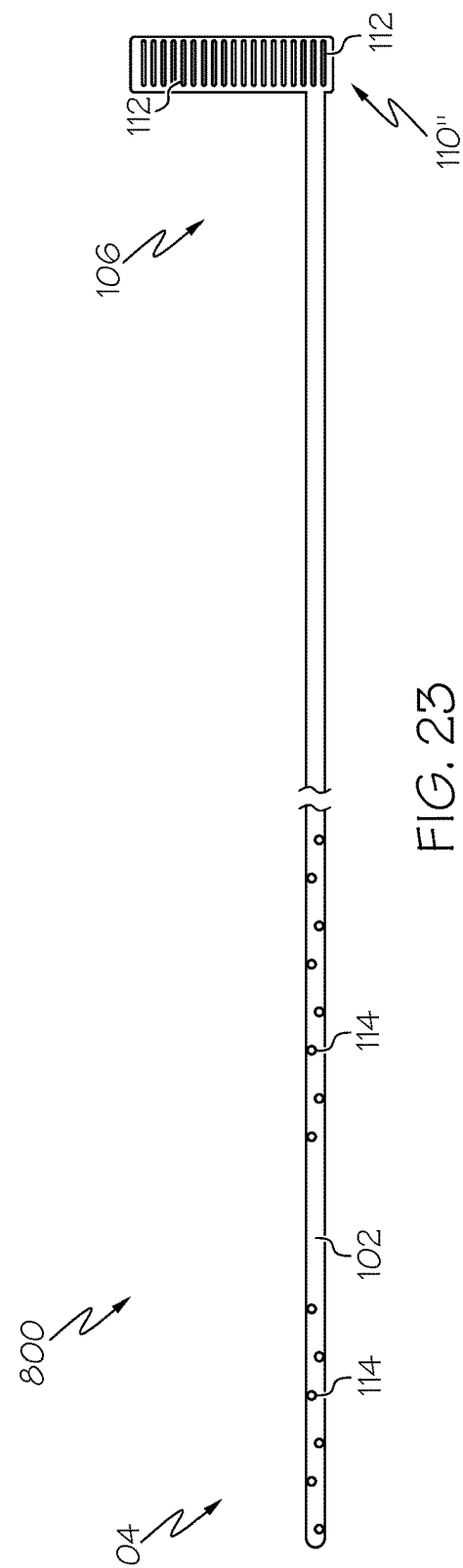
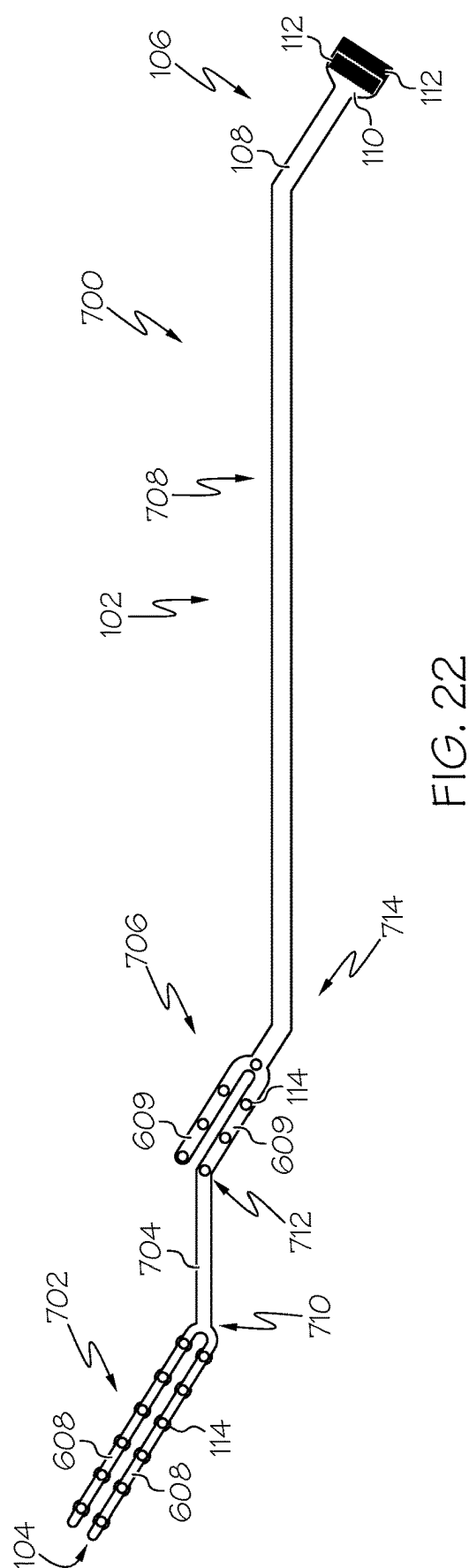


FIG. 21



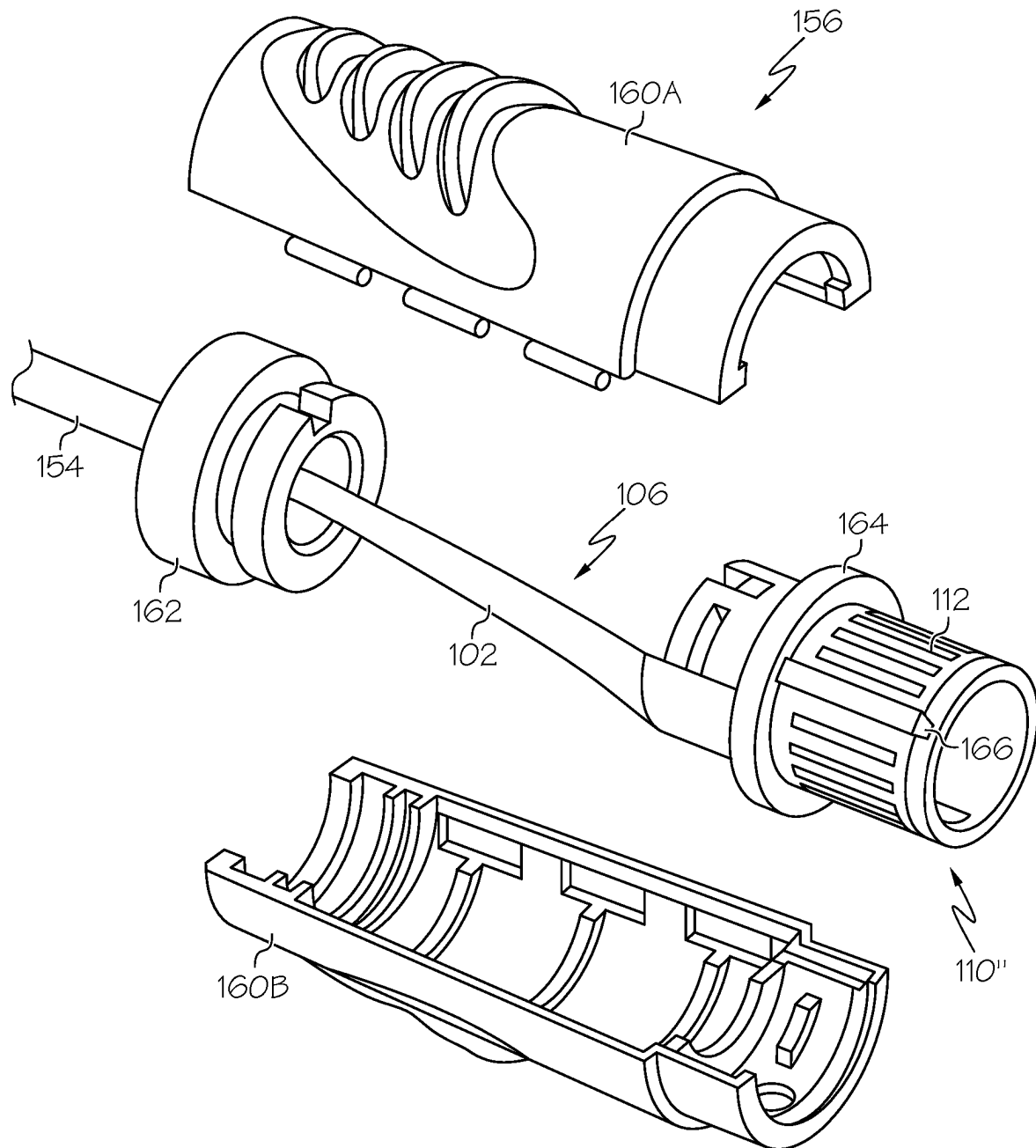


FIG. 24

1

CIRCUITRY FOR MEDICAL STIMULATION SYSTEMS

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. application Ser. No. 16/898,157, filed on Jun. 10, 2020, which claims priority to U.S. Provisional Application No. 62/860,452, filed on Jun. 12, 2019, each of which is hereby incorporated by reference in its entirety. All publications, patent applications, and patents mentioned in this specification are herein incorporated by reference in their entirety to the same extent as if each individual document was specifically individually indicated to be incorporated by reference.

TECHNICAL FIELD

Embodiments of this disclosure relate to systems, devices, and apparatuses for providing electrical circuits, leads, and/or electrodes on medical devices configured for stimulation of a body. Embodiments of this disclosure generally relate to medical devices (including systems) for the stimulation of nerves and/or muscles, including a flexible circuitry construction for use during the stimulation of muscles and/or nerves. In embodiments, systems may restore, enhance, and/or modulate diminished neurophysiological functions using electrical stimulation. Some embodiments provide a flexible layer disposed over a medical device for positioning electrodes external of the device and adjacent to one or more target nerves during a procedure.

BACKGROUND

Critical care patients, particularly those requiring invasive mechanical ventilation (MV), may generally experience higher levels of diaphragm, lung, brain, heart, and/or other organ injury. The respiratory muscles (e.g., diaphragm, sternocleidomastoid, scalenes, pectoralis minor, external intercostals, internal intercostals, abdominals, quadratus, etc.) may rapidly lose mass and strength during MV. A patient's lungs may suffer from ventilator-induced trauma, including high and low pressure injuries. Cognitive effects of MV may be caused by several factors, including aberrant neuro-signaling and inflammatory responses. Limiting a duration for which patients are subjected to MV may contribute toward minimizing such negative side effects. However, rapid respiratory muscle atrophy in MV patients makes it challenging to transition patients away from a dependency on MV. Options may be limited for strengthening the respiratory muscles of critical care patients, particularly for those that are on MV, so that they may regain an ability to breathe without external respiratory support.

SUMMARY

Embodiments of the present disclosure relate to, among other things, systems, devices, and methods for applying stimulation to one or more anatomical targets. Embodiments of flexible circuitry apparatuses for the systems and devices described herein may be used with MV, such as, for example, stimulation of respiratory nerves and/or respiratory muscles. Each of the embodiments disclosed herein may include one or more of the features described in connection with any of the other disclosed embodiments.

In one example, a medical device may include a flexible body having a proximal end and a distal end and an electrode

2

positioned on the body proximate the distal end. The electrode is configured to provide an electrical charge for stimulating tissue. The medical device may include an electrical connection positioned on the body proximate the proximal end. The electrical connection is configured to electrically couple the electrode to a power source. The medical device may include an electrical lead connecting the electrode to the electrical connection, wherein the lead is on or in the flexible body. The body is configured to have a first configuration prior to being secured to an exterior of a tube and a second configuration having a shape that conforms to a profile of the tube with the electrode secured to an exterior of the tube.

Any of the systems or methods disclosed herein may include any of the following features. The medical device may include the tube. The flexible body is helically wound about the exterior of the tube in a plurality of winds, to achieve the second configuration. The medical device may include the tube, and securing the flexible body about the exterior of the tube causes the tube to have regions of varying flexibility at locations of the tube having the flexible body. The flexible body includes a dielectric substrate film disposed between the electrode and the electrical lead. The flexible body further includes a hole through the dielectric substrate film, and the hole is filled with a conductive material electrically connecting the electrode to the electrical lead. The medical device includes a sensor, an electronic chip, or an integrated circuit disposed within the dielectric substrate film and between the electrode and the electrical lead. The flexible body, in the first configuration, includes at least one segment having an enlarged width relative to an adjacent portion of the flexible body, wherein the at least one segment is less flexible than the adjacent portion of the flexible body. The flexible body, in the first configuration, includes at least one segment having a nonlinear configuration adjacent to a portion of the flexible body having a linear configuration, wherein the at least one segment is more flexible than the portion. The medical device includes the tube. The flexible body, in the first configuration, includes at least one bend between the proximal end and the distal end such that a longitudinal axis of a proximal portion proximal to the bend is transverse to a longitudinal axis of a distal portion distal to the bend. A flexibility of a segment of the medical device having the proximal portion differs from a flexibility of a segment of the medical device having the distal portion, due to the arrangements of the proximal and distal portions on the tube. The at least one bend causes a change in direction of a helical wind of the flexible body about the tube, when the flexible body is on the second configuration. The flexible body includes at least one segment having a stiffening structure that is configured to increase a stiffness of the flexible body relative to an adjacent portion of the flexible body. The flexible body includes at least one segment at the distal end having a plurality of branches, wherein each of the plurality of branches includes an array of electrodes. The at least one segment is angled relative to the proximal end such that such that a longitudinal axis of the plurality of branches is transverse to a longitudinal axis of the flexible body. The flexible body includes a second segment proximal of the distal end and having a plurality of second branches, wherein each of the plurality of second branches includes a second array of electrodes.

In another example, a medical device may include a tube having a longitudinal length and at least one lumen. The tube is configured to receive at least one of a device and a fluid in the at least one lumen. The medical device may include

3

a flexible body having a stimulation array configured to provide an electrical charge for stimulating tissue. The flexible body includes a proximal portion and a distal portion separated from the proximal portion by at least one bend. The flexible body is configured to cover at least a portion of the longitudinal length of the tube such that the stimulation array is disposed about an exterior of the tubular body. The at least one bend is configured to arrange one of the proximal portion and the distal portion in a linear configuration and the other of the proximal portion and the distal portion in a nonlinear configuration relative to the tube, when the flexible body is wound about the exterior of the tube.

Any of the systems or methods disclosed herein may include any of the following features. The stimulation array includes a distal electrode array positioned on the flexible body proximate to the distal portion. The distal electrode array is configured to provide the electrical charge for stimulating tissue at a first location. The stimulation array includes a proximal electrode array positioned on the flexible body proximate to the proximal portion. The proximal electrode array is configured to provide the electrical charge for stimulating tissue at a second location that is different than the first location. The stimulation array includes an electrical connection positioned on the flexible body proximate to the proximal portion and configured to electrically couple the distal electrode array and the proximal electrode array to a power source. The at least one bend causes a change in direction of a helical wind of the flexible body about the tube, and a longitudinal axis of the proximal portion proximal to the at least one bend is transverse to a longitudinal axis of the distal portion distal to the at least one bend. A flexibility of a portion of the medical device having the proximal portion differs from a flexibility of a portion of the medical device having the distal portion, due to the arrangements of the proximal portion and distal portion on the tube.

In another example, a medical device may include a tube and a circuit having a flexible body including a proximal portion and a distal portion that is angled relative to the proximal portion. The distal portion having a plurality of branches each including an array of electrodes configured to provide an electrical charge for stimulating tissue. The flexible body is configured to engage the tube with the proximal portion disposed over the tube in a helical configuration, and the distal portion is disposed over the tube in a linear configuration, such that the plurality of branches is arranged substantially parallel to a longitudinal axis of the tube.

Any of the systems or methods disclosed herein may include any of the following features. The plurality of branches are positioned about a circumference of the tube at spaced intervals. The circuit is configured to modify a flexibility of the tube when the flexible body engages the tube, with the flexibility varying between a segment of the tube having the proximal portion and a segment of the tube having the distal portion.

It may be understood that both the foregoing general description and the following detail description are exemplary and explanatory only and are not restriction of the invention, as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate non-

4

limiting embodiments of the present disclosure and together with the description serve to explain the principles of the disclosure.

FIG. 1 illustrates a side view of an exemplary circuit device including a series of electrical and conductive components, according to various embodiments of the present disclosure;

FIG. 2 illustrates a partial side view of the circuit device of FIG. 1 including electrical circuitry associated with the electrical and conductive components, according to one or more embodiments of the present disclosure;

FIG. 3 illustrates a perspective view of the circuit device of FIG. 1 attached to a medical device in a helical configuration, according to various embodiments;

FIG. 4A illustrates a cross-sectional view of the circuit device of FIG. 1 attached to the medical device of FIG. 3 and including an electrode, according to one or more embodiments;

FIG. 4B illustrates a cross-sectional view of the circuit device of FIG. 1 attached to the medical device of FIG. 3 and including a device component, according to one or more embodiments;

FIG. 5 illustrates a perspective view of the circuit device of FIG. 1 including a device component, according to one or more embodiments of the present disclosure;

FIG. 6 illustrates a cross-sectional view of the circuit device of FIG. 1 with the device component received within a slot of the medical device of FIG. 3, according to one or more embodiments of the present disclosure;

FIG. 7A illustrates a side view the circuit device of FIG. 1 printed on a planar sheet and including one or more curved bends, according to one or more embodiments of the present disclosure;

FIG. 7B illustrates a side view of the circuit device of FIG. 1 printed on a planar sheet and including one or more angular bends, according to one or more embodiments of the present disclosure;

FIG. 8 illustrates a perspective of the circuit device of FIG. 1 attached to the medical device of FIG. 3 and including a hub; according to an embodiment of the present disclosure;

FIG. 9 illustrates a partial perspective view of the circuit device of FIG. 1 electrically coupled to the hub of FIG. 8 in a first configuration, according to one or more embodiments of the present disclosure;

FIG. 10 illustrates a partial perspective view of the circuit device of FIG. 1 electrically coupled to the hub of FIG. 8 in a second configuration, according to one or more embodiments of the present disclosure;

FIG. 11 illustrates a partial perspective view of the circuit device of FIG. 1 electrically coupled to a connector assembly, according to one or more embodiments of the present disclosure;

FIG. 12 illustrates of a partial perspective view of the circuit device of FIG. 1 electrically coupled to the hub of FIG. 8 and including a control unit, according to an embodiment of the present disclosure;

FIG. 13 illustrates a side view of another exemplary circuit device including a widened segment having reduced flexibility, according to one or more embodiments of the present disclosure;

FIG. 14 illustrates a side view of another exemplary circuit device including a non-linear segment having enhanced flexibility, according to one or more embodiments of the present disclosure;

5

FIG. 15 illustrates a side view of another exemplary circuit device including angular bends, according to one or more embodiments of the present disclosure;

FIG. 16 illustrates a perspective view of the circuit device of FIG. 15 attached to the medical device of FIG. 3 and including a flexible segment (low density) and a rigid segment (high density) between the angular bends to provide variable flexibility and other attributes, according to one or more embodiments of the present disclosure;

FIG. 17 illustrates a perspective view of another exemplary circuit device attached to the medical device of FIG. 3 and including a distal segment extending parallel to the medical device, according to one or more embodiments of the present disclosure;

FIG. 17A illustrates a partial side view of the circuit device of FIG. 17 with the distal segment including a bend, according to one or more embodiments of the present disclosure;

FIG. 18 illustrates a side view of another exemplary circuit device including a series of stiffener mechanisms, according to one or more embodiments of the present disclosure;

FIG. 19 illustrates a side view of another exemplary circuit device including an angled distal segment formed of a pair of branches, according to one or more embodiments of the present disclosure;

FIG. 20 illustrates a side view of another exemplary circuit device including an angled distal segment formed of a plurality of branches, according to one or more embodiments of the present disclosure;

FIG. 21 illustrates a perspective view of the circuit device of FIG. 20 attached to the medical device of FIG. 3, according to one or more embodiments of the present disclosure;

FIG. 22 illustrates a side view of another exemplary circuit device including an angled intermediate segment formed of a pair of branches and an angled distal segment formed of a pair of branches, according to one or more embodiments of the present disclosure;

FIG. 23 illustrates a side view of an exemplary circuit device including an electrical circuit at a proximal end, according to one or more embodiments of the present disclosure; and

FIG. 24 illustrates a perspective view of the circuit device of FIG. 23 with the electrical circuit received within a connector assembly, according to one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

Throughout the following description, specific details are set forth to provide a more thorough understanding to persons skilled in the art. The following description of examples of the technology is not intended to be exhaustive or to limit the system to the precise forms of any example embodiment. Accordingly, the description and drawings are to be regarded in an illustrative sense, rather than a restrictive sense.

All publications and patent applications mentioned in this specification are herein incorporated by reference in their entirety to the same extent as if each individual publication or patent application was specifically indicated to be incorporated by reference. For example, embodiments of the present disclosure may be used in combination with one or more systems, catheters, apparatuses, and electrodes described in U.S. Pat. Nos. 8,571,662, 9,242,088, 9,333,363, 9,776,005, 10,039,920, 10,293,164, U.S. Pat. Pub. 2015/

6

0045810, U.S. Pat. Pub. 2019/0001126, U.S. Pat. Pub. 2019/0175908, U.S. Pat. Pub. 2019/0038894, and/or U.S. Pat. Pub. 2020/0147364; the disclosures of all of which are hereby incorporated by reference.

Further aspects of the disclosures and features of example embodiments are illustrated in the appended drawings and/or described in the text of this specification and/or described in the accompanying claims. It may be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention, as claimed.

As used herein, the terms “comprises,” “comprising,” “including,” “having,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus.

Additionally, the term “exemplary” is used herein in the sense of “example,” rather than “ideal.” As used herein, the term “proximal” means a direction closer to an operator and the term “distal” means a direction further from an operator. The term “approximately” or like terms (e.g., “about,” “substantially”) encompass values within 10% of the stated value.

Reference will now be made in detail to examples of the present disclosure described above and illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

FIG. 1 shows a circuit device 100 according to an example of the present disclosure. Circuit device 100 may include a flexible body 102 having a longitudinal length defined between a distal end 104 and a proximal end 106. Flexible body 102 may include a flexible, planar substrate defined by an exterior surface 101 and an interior surface 103 (FIG. 2). In the example, flexible body 102 may generally include a linear configuration between distal end 104 and an intermediate portion of flexible body 102 distal of proximal end 106. Additionally, flexible body 102 may have a substantially constant and nominal width between distal end 104 and proximal end 106.

Flexible body 102 may include a bend 108 adjacent to proximal end 106 such that proximal end 106 may have a transverse alignment relative to the intermediate portion of flexible body 102 and distal end 104 (the axis of end 106 is transverse to the axis of the intermediate portion). As described further herein, bend 108 may form an angle that aligns a longitudinal axis of proximal end 106 with a central axis of a medical device. Flexible body 102 may further include a proximal tab 110 at proximal end 106 that may have a width that is relatively greater than a width of flexible body 102 at distal end 104 and/or the intermediate portion. Proximal tab 110 may be sized and/or shaped to receive one or more circuitry components thereon, such as conductive pads 112.

Flexible body 102 may be formed of a flexible material having pliability properties such that circuit device 100 may be configured to deform a shape, geometry, and/or configuration of flexible body 102. For example, flexible body 102 may be formed of Liquid Crystal Polymer (LCP), thermoplastics, polyimide, PEEK polyester, silicone, other elastomers, and the like. Flexible body 102 may be construed of multiple layers of materials, such as, for example, electrically conductive and/or non-conductive materials. As described in further detail below, circuit device 100 may be configured to engage a medical device and flexible body 102

may be configured to conform to a profile of the medical device when attached thereto.

Still referring to FIG. 1, circuit device 100 is an electrical connection device that may include a flexible printed circuit board and/or a flex ribbon circuit. For example, circuit device 100 may include flexible circuitry on flexible body 102, such as one or more (e.g., a plurality) electrical and/or conductive components including conductive pads 112 and electrodes 114. In the present example, circuit device 100 includes a pair of electrodes 114 patterned on exterior surface 101 adjacent to distal end 104 and a pair of conductive pads 112 patterned on exterior surface 101 adjacent to proximal end 106. Conductive pads 112 are included on proximal tab 110, and circuit device 100 may include at least one conductive pad 112 for each electrode 114 on flexible body 102.

Conductive pads 112 may be configured to connect electrodes 114 to an ancillary device (e.g., a programmable energy delivery system, a power source, etc.) for transmission of an electric current thereto. Electrodes 114 may be configured to treat a target site of a subject (e.g., a patient) when receiving the electrical current from the ancillary device. For example, electrodes 114 may be operable to stimulate a target site (e.g., a nerve, ganglia, etc.), deliver energy to tissue (e.g., thermal, electrical, etc.), and/or sense a physiologic parameter from or at the target site when distal end 104 is positioned at the target site.

Still referring to FIG. 1, electrodes 114 may form one or more stimulation arrays that are wired or wirelessly connected to one or more circuits, processors, devices, systems, applications, units, or controllers. Each electrode 114 of the stimulation array may include one or more nodes for delivering stimulation. As described in further detail herein, electrodes 114 may be supported on a medical device (e.g., an intravenous catheter, a guidewire, a stimulation lead, a needle, a probe, a lead, a patch, etc.) and may be configured to be placed on, or inserted into, a patient (e.g., a neck, torso, blood vessel, etc.) or placed on the surface of the skin. It should be understood that additional and/or fewer electrodes 114 of varying shapes and/or sizes may be included on flexible body 102 than those shown and described herein.

Electrodes 114 may be positioned along exterior surface 101 in various complex arrangements relative to one another to form high current densities along flexible body 102. It should be appreciated that flexible body 102 may be configured to permit enhanced variety in positioning electrodes 114 axially and radially along a circumference of a medical device (e.g., a catheter) when circuit device 100 is attached thereto. Stated differently, a position of electrodes 114 may not be limited by a structure of a medical device, such as, for example, a geometry of an inner lumen of the medical device, due to the incorporation of electrodes 114 on circuit device 100 in lieu of the medical device.

Electrodes 114 may be configured to deliver stimulation as monopolar, bipolar, tripolar, or other multipolar electrodes. As used herein, a stimulation transmitted by electrodes 114 may refer to an electrical signal transmitted by a single electrode 114 (anode) to the ground reference (cathode) that is positioned away from the anode to generate a larger electrical field (monopolar electrical stimulation); an electrical signal transmitted from an anode electrode 114 to a cathode electrode 114 (bipolar stimulation); or an electrical signal transmitted from one or more electrodes 114 to one or more other electrodes 114 (multipolar electrical stimulation) (e.g., from a cathode to two anodes, from two cathodes to an anode, or from a cathode to three or more anodes, or from two cathodes to two anodes, etc.). As any of these electrode

combinations and configurations are contemplated, for clarity when referring to one or more processes, methods, or modes of operation, such types of electrical signal transmission may all be referred to as a stimulation transmitted by one electrode 114 or a combination of electrodes 114.

Referring now to FIG. 2, interior surface 103 of flexible body 102 is shown and a relative position of conductive pads 112 and electrodes 114 on exterior surface 101 are schematically depicted for illustrative purposes. Each conductive pad 112 may be connected to a corresponding electrode 114 via a conductive (electrical) lead 116 extending along interior surface 103. Conductive leads 116 may include circuit traces patterned on interior surface 103, and conductive leads 116 may define a conductive pathway between conductive pads 112 and electrodes 114.

Conductive leads 116 may be configured to negate a necessity of including wires to connect electrodes 114 to a power source, thereby simplifying a circuitry construction of circuit device 100. Each conductive lead 116 may include a distal end positioned adjacent to distal end 104 at a location of electrode 114, and a proximal end positioned adjacent to proximal end 106 at a location of conductive pad 112.

It should be understood that conductive pads 112 and electrodes 114 may be patterned on exterior surface 101 opposite of conductive leads 116 on interior surface 103. Circuit device 100 may include at least one conductive lead 116 for each corresponding pair of conductive pads 112 and electrodes 114 on flexible body 102. An electrical current received by circuit device 100 from an ancillary device (e.g., a power source, etc.) at conductive pads 112 may be delivered to electrodes 114 via conductive leads 116.

In some embodiments, flexible body 102 may include a shape memory alloy wire (not shown) disposed along exterior surface 101 and/or interior surface 103. The wire may be configured to selectively manipulate a shape of flexible body 102, such as, for example, in response to receiving a current and/or an increase in temperature (e.g., from a subject's body, etc.). As described in further detail below, the wire may be operable to adjust a shape of a medical device when circuit device 100 is attached thereto. Control of a shape and/or orientation of the medical device may allow for control of a stimulation field generated by electrodes 114. As noted above, a shape and intensity of a stimulation field may be further controlled by a shape, size, charge, and/or arrangement of electrodes 114 relative to one another along flexible body 102.

Referring to FIG. 3, circuit device 100 is shown disposed over and attached to a medical device 10. Medical device 10 may include a tubular body 12 having a longitudinal length defined between a distal tip 14 and a proximal tip 16. In some embodiments, medical device 10 may include a catheter, a flexible tube, a sheath, a scope, a lead, a probe, a hypotube, a patch, a cable, and/or various other flexible instruments suitable for intravascular, percutaneous, or transcutaneous delivery. In the example, medical device 10 may include an intravenous catheter having a flexibility sufficient to traverse a subject's blood vessels, esophagus, nasal cavity, and/or various other anatomical lumens, passageways, etc.

In some examples, tubular body 12 may be formed of an extruded polymer having homogenous properties (e.g., a constant thickness, etc.) between distal tip 14 and proximal tip 16 such that tubular body 12 may include a substantially consistent flexural strength along the longitudinal length. In other embodiments, circuit device 100 may be configured to increase a material thickness along certain portions of tubular body 12 and/or include one or more structures that

modify the properties of tubular body 12 (e.g., increase or decrease rigidity or flexibility, affect shape memory, etc.) when flexible body 102 is attached thereto.

Tubular device 12 may include one or more lumens (see FIG. 6) defined between distal tip 14 and proximal tip 16. The one or more lumens may be sized, shaped, and configured to receive one or more devices therein, including, but not limited to, an irrigation device, an ultrasound device, an optical device, an imaging device, a sensing device, an illumination device, a guidewire, a diagnostic or therapeutic device, etc., and/or receive a fluid. Distal tip 14 may include an opening in communication with the one or more lumens of tubular body 12 such that the one or more devices received within the lumens may extend outwardly from distal tip 14 through the opening. Distal tip 14 may have an atraumatic configuration to minimize injury to tissue when traversed through a subject.

Circuit device 100 may be disposed about medical device 10 and secured to an outer surface of tubular body 12. Flexible body 102 is configured to have a first configuration prior to being secured to an exterior of tubular body 12 and a second configuration having a shape that conforms to a profile of the tubular body 12 with electrodes 114 secured to an exterior of tubular body 12. Accordingly, flexible body 102 may be configured to conform to a geometry of tubular body 12 (e.g., cylindrical). In the example, distal end 104 may be secured to a distal portion of tubular body 12 adjacent to and proximal of distal tip 14. Proximal end 106 may be secured to a proximal portion of tubular body 12 adjacent to and distal of proximal tip 16.

Circuit device 100 may be attached to medical device 10 at one or more locations and by various methods, including, for example, bonding flexible body 102 to an exterior of tubular body 12 (e.g., thermal bonding, thermal welding, thermal lamination, ultrasonic welding, laser welding, RF welding, adhesive bonding, solvent bonding, chemical reaction, and/or solvent bonding). It should be appreciated that a resulting flexural property of tubular body 12 may vary based on various factors, including but not limited to, the method of bonding flexible body 102 thereto, a configuration/arrangement of flexible body 102 relative to tubular body 12, and/or a material composition of flexible body 102. In some embodiments, circuit device 100 may be treated to modify the surface properties of flexible body 102 prior to limitation or bonding onto tubular body 12. Examples of surface treatments may include, but are not limited to, micro-blasting, sand blasting, plasma treatment, chemical etching, filing, grinding, electroplating, thermal spraying, physical vapor deposition, electron beam heating, and more.

The resulting flexural properties of the combined assembly of medical device 10 and circuit device 100 may influence the performance of tubular body 12 during a procedure. For example, the flexural properties may be decreased in specific regions along a longitudinal length of tubular body 12 to influence where tubular body 12 may bend during a procedure, such as when navigating through anatomical structures of a subject. The flexural properties may be decreased to further reduce stress along specific regions of tubular body 12 that may be maintained in a bent configuration when positioned at a target site within a subject. In some embodiments, the flexural properties may be configured to influence a direction of bending tubular body 12.

Still referring to FIG. 3, flexible body 102 may be disposed about tubular body 12 in various arrangements and/or configurations to position electrodes 114 at a desired location relative to tubular body 12. It should be appreciated

that electrodes 114 may be positioned at a variety of axial locations along tubular member 12, and/or various radial positions around tubular member 12, to achieve varying physical properties of medical device 10. For example, flexible body 102 may be attached to tubular body 12 in a manner that may be parallel or transverse (e.g., perpendicular) to a central axis of tubular body 12.

In the present example, flexible body 102 may be attached to tubular body 12 at an angle that is transverse to a central axis of tubular body 12 such that circuit device 100 is secured to medical device 10 in a helical configuration. Flexible body 102 may be wound about the central axis of tubular body 12 in a plurality of helical winds 115 (i.e., helix turns). A pair of adjacent helical winds 115 may form a gap 117 therebetween. Electrodes 114 may be wrapped around a circumference of tubular body 12 and flexible body 102 may conform to a profile of tubular body 12.

Still referring to FIG. 3, flexible body 102 may be helically wrapped with a substantially constant pitch and uniform gaps 117 between adjacent helical winds 115. Accordingly, circuit device 100 may have constant flexural properties along a longitudinal length of flexible body 102. In other embodiments, flexible body 102 may be helically wrapped to have varying pitches and/or gaps (see FIGS. 16-17), including no gaps, to selectively adjust the flexural properties of circuit device 100 at various locations along tubular body 12.

A substantial length of tubular body 12 may be disposed underneath flexible body 102. As described above, bend 108 may form an angle between an intermediate portion of flexible body 102 and proximal end 106 to align proximal end 106 with the central axis of tubular body 12. Accordingly, the plurality of helical winds 115 may terminate at bend 108 such that proximal end 106 may extend substantially parallel to tubular body 12. Stated differently, the helical configuration of flexible body 102 may extend from distal end 104 to bend 108 such that proximal portion 106 may be maintained in a non-helical (e.g., linear) configuration that is parallel to the central axis of tubular body 12.

Still referring to FIG. 3, distal tip 14 may be positioned relatively distal of distal end 104, and proximal tip 16 may be positioned proximal of bend 108 and relatively distal of proximal tab 110. Accordingly, proximal tab 110 may extend proximally of proximal tip 16. As described in greater detail below, proximal tab 110 may extend beyond proximal tip 16 for receipt within a hub 140 (see FIG. 9).

FIGS. 4A-4B are exemplary partial cross-sectional views of FIG. 3, taken at line 4-4. Referring now to FIG. 4A, an exemplary flexible circuitry construction of flexible body 102 is schematically depicted. For example, flexible body 102 may be formed of one or more electrically conductive and/or insulating materials that form conductive pathways between distal end 104 and proximal end 106. Electrode 114 may define a first conductive layer and conductive lead 116 may define a second conductive layer. Circuit device 100 may further include a base layer 120 formed of one or more dielectric substrate films disposed between the pair of electrically conductive layers. Electrode 114 and conductive lead 116 may be affixed to base layer 120 by, for example, depositing, adhering, and/or bonding the layers together.

In some examples, base layer 120 may have a thickness ranging from about 125 microns to about 125 microns. The one or more dielectric substrate films forming base layer 120 may comprise polymer materials, such as, for example, polyimides, polyesters, liquid crystal polymer (LCP), polyethylene naphthalate, polyvinylidene fluoride, polyetherim-

11

ide, various fluoropolymers (FEP), etc. In some examples, the polymer materials comprising base layer 120 may be biocompatible with tissue.

Still referring to FIG. 4A, the one or more electrically conductive materials forming electrode 114 and/or conductive lead 116 may comprise, for example, copper, gold, graphene, silver, nickel, platinum, platinum-iridium alloy, iridium oxide, titanium nitride, tungsten, alloys, and/or combinations thereof. In other embodiments, conductive lead 116 may be patterned to form various other electrical circuit elements.

For example, the electrically conductive layer forming conductive lead 116 may be patterned using additive processes (e.g., spraying, sputtering, sputter coating, screen printing, stenciling, electrodeposition, electroplating, atomic layer deposition, and/or digital printing, etc.) and/or subtractive processes (e.g., chemical etching, laser etching, and/or laser machining, etc.). Stated differently, flexible body 102 may include the desired electrical circuit traces and/or other electrical circuit elements by selectively removing a portion of conductive lead 116 via one or more of the additive and/or subtractive processes described above.

Still referring to FIG. 4A, electrode 114 may be affixed to an opposing side of base layer 120 as conductive lead 116. Further, electrode 114 may be interconnected with conductive lead 116 and through base layer 120 by a via 124. Via 124 may be a hole formed through base layer 120 by, for example, drilling, laser cutting, and/or mechanically puncturing base layer 120 and/or the pair of opposing conductive layers (e.g., electrode 114, conductive lead 116). In some embodiments, circuit device 100 may include a plurality of vias 124, each of which may be filled with a conductive material configured to connect electrode 114 and conductive lead 116 through base layer 120.

In some embodiments, an exterior surface of electrode 114 and/or conductive lead 116 may have one or more materials and/or finishes included thereon, such as, for example, via additive processing (e.g., electrodeposition, electroplating, electroless plating, sputtering, sputter coating, etc.). The one or more materials and/or finishes included on electrode 114 and/or conductive lead 116 may include, for example, nickel, silver, gold, platinum, platinum-iridium, titanium nitride, and the like. The one or more materials and/or finishes may be operable to enhance a conductivity, corrosion resistance, and/or a biocompatibility of circuit device 100.

Additionally, an insulative or non-conductive substrate, layer, coating, film, and/or finish may be attached (e.g., bonded, soldered, etc.) to flexible body 102 to provide insulation of electrode 114, conductive lead 116, and/or base layer 120. Such materials may include, for example, tetrafluoroethylene (TFE), polytetrafluoroethylene (PTFE), polyimide, polyetheretherketone (PEEK), polyester, silicon, and other various flexible substrates.

Still referring to FIG. 4A, a protective cover layer may be included on circuit device 100 and bonded to base layer 120 and/or conductive lead 116. The protective cover layer may include a film and/or finishing (e.g., laminate adhesive layer, etc.) configured to protect the conductive lead 116 (and/or other electrical components) from environmental exposure to moisture, contaminants, mechanical damage, etc., during use of circuit device 100 in a procedure. In the present example, circuit device 100 may include a coverlay film 122 bonded to a surface of conductive lead 116 and base layer 120 with an adhesive 126. Coverlay film 122 may be configured and operable to protect conductive leads 116.

12

Referring now to FIG. 4B, another exemplary flexible circuitry construction of flexible body 102 is schematically depicted. In this example, flexible body 102 includes a plurality of base layers 120 (e.g., dielectric substrate films) secured to one another by at least one adhesive layer 126 positioned between a pair of adjacent base layers 120. Flexible body 102 may further include a plurality of electrodes 114 disposed over an upper-most base layer 120, with each electrode 114 including a corresponding conductive lead 116 disposed under a lower-most base layer 120 (adjacent to tubular body 12).

Each electrode 114 may be interconnected with conductive lead 116 by a respective via 124 that extends through the plurality of base layers 120 disposed therebetween. Each via 124 may be filled with a conductive material configured to connect the conductive materials (e.g., electrode 114 and conductive lead 116) to one another through the plurality of base layers 120 and adhesive layers 126 positioned therebetween. In the present example, flexible body 102 may include three base layers 120 and one coverlay film 122 disposed beneath base layers 120.

Still referring to FIG. 4B, flexible body 102 may further include a device component 130 embedded between the plurality of base layers 120. Device component 130 may be positioned in alignment with a center base layer 120 and between an upper-most base layer 120 and a lower-most base layer 120. Flexible body 102 may further include one or more conductive layers 132 in contact with device component 130 and one or more (e.g., a plurality) vias 124 (filled with conductive material) connecting conductive layers 132 with corresponding conductive leads 116. It should be understood that circuit device 100 may include a respective conductive pad 112 at proximal end 106 corresponding to each of the plurality of electrodes 114 and device component 130.

Device component 130 may include various systems, including, for example, a sensor, an electronic chip, an integrated circuit, and the like. In some embodiments, device component 130 may be fully enclosed between the plurality of base layers 120; while in other embodiments, at least a portion of device component 130 may be exposed from the plurality of base layers 120. In this instance, device component 130 may be accessible to a subject's tissue and/or fluids during use of circuit device 100 in a procedure. For example, device component 130 may include a sensing system configured to detect a signal (e.g., cardiac signal, impedance, temperature, flow, pressure, chemical composition, blood gas, etc.) from a subject (i.e., a patient) to facilitate determining a position of circuit device 100 therein, and/or to detect stimulation of an anatomical target based on a physical response of the subject.

Although not shown in FIGS. 4A-4B, in some embodiments, an adhesive, a polymer, or other filling material may be included to provide a smooth transition between one or more edges of flexible body 102 and a surface of tubular body 12. By providing a smooth transition, medical device 10 and circuit device 100 may be inserted and/or removed from the subject with minimal resistance and potential for tissue trauma.

In other embodiments, device component 130 may include flex detection circuitry that may be configured to detect the flexure of flexible body 102 during a procedure at various regions along the longitudinal length of tubular body 12. Such information may guide a placement of tubular body 12 and/or provide anatomical information to an operator of medical device 10. For example, device component 130 may include a strain circuit, a piezo circuit, and/or various other

13

sensors configured to measure a compression and/or expansion (e.g., lateral extension, longitudinal elongation, etc.) of opposite surfaces of tubular body 12 to determine a two or three-dimensional spatial orientation (e.g., distance, rotation, etc.) of tubular body 12 relative to a subject.

Additionally and/or alternatively, one or more integrated circuits may be directly attached to flexible body 102 to provide further localized smart circuitry. For example, as shown in FIG. 5, one or more device components 130 may be attached (e.g., soldered) to interior surface 103 at various positions along flexible body 102, such as adjacent to distal end 104. In this instance, device components 130 extend laterally outward from interior surface 130, thereby forming a protruding profile relative to interior surface 103.

Circuit device 100 may include a circuit trace 134 along interior surface 103 for each device component 130 included on flexible body 102. Circuit traces 134 may interconnect device components 130 with a corresponding conductive pad 112 at proximal end 106 (FIG. 1). In the present example, medical device 10 may include a longitudinal slot, channel, and/or groove formed along an outer perimeter of tubular body 12 that is sized and shaped to receive the one or more device components 130 when flexible body 102 is secured thereto. The one or more slots on tubular body 102 may receive device components 130 therein to accommodate a flush engagement of flexible body 102 with tubular body 12.

Referring now to FIG. 6, at least one slot 15 may be formed along at least one surface of tubular body 12. Slot 15 may extend through tubular body 12 and into at least one lumen 13 of tubular body 12. Accordingly, lumen 13 may be exposed to an exterior of tubular body 12 via slot 15. It should be understood that tubular body 12 may include a plurality of other lumens 11 that are separate from the at least one lumen 13 aligned with slot 15. Accordingly, the plurality of lumens 11 are not in fluid communication with lumen 13 or with one another. In other examples, two or more of the plurality of lumens 11 may be interconnected with one another and/or with lumen 13 at one or more locations.

Flexible body 102 may be secured to tubular body 12 such that device component 130 may be aligned with and received in slot 15. Thus, interior surface 103 may be flush against an exterior surface of tubular body 12. It should be appreciated that flexible body 102 may cover and/or encapsulate a majority of tubular body 102. Accordingly, slot 115 may be covered when flexible body 102 is attached to tubular body 12, thereby sealing lumen 13 with interior surface 103. Device component 130 may include, for example, a thermistor sensor, an electronic chip, a resistor, a diode, a capacitor, a strain gauge, a piezoelectric sensor, an integrated circuit, and various other electronic devices.

By way of example, device component 130 may be configured to measure a temperature of a fluid, such as, for example, a fluid disposed within one of the plurality of lumens 11, 13 and/or positioned external of medical device 10. Circuit device 100 may be configured to measure a change in temperature based on the temperature data detected by device component 130. In some examples, circuit device 100 may be configured to measure a flow rate and/or other characteristics of the fluid based on the temperature data sensed by device component 130. In other examples, device component 130 may be configured to heat fluid (e.g., blood, etc.) located external of circuit device 100 and measure a temperature change of the fluid at one or more locations along a longitudinal length and/or lateral width of circuit device 100. For example, at least one location for

14

measuring a temperature change of the fluid may be positioned relatively distal to the heat source (e.g., distal component 130) for determining a flow rate or other properties of the fluid.

In some embodiments, as seen in FIG. 7A, circuit device 100 may be printed on a dielectric substrate film in sheet format, such as on a planar sheet 50. Flexible body 102 may be cut from sheet 50 in various desired sizes, shapes, and/or configurations. For example, flexible body 102 may be printed to have a substantially constant width between distal end 104 and proximal end 106. Further, flexible body 102 may be printed to have various suitable lengths between distal end 104 and proximal end 106 in accordance with a longitudinal length of tubular body 12.

Proximal end 106 may be printed on sheet 10 with various configurations, sizes, and/or shapes. For example, proximal end 106 may be printed to include bend 108 having a desired angle. It should be appreciated that bend 108 may define a plurality of angles and/or degrees of curvature between proximal end 106 and an intermediate portion of flexible body 102. Further, proximal end 106 may be printed to include a squared end defining proximal tab 110 (FIGS. 1-3). In other examples, proximal end 106 may be printed to omit bend 108 entirely and/or to include a circular end defining another exemplary proximal tab 110'.

Still referring to FIG. 7A, flexible body 102 may be printed on sheet 50 with one or more bends 105A to accommodate various longitudinal lengths. Bends 105A may be sized to minimize material use on sheet 50 when printing circuit device 100 thereon. For example, the one or more bends 105A may range from about 90 degrees to about 180 degrees. In other examples, as seen in FIG. 7B, flexible body 102 may be printed on sheet 50 with one or more bends 105B that range from about 1 degree to about 89 degrees, such as 45 degrees. In this instance, flexible body 102 may be more easily straightened for attachment onto tubular body 102 than when circuit device 100 is printed with the one or more bends 105A (e.g., bends having a greater angle).

Referring now to FIG. 8, hub 140 may be configured and operable to receive tubular member 12 and flexible body 102 therein. In some embodiments, tubular member 12 and flexible body 102 may be coupled to hub 140; while in other embodiments, hub 140 may be molded over tubular member 12 and flexible body 102. Hub 140 may extend proximally from proximal tip 16 and include one or more extension tubes 150 coupled and/or molded to a proximal end of hub 140. Hub 140 may be configured to interconnect lumens 11 (FIG. 6) with extension tubes 150.

Extension tubes 150 may include a coupling mechanism 152 (e.g., a luer connector, a female adapter, a fitting, and/or other fluid tight connector assemblies) at a distal end of each extension tube 150 for securely coupling an ancillary device (e.g., a tool, a sensor, a cable, a fiber, etc.). Extension tubes 150 may facilitate receipt of the one or more ancillary devices into lumens 11 via hub 140. In some examples, extension tubes 150 may be configured to couple hub 140 to a fluid source for delivering a fluid (e.g., drug, therapeutic agent, etc.) to lumens 11. Extension tubes 150 may be further configured to extract a fluid sample (e.g., blood) collected from medical device 10 and received at hub 140 via lumens 11.

In other examples, extension tubes 150 may receive an ancillary device for monitoring a pressure and/or a temperature (e.g., a sensor) and/or for positioning tubular body 12 within a patient (e.g., a guidewire). Other ancillary devices, such as, for example, optical fibers, ultrasound cameras,

15

intravascular imagers, filters, lasers, lights, and the like may be introduced into lumens 11 through hub 140 and from extension tubes 150.

Still referring to FIG. 8, hub 140 may include a protective sheath 154 secured thereto. Protective sheath 154 may include an electrical connector 156 at a proximal end of protective sheath 154 and may define a channel housing one or more wires and/or cables. As described in further detail herein, the wires may be coupled to electrical connector 156 at a proximal end and be received within hub 140 for connection to flexible body 102 at a distal end.

FIG. 9 shows hub 140 with at least a top wall of hub 140 omitted for illustrative purposes. Proximal end 106 may be received through an inlet 142 of hub 140 such that proximal tab 110 is disposed within hub 140. Hub 140 may further include a proximal wall 144, positioned opposite inlet 142, and one or more outlets 146 formed through proximal wall 144. Each extension tube 150 and protective sheath 154 may be coupled to at least one outlet 146. In some examples, hub 140 may receive one or more wires 118 from protective sheath 154 via at least one outlet 146.

It should be appreciated that proximal hub 140 may incorporate one or more features for facilitating an electrical connection to wires 118. The one or more wires 118 may be connected to conductive pads 112 to thereby establish electrical communication between wires 118 and electrodes 114 (e.g., via conductive leads 116). An electrical junction between conductive pads 112 and wires 118 may be formed by various suitable attachment methods, including, for example, solder, laser weld, conductive adhesive, hot bar, interference, friction fits, and/or various other mechanical connection methods.

Further, the electric junction may be reinforced with potting, overmolding, encapsulating, mechanical fasteners, and/or combinations thereof. Each wire 118 may extend through protective sheath 154 and may terminate at electrical connector 156. Electrical connector 156 may be selectively coupled to a power source configured to supply electrodes 114 with electrical power.

Still referring to FIG. 9, hub 140 may include a rigid body defining a fixed and stable interaction point between flexible body 102 and wires 118. Accordingly, hub 140 may be configured to provide strain relief to flexible body 102 and wires 118 at the juncture between proximal tab 110 and wires 118. In some embodiments, the rigid body of hub 140 may be formed of a thermoplastic material having dielectric properties. Thus, hub 140 may be configured to electrically isolate the connections of conductive pads 112 with wires 118.

FIG. 10 shows another configuration of circuit device 100 in which at least an intermediate portion of flexible body 102 may be received through inlet 142 and into the respective outlet 146 coupled with protective sheath 154. Accordingly, a portion of flexible body 102 may extend through protective sheath 154 and be received within electrical connector 156. It should be understood that wires 118 may be omitted entirely from the present example such that proximal tab 110 may be directly coupled to the power source.

Referring now to FIG. 11, an alternative exemplary proximal tab 110' is depicted at proximal end 106. In the example, proximal tab 110' may have a circular shape and may extend at an angle relative to proximal end 106. In some embodiments, a position and/or orientation of proximal tab 110' relative to proximal end 106 may be selectively adjustable. Proximal tab 110' may include a plurality of apertures 111 formed therethrough and a plurality of conductive pads 112 positioned about the plurality of apertures 111.

16

Stated differently, each conductive pad 112 is formed about at least one aperture 111 such that conductive pads 112 may contact a device received through aperture 111. Internal components of electrical connector 156 are depicted with an outer housing of electrical connector 156 omitted for illustrative purposes. Electrical connector 156 may include a socket 158 having a plurality of openings, and a plurality of pins 159 extending through the plurality of openings. Each of the plurality of pins 159 may have a proximal end extending proximally out of socket 158 and a distal end extending distally out of socket 158.

Still referring to FIG. 11, the distal ends of pins 159 may be disposed within the outer housing of electrical connector 156 when in a fully assembled state for connection with proximal tab 110'. The proximal ends of pins 159 may extend outwardly from the outer housing of electrical connector 156 when in a fully assembled state for connection with a power source. Accordingly, the proximal ends of the plurality of pins 159 may define a connector interface of electrical connector 156.

Each of the plurality of apertures 111 may be sized and shaped to receive at least one of the plurality of pins 159. It should be appreciated that a quantity and position of apertures 111 relative to proximal tab 110' may be based on a number and position of pins 159 on socket 158. Circuit device 100 may establish connection with the power source in response to conductive pads 112 contacting pins 159. In some examples, proximal tab 110' may be soldered to socket 158 thereby securing an electrical connection between conductive pads 112 and pins 159.

In other embodiments, referring now to FIG. 12, circuit device 100 may include an electronic component 135 mounted to flexible body 102 at proximal tab 110. Electronic component 135 may be positioned along exterior surface 101 and encapsulated within hub 140. Electronic component 135 may be secured to flexible body 102 (e.g., at proximal tab 110) via various suitable configurations, including, for example, by a pin and via, an overmolding, potting, and the like. In other embodiments, electronic component 135 may be secured directly to hub 140.

Electronic component 135 may include various suitable devices, including, for example, computer readable storage media such as various types of volatile and non-volatile storage media, including but not limited to, static memory, dynamic memory, a read-only memory chip, a read-write memory chip, an electrically erasable programmable read-only memory (EEPROM), random access memory, flash memory, and the like. In other embodiments, electronic component 135 may include a sensing circuit, a radio-frequency identification (RFID) tag, and other various devices.

For example, electronic component 135 may be configured to store product specific data relating to circuit device 100, such as a serial number, lot number, product configuration, expiration date, calibration constants for device component 130, etc. By way of further example, electronic component 135 may be configured to record use data of circuit device 100 and/or medical device 10, such as dates of activation, variable data from therapy procedure sessions, control unit function algorithms, sensor data of device component 130, etc.

In some embodiments, electronic component 135 may further include a computer system, such as, for example a processor (e.g., a central processing unit (CPU), a graphics processing unit (GPU), or both) in conjunction with the memory described above. For example, the processor may be part of a standard personal computer or a workstation.

The processor may be one or more general processors, digital signal processors, application specific integrated circuits, field programmable gate arrays, servers, networks, digital circuits, analog circuits, or combinations thereof, for analyzing and processing data. The processor may implement a software program, such as programmable code. In other embodiments, the memory may be an external storage device or database separate from electronic component 135.

Still referring to FIG. 12, electronic component 135 may include a controller that is generally configured to accept information from one or more other components of circuit device 100 (e.g., electrodes 114, device component 130, etc.), and process the information according to various algorithms to produce control signals for controlling the other components of circuit device 100.

For example, the controller may accept information from the system and system components, process the information according to various algorithms, and produce information signals that inform a user of circuit device 100 of a procedure status or other information that is being monitored by circuit device 100. The controller may be a digital IC processor, analog processor, or any other suitable logic or control system that carries out control algorithms. Electronic component 135 may be powered by a battery, a direct wire energy, or other suitable power devices. In some examples, electronic component 135 may be configured to communicate with the controller or other components of circuit device 100 via a wired and/or wireless connection.

According to an exemplary method of using circuit device 100 during a procedure, flexible body 102 may be initially bonded to tubular body 12 in one of a plurality of configurations as shown and described above. Proximal tab 110 may be received within hub 140 and coupled to a power source from a connection of conductive pads 114 with electrical connector 156. With circuit device 100 secured to medical device 10, tubular body 12 may be received within a subject (e.g., patient) and navigated toward a target treatment site.

Medical device 10 may be configured to traverse through blood vessels located relatively proximate to one or more nerves (e.g., phrenic nerves, vagus nerves, etc.), muscles, and/or other tissues. A flexibility (or lack thereof) of tubular body 12 may be at least partially determined by the flexural properties of circuit device 100 based on an attachment configuration of flexible body 102 and/or a material composition of flexible body 102. In other words, a geometry and arrangement of circuit device 100 may influence the flexural properties of medical device 10. Medical device 10 may be guided to a target treatment site using one or more devices received within lumens 11 (FIG. 6) via extension tubes 150 and hub 140. For example, medical device 10 may receive a stiffening element (e.g., polymer and/or metallic mandrels), a guidewire, a sensor, an imaging device, an illumination device, a shape memory material (e.g., Nitinol, etc.), and/or other therapeutic or diagnostic medical devices to facilitate placement of circuit device 100.

Accordingly, tubular member 12 may be configured to guide placement of circuit device 100 relative to the target treatment site. Upon arriving at the target treatment site, tubular body 12 may be moved (e.g., translated, rotated, etc.) to position electrodes 114 at a desired location relative to a target object (e.g., a nerve, a muscle, a tissue, etc.). With electrodes 114 positioned adjacent to the target object, activation of the power source may deliver an electrical current through wires 118, conductive pads 112, and to electrodes 114 via conductive leads 116. Electrodes 114 may be configured to stimulate the target object (e.g., a stimula-

tion sufficient to cause a desired physiological result) upon receiving the electrical current from conductive leads 116.

Electrodes 114 may generate therapeutic effects for specific anatomical targets (e.g., nerve stimulation) and a relative position of electrodes 114 to one another on exterior surface 101 may generate one or more charge fields along tubular body 12 for certain therapeutic effects (e.g., asymmetric charge fields). Electrodes 114 may generate a sufficient charge to cause contraction of a diaphragm or other lung-accessory. In some embodiments, electrodes 114 may be positioned along tubular body 12 to form a sensor array antenna configured and operable to detect various forms of stimulation, including, for example, nerve stimulation, muscle stimulation, cardiac stimulation, phrenic nerve activation, diaphragm muscle activation, and more.

In examples where circuit device 100 includes device component 130 (e.g., a sensor, etc.), device component 130 (FIGS. 4B-6) may be positioned toward the target object and activated to detect and/or record one or more characteristics or properties of the target object (e.g., input on whether the requisite stimulation of an anatomical target occurred). Device component 130 may be communicatively coupled to one or more computing systems, such as, for example, via electrical component 135 (FIG. 12) by a wired or wireless connection.

It should be understood that with circuit device 100 secured to an exterior of medical device 10, lumens 11 may be utilized to receive other devices than those included in circuit device 100. Stated differently, circuit device 100 may be operable to maximize an available space within lumens 11 to receive other ancillary devices. For example, circuit device 100 may omit a necessity to route wires internally through tubular body 12 for connecting electrodes 114 to a power source, thereby allowing tubular body 12 to be sized with a smaller minimum diameter.

Further, circuit device 100 includes conductive leads 116 such that use of wires to connect electrodes 114 to a power source may be omitted entirely, whether received within and/or disposed over tubular body 12, thereby further reducing a complexity of medical device 10. Accordingly, tubular body 12 may include a smaller cross-sectional profile than other medical devices (e.g., catheters) sized and shaped to accommodate electrical components (e.g., wires) for including one or more electrodes on the device.

In some embodiments, medical device 10 may sense physiologic parameters of the target treatment site using one or more devices (e.g., intravascular ultrasound (IVUS)) received within lumens 11 (FIG. 6) via extension tubes 150 and hub 140. In other embodiments, medical device 10 may receive devices that vary the flexural properties of tubular body 12 to create zones of stiff and/or flexible regions. For example, a Nitinol wire (not shown) may be received within lumen 11 to alter or control the shape of tubular body 12 in response to the wire receiving a current applied thereto. In this instance, the wire may apply a force onto tubular body 12 from within lumen 11 to a fixed/desired configuration. Removal of the current may terminate the force applied to tubular body 12 by the wire, thereby permitting tubular body 12 to return to a flexible state. In other examples, the shape memory wire may be activated by the heat produced by the subject's body.

FIGS. 13-24 show alternative exemplary circuit devices that may be substantially similar to circuit device 100. A geometry, orientation, and layer construction of the circuit devices relative to an axis of tubular body 12 may influence the flexural properties of medical device 10 in specific regions along the longitudinal length of tubular body 12. It

19

should be understood that the circuit devices described in the examples below may be configured and operable similar to circuit device **100** except for the differences explicitly noted herein, and therefore like reference numerals are used to identify like components. It should further be appreciated that the exemplary circuit devices below may be readily incorporated onto medical device **10**.

Referring specifically to FIG. **13**, an exemplary circuit device **200** is shown in accordance with an example of this disclosure. Circuit device **200** may include a varying geometry and/or width along a longitudinal length of flexible body **102**. For example, flexible body **102** may include one or more widened segments **203** between distal end **104** and proximal end **106**. In the present example, circuit device **200** may include a pair of widened segments **203** separated from one another along the longitudinal length of flexible body **102** by an intermediate segment **205** having a relatively smaller width.

Flexible body **102** may include layers having a greater thickness along widened segments **203** relative to intermediate segment **205**. Widened segments **203** may be configured to impart reduced flexibility on flexible body **102**. In the example, widened segments **203** have a relatively smaller longitudinal length than intermediate segment **205**, however, it should be understood that various other relative lengths and/or configurations of segments **203**, **205** may be included without departing from a scope of this disclosure. In other embodiments, additional and/or fewer widened segments **203** and/or intermediate segments **205** may be included along flexible body **102**.

Still referring to FIG. **13**, flexible body **102** may include a plurality of electrodes **114** along the pair of widened segments **203** and intermediate segment **205**. Thus, proximal tab **110** may include a plurality of conductive pads **112** at proximal end **106** in accordance with a quantity of electrodes **114** on flexible body **102**. In some embodiments, one or more of widened segments **203** and/or intermediate segment **205** may exclude any electrodes **114** thereon. In use, widened segments **203** may be configured and operable to reduce the flexural properties of tubular body **12** along the respective portions of tubular body **12** that align with widened segments **203** when circuit device **200** is attached to medical device **10**.

Referring now to FIG. **14**, an exemplary circuit device **300** is shown in accordance with another example of this disclosure. Circuit device **300** may include an irregular segment **303** along a longitudinal length of flexible body **102**. Irregular segment **303** may be included between distal end **104** and proximal end **106** and may have a geometry and/or configuration that is transverse to a remaining linear geometry of flexible body **102**. For example, irregular segment **303** may have a sinusoidal-shape (e.g., S-profile) including a plurality of convex waves **305** (e.g., peaks) and a plurality of concave waves **307** (e.g., troughs). The plurality of waves **305**, **307** may have various suitable lengths and/or amplitudes relative to a longitudinal axis of flexible body **102**.

In the present example, irregular segment **303** includes a pair of convex waves **305** and a pair of concave waves **307**, however, it should be understood that additional and/or fewer waves **305**, **307** may be included on irregular segment **303**. In other embodiments, flexible body **102** may include additional irregular segments **303** between distal end **104** and proximal end **106**. In the present example, irregular segment **303** may be formed along flexible body **102** adjacent to distal end **104** and may include a constant width with a remaining portion of flexible body **102**. Further, irregular

20

segment **303** may be configured to increase a flexibility of flexible body **102** relative to the remaining portion of flexible body **102**.

Still referring to FIG. **14**, flexible body **102** may include a plurality of electrodes **114** along portions of flexible body **102** proximal to and distal of irregular segment **303** such that irregular segment **303** excludes electrodes **114** thereon. In some embodiments, one or more of convex waves **305** and/or concave waves **307** may include electrodes **114** thereon. In use, irregular segment **303** may be configured and operable to relatively increase the flexural properties of tubular body **12** along the respective portions of tubular body **12** that align with irregular segment **303** when circuit device **300** is attached to medical device **10**.

Referring now to FIG. **15**, an exemplary circuit device **400** is shown in accordance with another example of this disclosure. Circuit device **400** may include one or more angular changes/bends along a longitudinal length of flexible body **102**. For example, flexible body **102** may include a plurality of segments **404**, **406**, **408** separated from one another by at least one bend **410**, respectively. Accordingly, each of the plurality of segments **404**, **406**, **408** may be positioned and aligned relatively transverse to one another.

In the present example, a distal segment **404** is positioned adjacent to distal end **104**, a proximal segment **408** is positioned adjacent to proximal end **106**, and an intermediate segment **406** is disposed between distal segment **404** and proximal segment **408**. Flexible body **102** may include a pair of bends **410**, at least one of which is disposed between distal segment **404** and intermediate segment **406** and another of which is disposed between intermediate segment **406** and proximal segment **408**.

Still referring to FIG. **15**, each of the plurality of segments **404**, **406**, **408** may have varying flexural properties relative to one another based on an angle of bends **410** between an adjacent pair of segments. Bends **410** may range from about 1 degree to about 60 degrees and each bend **410** may extend an adjacent pair of segments in a different (or similar) direction and/or angle relative to one another. Further, each bend **410** may include varying angles relative to the one or more other bends **410** on flexible body **102**.

In other embodiments, circuit device **400** may include additional and/or fewer segments along flexible body **102** that may be separated by additional and/or fewer bends **410**, respectively. Each of the plurality of segments **404**, **406**, **408** may include one or more electrodes **114**; while in some embodiments, segments **404**, **406**, **408** may omit electrodes **114** entirely.

Referring now to FIG. **16**, each of the plurality of segments **404**, **406**, **408** may cause the corresponding portion of the overall medical device to have varying flexural properties relative to one another based on an angle of helical wrap of each segment **404**, **406**, **408** about tubular body **12**. In other words, the angle, pitch, and/or spacing between windings that flexible body **102** helically winds about tubular body **12** may vary at each segment **404**, **406**, **408**. For example, a pitch of helical wind of flexible body **102** may be relatively greater at distal segment **404** and proximal segment **408** than at intermediate segment **406**. In this instance, flexible body **102** may be substantially flexible along distal segment **404** and proximal segment **408** relative to intermediate segment **406**.

By way of further example, the pitch or angle that flexible body **102** helically winds about tubular body **12** may be relatively smaller along intermediate segment **406** such that flexible body **102** may be substantially rigid along intermediate segment **406**. It should be appreciated that segments

21

404, 406, 408 may include various other angles and pitches of helical wrap than those shown and described herein, and relative to one another, than those shown and described herein dependent on a desired configuration of circuit device 400.

Still referring to FIG. 16, a direction of the helical wrap of flexible body 102 may change directions between one or more of the plurality of segments 404, 406, 408. In the present example, distal segment 404 and proximal segment 408 may be wound about tubular body 12 in a first direction A (e.g., clockwise direction) and intermediate segment 406 may wrap around tubular body 12 in a second direction B (e.g., counter clockwise direction) that is different than the first direction A. Further, distal segment 404 and/or proximal segment 408 may helically wind about tubular body 12 at a first helical pitch α and intermediate segment 406 may helically wind about tubular body 12 at a second helical pitch β that is different than first helical pitch α . For example, first helical pitch α may be relatively less than second helical pitch β such that distal segment 404 and/or proximal segment 408 may include fewer helical winds than intermediate segment 406. It should be appreciated that segments 404, 406, 408 may include various other helical pitches and/or winds relative to one another than those shown and described herein.

Circuit device 400 may be configured to form a gap 405 between adjacent segments 404, 406, 408 that change direction relative to one another when flexible body 102 is attached to tubular body 12. Circuit device 400 may not cover at least a portion of tubular body 12 when the helical wrap of flexible body 102 changes directions, thereby exposing a portion of tubular body 12 disposed beneath flexible body 102.

It should be appreciated that the flexural properties of tubular body 12 may vary along the portions that are exposed along gaps 405 relative to other portions that receive flexible body 102. For example, tubular body 12 may have a relatively decreased rigidity (i.e., an increased flexibility) along regions that omit flexible body 102 thereon. Further, an assembly of medical device 10 and circuit device 100 may provide areas having a greater material thickness along portions where flexible body 102 is attached to tubular body 12, thereby providing asymmetric properties along the longitudinal length of tubular body 12.

Still referring to FIG. 16, one or more features of medical device 10 on tubular body 12 (e.g., access holes 18) may be revealed and accessible along gap 405 when flexible body 102 is wound about tubular body 12. A size of gap 405 may vary and may be dependent on, for example, the angle of the helical wrap between the adjacent pair of segments defining gap 405. Access holes 18 may be in fluid communication with one or more lumens 11 (FIG. 6) such that a fluid and/or a device (e.g. a guidewire) may be delivered outwardly from tubular body 12 and/or a received into tubular body 12 via access holes 18. In some examples, access holes 18 may provide sensing portals to allow a device (e.g., fiber optic sensor, etc.) to measure one or more parameters, such as, for example, oxygen levels of a fluid (e.g., blood, etc.), central venous pressure, and more. Further, access holes 18 may provide deployment portals that allow one or more instruments to extend externally from tubular body 12. Although not shown, in some embodiments, access holes 18 may facilitate deployment of a Nitinol positioning device from tubular body 12 for contacting a vessel wall, thereby biasing tubular body 12 to a preferred configuration and stabilizing medical device 10 relative to a target treatment site.

22

It should be appreciated that an angle and/or a pitch of helical wrap of each the plurality of segments 404, 406, 408 may remain constant and/or vary relative to one another when a direction of the helical wind changes. In some embodiments, bend 410 may match an angle of the helical wrap of flexible body 102. In this instance, at least one of the plurality of segments 404, 406, 408 may have an elongated length extending along a longitudinal axis that is transverse to the remaining segments of flexible body 102.

For example, as seen in FIG. 17, distal segment 404 may extend parallel to a central axis of tubular body 12 when circuit device 400 is secured to medical device 10, and an angular change at bend 410 between distal segment 404 and intermediate segment 406 may be determinative of an angle of the helical wrap of flexible body 102 about tubular body 12. Stated differently, an angle of bend 410 may define, or correspond to, a helix angle of flexible body 102 wrapping about the central axis of tubular body 12.

As seen in FIG. 17A, bend 410 may define a bend angle C between distal segment 404 of flexible body 102 and intermediate segment 406. In some embodiments, bend angle C may range from about 1 degree to about 179 degrees, and may at least partially define a configuration of the plurality of helical winds (of flexible body 102) relative to tubular body 12. For example, relative to an axis T that is transverse to a central axis L of tubular body 12, the plurality of helical winds along intermediate segment 406 may include a helix angle D. Helix angle D may have a relationship relative to bend angle C such that bend angle C may be substantially equal to helix angle D plus about 90 degrees, when distal segment 404 is aligned with, or substantially parallel to, axis of tubular body 12, ($C=D+90$ degrees).

It should be understood that bend angle C and helix angle D of the present example are merely illustrative such that bend 410 and the plurality of helical winds along intermediate segment 406 may have various other suitable relationships relative to one another than those shown and described herein. Additionally, a helical pitch of intermediate segment 406 (e.g., see FIG. 16, first helical α , second helical pitch β , etc.) may be at least partially defined by, or otherwise correspond to, bend angle C and/or helix angle D of flexible body 102. Thus, intermediate segment 406 may be wrapped in a helical configuration about tubular body 12. In the present example, circuit device 400 only includes a single bend 410 such that intermediate segment 406 extends to and terminates at bend 108.

Flexible body 102, and the portion of the medical device corresponding to it, may have varying flexural properties along distal segment 404 and intermediate segment 406 based on a respective geometry of each segment 404, 406. In the present example, distal segment 404 may be oriented parallel to a central axis of tubular body 12 such that the portion of the medical device corresponding to it distal segment 104 may have radial asymmetric flexural properties. Medical device 10 may experience greater deflection along a portion corresponding to distal segment 404 when a force that is transverse to a longitudinal axis of flexible body 102 and parallel to a central axis of tubular body 12 is applied, than when the force is perpendicular to both flexible body 102 and tubular body 12.

Still referring to FIG. 17, a radial flexural asymmetry of distal segment 404 may provide a greater rigidity on tubular body 12, such as, for example, in a specific direction when tubular body 12 is bent during use in a procedure. For example, a distal portion of tubular body 12 may be bent in one of a plurality of directions relative to a target treatment site, such as, for example, to contact a vessel wall within a

23

subject. The distal portion of tubular body 12 may maintain contact with the vessel wall due to an enhanced rigidity of the distal portion provided by the radial flexural asymmetry of distal segment 404. In contrast to the radial asymmetric flexural properties of distal segment 404, intermediate segment 406 may have radial symmetric flexural properties based on the geometry of intermediate segment 406.

Referring now to FIG. 18, another exemplary circuit device 500 is shown in accordance with an example of this disclosure. Circuit device 500 may include one or more segments 504 having a stiffening structure formed therein, and one or more segments 506 omitting a stiffening structure. In the present example, circuit device 500 may include a pair of stiffened segments 504 and a pair non-stiffened segments 506 positioned between distal end 104 and proximal end 106. Stiffened segments 504 alternate with non-stiffened segments 506 along the length of circuit device 500. In other embodiments, circuit device 500 may include additional and/or fewer segments 504, 506 along flexible body 102 in various other suitable arrangements than those shown and described herein.

The stiffening structures may include components (e.g., additional layers, adhesive used for lamination, stiffening wires, etc.) made of materials of varying stiffness and/or thickness and that may be incorporated into regions of flexible body 102 to increase and/or decrease a stiffness of circuit device 500. Stiffeners may include glass-reinforced epoxy laminate materials (FR4), polyimide, polyester, polymers, and/or stainless steel. The stiffening structures may be formed of materials having a relatively high stiffness to increase a rigidity of tubular body 12 in the specific areas aligned with stiffening segments 504 when flexible body 102 is attached thereto.

Stated differently, flexible body 102 may be configured to reduce a flexibility of tubular body 12 along areas aligned with stiffening segments 504. The stiffening structures may provide additional layers on tubular body 12, thereby varying a width of the assembly of medical device 10 and circuit device 500 along a longitudinal length of tubular body 12. Accordingly, the flexural properties of tubular body 12 may vary at the selective regions corresponding to a location of stiffened segments 504 thereon. In other embodiments, a stiffening structure may be disposed between an exterior surface of tubular body 12 and interior surface 103 of flexible body 102 prior to attaching circuit device 100 to medical device 10. In this instance, the stiffening structure may be encapsulated between tubular body 12 and flexible body 102.

Referring now to FIG. 19, another exemplary circuit device 600 is shown in accordance with an example of this disclosure. Circuit device 600 may include a distal segment 604 and an intermediate segment 606 defining a longitudinal length of flexible body 102 between distal end 104 and proximal end 106. Distal segment 604 may include a pair of branch segments 608 extending distally from intermediate segment 606 such that distal end 104 may be defined by the pair of branch segments 608. Each branch segment 608 may be in parallel alignment with one another and may have a longitudinal length that is equal to, greater than, and/or less than one another. In the present example, branch segments 608 may have a substantially similar length that is less than a longitudinal length of intermediate segment 606. Each of branch segments 608 may have a width that is relatively less than a width of intermediate segment 606.

Still referring to FIG. 19, flexible body 102 may further include a bend 610 at a proximal end of the pair of branch segments 608 such that branch segments 608 may be angled

24

relative to intermediate segment 606. Thus, each branch segment 608 may have a longitudinal axis that is transverse to an axis of intermediate segment 606. In some examples, an angular change of flexible body 102 at bend 610 may be determinative of an angle of a helical wrap of intermediate segment 606 about tubular body 12 when circuit device 600 is secured to medical device 10 (see, e.g., FIG. 17A, bend angle C).

Distal segment 604 and/or intermediate segment 606 may include one or more (e.g., a plurality) of electrodes 114. In the present example, each of the pair of branch segments 608 may define an array of electrodes 114 along distal segment 604, thereby providing a complex circuit density at distal segment 604. It should be appreciated that circuit device 600 may be configured to generate greater current and/or fields of coverage with the array of electrodes 114 on branch segments 608 for providing enhanced treatment therapies to a target site. The array of electrodes 114 may provide medical device 10 with complex electrode geometries capable of selectively generating greater current densities and/or concentrated density fields.

Circuit device 600 may be assembled onto medical device 10 such that the longitudinal axis of each branch segment 608 may be parallel to a center, longitudinal axis of tubular body 12. Stated differently, distal segment 604 may be secured onto a distal portion of tubular body 12 in a linear arrangement such that branch segments 608 extend parallel to the center axis of tubular body 12. Further, an axis of intermediate segment 606 may be transverse to the center axis of tubular body 12 such that flexible body 102 may be helically wound about an exterior of tubular body 12 along intermediate segment 606. Accordingly, circuit device 600 may be configured to impart different flexural properties onto tubular body 12 along a portion receiving distal segment 604 and intermediate segment 606, respectively.

Still referring to FIG. 19, branch segments 608 may be spaced apart from one another by an angle and/or distance when flexible body 102 is mounted onto tubular body 12. For example, each branch segment 608 may be positioned about 180 degrees apart from another when secured to a distal portion of tubular body 12. Thus, circuit device 600 may be configured to provide a pair of opposing electrode arrays that are offset from one another and positioned about a circumference of tubular body 12. For example, branch segments 608 may be positioned along a top and bottom (e.g., or a left and right) exterior surface of tubular body 12 when circuit device 600 is attached to medical device 10.

It should be appreciated that branch segments 608 may be selectively spaced apart from one another at various other relative degrees and/or distances dependent on a desired therapeutic effect to be provided by circuit device 600 and medical device 10. In other embodiments, circuit device 600 may include additional branch segments 608 at distal segment 604, intermediate segment 606, and/or at various other portions of flexible body 102.

For example, as shown in FIG. 20, circuit device 600 may include a second pair of branch segments 609 extending distally from intermediate section 606 and angled relative thereto by bend 610. In the present example, the pair of branch segments 609 may extend parallel to branch segments 608 and may have a longitudinal length that is equal to, greater than, and/or less than a longitudinal length of branch segments 608. In the example, branch segments 609 may have a smaller length than branch segments 608 and may be positioned in an alternating configuration with branch segments 608. Thus, each branch segment 608 may

25

be separated from one another by at least one branch segment 609 positioned therebetween.

Each branch segment 609 may include one or more (e.g., a plurality) of electrodes 114, thereby defining an array of electrodes 114 on each branch segment 609. In some examples, electrodes 114 may be arranged along each branch segment 609 to be longitudinally aligned with one another, and longitudinally offset from the plurality of electrodes 114 along branch segments 608.

Still referring to FIG. 20, branch segments 609 may be spaced apart from one another, and/or from branch segments 608, by an angle and/or distance when flexible body 102 is mounted onto tubular body 12. For example, each branch segment 609 may be positioned about 180 degrees from one another and about 90 degrees from each branch segment 608 when secured to a distal portion of tubular body 12. Thus, circuit device 600 may provide two pairs of opposing electrode arrays that are offset from one another and positioned about a circumference of tubular body 12.

For example, FIG. 21 shows branch segments 608 positioned along a top and bottom exterior surface of tubular body 12 (only the top can be seen in the Figure), and branch segments 609 positioned along a left and right side surface (only the left side can be seen in the Figure) when circuit device 600 is attached to medical device 10. It should be understood that branch segments 608 are spaced apart from one another by 180 degrees on tubular body 12 and branch segments 609 are spaced apart from one another by 180 degrees. Further, each branch segment 608 is positioned apart from branch segment 609 by 90 degrees on tubular body 12. In other embodiments, circuit device 600 may include additional and/or fewer branch segments 608, 609 at distal segment 604, intermediate segment 606, and/or at various other portions of flexible body 102.

Referring now to FIG. 22, an exemplary circuit device 700 is schematically depicted in accordance with another example of this disclosure. Circuit device 700 may include flexible body 102 having a longitudinal length defined by a distal segment 702, a first intermediate segment 704, a second intermediate segment 706, and a proximal segment 708. Distal segment 702 may terminate at distal end 104 and extend distally relative to first intermediate segment 704. Distal segment 702 may include the pair of branch segments 608 extending at an angle relative to first intermediate segment 702 by a bend 710.

First intermediate segment 704 may extend distally relative to second intermediate segment 706 such that first intermediate segment 704 may be disposed between distal segment 702 and second intermediate segment 706. First intermediate segment 704 may have a longitudinal length (an axis) extending transverse to the longitudinal axes of each of distal segment 702 and second intermediate segment 706. In some examples, first intermediate segment 704 may include one or more (e.g., a plurality) of electrodes 114; while in other examples, first intermediate segment 704 may exclude electrodes 114 thereon.

Still referring to FIG. 22, flexible body 102 may further include a second bend 712 between first intermediate segment 704 and second intermediate segment 706 such that first intermediate segment 704 may be configured to extend at an angle relative to second intermediate segment 706 by bend 712. Second intermediate segment 706 may include the pair of branch segments 609, with first intermediate segment 704 extending distally from a distal end of at least one of the pair of branch segments 609. Stated differently, each branch segment 608, 609 has a free distal end except for at least one

26

of branch segments 609 which is secured to a proximal end of first intermediate segment 704.

Accordingly, the pair of branch segments 609 may be positioned proximal to, and at angle from, the pair of branch segments 608. Thus, each branch segment 609 may have a longitudinal axis that is transverse to an axis of the pair of branch segments 608. Second intermediate segment 706 may extend distally relative to proximal intermediate segment 708 such that second intermediate segment 706 may be disposed between first intermediate segment 704 and proximal segment 708. Second intermediate segment 706 may have a longitudinal length (an axis) extending transverse to a longitudinal axis of each of first intermediate segment 704 and proximal segment 708. In some examples, first intermediate segment 704 may have an axis that is parallel to an axis of proximal segment 708.

Still referring to FIG. 22, second intermediate segment 706 may include one or more (e.g., a plurality) of electrodes 114 on one or more of the pair of branch segments 609; while in other examples, second intermediate segment 706 may exclude electrodes 114 entirely. Flexible body 102 may include a bend 714 between second intermediate segment 706 and proximal segment 708 such that second intermediate segment 706 may be configured to extend at an angle relative to proximal segment 708 by bend 714.

Distal segment 702 and second intermediate segment 706 may extend parallel to each other and to a central longitudinal axis of tubular body 12, and first intermediate segment 704 and proximal segment 708 may be helically wrapped about tubular body 12, when circuit device 700 is attached to medical device 10. Accordingly, varying flexural properties may be imparted on tubular body 12 along those portions receiving segments 702, 706 thereon compared to segments 704, 708.

The pair of branches 608 may be secured to tubular body 12 along a similar or different radial location about the axis of body 12 than the pair of branches 609 (i.e., either spaced 180 degrees or 90 degrees apart, respectively, or any other suitable angle apart). An angular change of flexible body 102 at bends 710, 712, 714 may be determinative of an angle of a helical wrap of first intermediate segment 704 and proximal segment 708 about tubular body 12 when circuit device 700 is secured to medical device 10 (see, e.g., FIG. 17A, bend angle C). Accordingly, one or more gaps may be formed between branch segments 608 and branch segments 609.

Referring now to FIG. 23, an exemplary circuit device 800 is schematically depicted in accordance with another example of this disclosure. Circuit device 800 may include an alternative exemplary proximal tab 110" at proximal end 106. Proximal tab 110" may include a rectangular shape having one or more (e.g. a plurality) conductive pads 112 positioned thereon. In the example, proximal tab 110" may define a connector interface of circuit device 800 for connection with a power source (not shown). Stated differently, proximal tab 110" may be configured to directly connect with a power source in lieu of the circuit device establishing connection with the power source via one or more wires 118 (FIGS. 8-9).

FIG. 24 shows electrical connector 156 in a disassembled state for illustrative purposes. As shown, flexible body 102 may be received within electrical connector 156 via protective sheath 154. In the example, protective sheath 154 may be secured to electrical connector 156 at an inlet port 162 of electrical connector 156. Further, electrical connector 156 may include a top housing 160A and a bottom housing 160B defining an interior cavity for receiving proximal end 106. In

the example, electrical connector **156** may include an outlet connector **164** in lieu of socket **158** and pins **159** (see FIG. **11**).

Outlet connector **164** may extend proximally outward from housings **160A**, **160B** when electrical connector **156** is in a fully assembled state. Outlet connector **164** may define a connector interface of electrical connector **156** for connection with a power source and may be configured to receive proximal tab **110"**. Proximal tab **110"** may be disposed about an exterior of outlet connector **164** with the plurality of conductive pads **112** facing radially outward. Outlet connector **164** may include a keying feature **166** configured to facilitate an alignment and attachment of proximal tab **110"** to outlet connector **164**. For example, keying feature **166** may engage the terminal ends of proximal tab **110"** and/or designate a starting and ending point for wrapping proximal tab **110"** about a circumference of outlet connector **164**.

Still referring to FIG. **24**, proximal tab **110"** may be secured to outlet connector **164** by various suitable mechanisms, including, for example, by an adhesive. In exemplary use, conductive pads **112** may be exposed from housings **160A**, **160B** when proximal end **106** is received within electrical connector **156** and proximal tab **110"** is secured to outlet connector **164**. Conductive pads **112** may connect directly to the power source when electrical connector **156** is coupled thereto and outlet connector **164** is received within the power source, thereby providing electrical power to circuit device **800**.

Each of the aforementioned systems, devices, assemblies, and methods may be used to provide electrical circuitry at a target treatment site in a subject (e.g., a patient) that may be capable of providing diagnostic or therapeutic effects. By providing a medical device including a circuit device disposed at the exterior of the medical device (over the sheath or tube), a user may utilize the lumens of the medical device for receipt of others devices and/or materials. With the circuit device having a minimal size and flexible electrical circuitry, the circuit device may minimize an overall physical profile of the medical device while maximizing a selective positioning of the electrical components at various locations along the medical device. In this instance, a user may increase efficiency of procedures, reduce overall procedure time, and/or avoid unnecessary harm to a subject's body caused by the medical device having an invasive profile for accommodating electrical circuitry devices within the lumens.

It will be apparent to those skilled in the art that various modifications and variations may be made in the disclosed devices and methods without departing from the scope of this disclosure. Other aspects of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the features disclosed herein. It is intended that the specification and examples be considered as exemplary only.

What is claimed is:

1. A medical device comprising:
 - a tube having a longitudinal length and at least one lumen;
 - a flexible body including a stimulation array configured to provide an electrical charge for stimulating tissue;
 - an electrical connection positioned on the flexible body proximal to the stimulation array, wherein the electrical connection is configured to electrically couple the stimulation array to a power source; and
 - an electronic component positioned between the stimulation array and the power source, wherein the electronic

component includes a memory and a processor configured to implement programmable code stored on the memory;

wherein the flexible body is configured to cover at least a portion of the longitudinal length of the tube such that the stimulation array is disposed about an exterior of the tube.

2. The medical device of claim **1**, wherein the memory is configured to store data relating to a serial number, a lot number, a product configuration, an expiration date, a calibration constant, sensor data, or a combination thereof.

3. The medical device of claim **1**, wherein the memory is an electrically erasable programmable read-only memory.

4. The medical device of claim **1**, wherein the flexible body includes (1) a lead that electrically couples the electrical connection to the stimulation array and (2) a dielectric substrate film between the lead and the stimulation array.

5. The medical device of claim **4**, wherein the flexible body further includes a sensor disposed within the dielectric substrate film; and

the processor of the electronic component is configured to:

receive information from the sensor; and

process the information received from the sensor to produce a control signal for controlling operation of the medical device.

6. The medical device of claim **1**, wherein the flexible body is helically wound about the exterior of the tube in a plurality of winds, and a first position of the tube that contacts the helically wound flexible body has a different flexibility than a second position of the tube that contacts the helically wound flexible body.

7. A medical device comprising:

a flexible body;

an electrode positioned on the flexible body, wherein the electrode is configured to provide an electrical charge for stimulating tissue;

an electrical connection including a proximal end and a distal end, wherein the electrical connection is configured to electrically couple the electrode to a power source;

an electrical lead connecting the electrode to the electrical connection, wherein the electrical lead is on or in the flexible body;

an electronic component between the proximal end of the electrical connection and the electrode, wherein the electronic component includes a memory, a sensing circuit, a radio-frequency identification tag, a processor, or a combination thereof; and

a dielectric substrate film disposed between the electrode and the electrical lead, wherein the dielectric substrate film includes a hole filled with a conductive material electrically connecting the electrode to the electrical lead.

8. The medical device of claim **7**, wherein the electronic component includes a memory, the memory is an electrically erasable programmable read-only memory (EEPROM), and the EEPROM is configured to store data related to an expiration date, a date of activation, and/or data from a therapy session.

9. The medical device of claim **7**, wherein the distal end of the electrical connection is proximate to a proximal end of the flexible body.

10. The medical device of claim **7**, wherein the electronic component is mounted on the flexible body.

29

11. The medical device of claim 10, further comprising a hub, wherein the electronic component is located within the hub.

12. The medical device of claim 11, wherein the hub further includes an inlet, and the flexible body passes through the inlet.

13. The medical device of claim 7, further including a sensor, an electronic chip, or an integrated circuit disposed within the dielectric substrate film and between the electrode and the electrical lead.

14. A medical device comprising:

a tube including a lumen; and

a circuit having a flexible body, wherein the circuit comprises:

a distal portion, wherein the distal portion includes an electrode configured to provide an electrical charge for stimulating tissue;

a proximal portion coupled to a distal end of an electrical connection, wherein the proximal portion is angled relative to the distal portion, and the electrical connection is configured to electrically couple the electrode to a power source;

an electronic component between a proximal end of the electrical connection and the distal portion of the circuit, wherein the electronic component includes a memory, a sensing circuit, a radio-frequency identification tag, a processor, or a combination thereof;

a conductive lead electrically coupled to the electrical connection and the electrode; and

a via between the conductive lead and the electrode, wherein the via provides an electrical connection between the conductive lead and the electrode.

30

15. The medical device of claim 14, wherein the electronic component is configured to record use data of the medical device.

16. The medical device of claim 14, wherein a length of the circuit is helically wound around the tube.

17. The medical device of claim 14, wherein the circuit includes at least one segment having an enlarged width relative to an adjacent portion of the circuit, and wherein the at least one segment is less flexible than the adjacent portion of the circuit.

18. The medical device of claim 14, wherein the medical device further includes a hub comprising:

an inlet, wherein a length of the circuit is positioned within the inlet; and

an extension tube configured to pass fluids to and from the lumen.

19. The medical device of claim 18, wherein the electrical connection is positioned within the hub.

20. The medical device of claim 14, wherein:

the electrode is a first electrode;

the distal portion of the circuit includes a first branch and a second branch;

the first and second branches are arranged substantially parallel to a longitudinal axis of the tube;

the first branch includes the first electrode and a second electrode; and

the second branch includes a third electrode and a fourth electrode.

* * * * *